

Inequality and Misallocation under Production Networks

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Abstract

Redistribution can change aggregate productivity when households' spending patterns differ in a distorted economy. I develop a heterogeneous-household accounting framework for production network economies in which factor-income, expenditure, and demand reallocations affect TFP through wedge-weighted centralities. The same framework delivers household incidence through positional terms of trade (PTT), whose expenditure-weighted average equals TFP growth. Applying the sufficient statistics to U.S. data, 1997–2021, I find that distributional reallocation explains about 20% of TFP-growth variation. Budget-neutral transfers from high- to low-expenditure-centrality occupations generate modest TFP gains and larger PTT gains for lower-earning workers. Redistribution need not mechanically trade equity for efficiency.

Keywords: Networks; allocative efficiency; heterogeneity; incidence.

1 Introduction

The classic equity-efficiency tradeoff treats redistribution as a cost in aggregate efficiency (Okun, 1975). This paper shows that this tradeoff is not mechanical in distorted production networks. When households differ in what they buy and where their income comes from, redistribution changes the composition of final demand. More generally, changes in who earns, who spends, and which firms and inputs expand recompose income and demand across the production network. Because firms differ in wedges and network position, these reallocations can attenuate or accentuate misallocation. Whether redistribution lowers or raises aggregate efficiency is therefore not a primitive property of redistribution; it is an empirical property of the production network, the wedges, and the household expenditure distribution.

This paper develops and estimates a general equilibrium accounting framework for measuring these effects. I study a distorted production network in which firms trade intermediate inputs and face price-cost wedges, such as markups, while households differ in preferences, factor-income exposure, and profit exposure. The main object of analysis is total factor productivity (TFP). In a first-best economy, the income-expenditure distribution is locally irrelevant for TFP. In the economy studied here, it is a first-order state variable for aggregate efficiency. Shifts in income sources, profit exposure, expenditure shares, and demand composition feed back into equilibrium prices, sectoral sales, factor payments, and TFP.

A first contribution is to derive a general equilibrium decomposition of real output and TFP that measures how income-demand recomposition shapes aggregate efficiency. The decomposition separates technology, wedges, and distributional reallocation. The distributional component is decomposed into factor-income shifts, expenditure redistribution across households, final-demand recomposition across firms, intermediate-demand recomposition, and factor-demand recomposition. This decomposition turns the question of whether distribution matters for productivity into a set of measurable margins.

The sufficient statistics are organized through a layered centrality system that separates factor-firm-household linkages. *Distortion centrality* (δ) ranks factors by the wedge between their value-added contribution and their income share. *Expenditure centrality* (M) aggregates this distortion exposure into household consumption baskets and governs the TFP effect of reallocating expenditure across households. *Revenue centrality* (F) aggregates the same exposure into firms' sales and governs the TFP effect of recomposing final and intermediate demand across firms. Together, these objects map each recomposition margin into TFP: δ governs factor-income and factor-demand shifts, M governs expenditure redistribution across households, and F governs final- and intermediate-demand shifts across firms.

The decomposition clarifies why redistribution is locally neutral in the first best but generically non-neutral in distorted production networks. In a first-best economy, redistribution affects who consumes but not, to a first order, how efficiently the economy transforms primary factors into real output. With wedges, this neutrality fails. A transfer does more than move purchasing power across households. It changes which goods are demanded, which firms expand, which input-output chains are activated, and which factors receive income. If the induced reallocation pulls resources away from high-marginal-product bottlenecks, TFP falls. If it shifts production toward supply chains where distortions had suppressed activity, TFP rises.

The central statistic for the equity-efficiency question is household expenditure centrality. This object, denoted M_h , ranks household group h by the local marginal TFP cost of an additional dollar of expenditure allocated to that group. A high- M_h household buys goods whose supply chains tend to intensify existing misallocation; a low- M_h household buys goods whose induced production tends to ease it. Hence, through the direct distributive channel, a budget-neutral transfer from household q to household h raises TFP when $M_q > M_h$. The classic equity-efficiency tradeoff becomes an empirical ranking problem: do progressive transfers move expenditure from high- M to low- M households, or in the opposite direction?

A second contribution is to connect aggregate efficiency to household incidence. The equity-efficiency question is not only whether TFP rises, but also who gains and who loses. Standard TFP decompositions do not answer this incidence question. Aggregate output can barely move while real consumption changes sharply across households, because households differ in expenditure baskets and income exposure. To measure these heterogeneous effects, I define *positional terms of trade* (PTT), a household-level incidence object that maps productivity shocks, wedge shocks, and income-demand recomposition into real consumption outcomes. PTT is the household analog of TFP: it summarizes how a household's position in the production network transforms aggregate and sectoral changes into real consumption. Aggregating across households links incidence back to aggregate efficiency $d \log TFP = \sum_h \chi_h d \log PTT_h$, where χ_h is household h 's final expenditure share. This identity makes the aggregate-incidence connection transparent: aggregate TFP growth is the expenditure-weighted average of household PTT growth. The same decomposition therefore delivers both the aggregate efficiency effect of income-demand recomposition and the household-level incidence of that recomposition.

The decompositions also clarify why a representative household or centralized economy can miss the relevant distributional channel. I study a constrained planner that takes wedges as given but centralizes household consumption and factor supply. The planner internalizes the general equilibrium effects of expenditure and factor-supply choices, restoring balancing conditions along the margins that generate inequality-driven misallocation. In this benchmark,

factor supplies are chosen to equalize distortion-weighted exposure across factors, and budget-neutral transfers neutralize the TFP effect of expenditure reallocation. The planner serves as a diagnostic benchmark: it shows which distributional margins are suppressed by centralization and why the income-expenditure distribution remains a first-order state variable for TFP in the decentralized economy.

I take the sufficient statistics to U.S. data from 1997 to 2021. The empirical implementation combines BEA input-output accounts and industry production accounts with household expenditure data and occupational earnings data. I first estimate the aggregate production-network objects—sectoral productivity, wedges, input-output linkages, and factor shares—and then introduce occupational heterogeneity by assigning occupation-specific earnings exposure across industries and income-conditioned consumption bundles. The empirical implementation is central to the paper: the theory says that the aggregate and distributional effects of recomposition depend on measurable δ -, M -, and F -rankings, and the data reveal what those rankings are.

Three empirical findings summarize the role of income-demand recomposition in U.S. aggregate efficiency. First, broad distributional reallocation is quantitatively relevant for TFP fluctuations. Across specifications, the distributional-reallocation component accounts for roughly 17–20% of the variance of U.S. TFP growth. Technology remains the dominant source of variation, but treating the income-expenditure distribution as irrelevant would miss an economically meaningful channel of TFP dynamics. Second, the large aggregate component is not driven only by household redistribution. In the time-series decomposition, household distributive and final-demand terms are modest, while intermediate-demand and factor-demand recomposition are larger. Third, the cross-sectional M -ranking identifies the household redistribution margin relevant for policy. Before 2021, higher-earning occupations tended to have higher expenditure centrality, so progressive expenditure shifts tended to move demand from high- M to low- M groups. By 2021, the mapping becomes less monotone and more middle-income oriented, implying that the productivity effects of redistribution depend on the changing structure of consumption and production.

I use these empirical rankings to study budget-neutral lump-sum transfers across occupational groups. The transfer experiments are designed to isolate the production-network channel. The first transfer rule taxes workers with above-median expenditure centrality and rebates to workers with below-median expenditure centrality. The second transfer rule imposes a progressivity constraint: transfers are allowed only from higher-income, higher- M workers to lower-income, lower- M workers. Under pure redistribution, both rules raise TFP in every year of the sample. When the same transfers are superimposed on the historical sequence of productivity and markup shocks, the TFP effect is positive in 21 of 24 years. The effects are

small but generally positive, showing that M -ranking transfers matter in general equilibrium.

The incidence effects are more pronounced than the aggregate TFP effects. The progressive rule shifts the distribution of occupation-level PTT growth to the right and delivers larger PTT gains to lower-earning occupations. Thus, the same expenditure reallocation that raises aggregate TFP also improves real-consumption incidence for lower-income workers, and the sign of the equity-efficiency trade-off is guided by the M -ranking of households. In the U.S. data, progressive transfers often move expenditure in the productivity-enhancing direction. The Okun tradeoff is therefore conditional, not mechanical.

The paper's main contribution is to make this conditional tradeoff measurable through a nonparametric accounting framework that links household heterogeneity, production networks, and misallocation; it derives centrality-based sufficient statistics for aggregate TFP and household incidence; and it implements these statistics in U.S. data to measure income-demand recomposition and evaluate redistribution. The empirical message is simple. In a distorted network economy, distributional changes do not merely move households along a fixed allocation. They change which supply chains expand, which factors are used, and how existing wedges distort the allocation of production. Whether this movement raises or lowers aggregate efficiency is an empirical property of the network, the wedges, and expenditure patterns.

Related Literature

This paper contributes to macroeconomic aggregation in production networks and to distributional macroeconomics. Aggregation in multisector input-output economies builds on [Hulten \(1978\)](#): in a perfectly competitive representative-household economy, first-order aggregate effects depend only on the sales distribution. A large literature studies how idiosyncratic sectoral shocks propagate through input-output linkages and generate aggregate comovement and volatility ([Horvath, 1998](#); [Dupor, 1999](#); [Horvath, 2000](#); [Foerster, Sarte, & Watson, 2011](#); [Acemoglu, Carvalho, Ozdaglar, & Tahbaz-Salehi, 2012](#); [Carvalho, Nirei, Saito, & Tahbaz-Salehi, 2021](#)). A parallel literature studies wedge propagation in distorted production networks, either under parametric structures ([Jones, 2011](#)) or through nonparametric sufficient-statistic decompositions ([Liu, 2019](#); [Bigio & La'O, 2020](#); [Baqae & Farhi, 2020](#)). The closest antecedents are [Baqae & Farhi \(2019b\)](#) and [Baqae & Farhi \(2024\)](#). The former develops the heterogeneous-agent input-output environment, while the latter provides country-level real-GDP and real-GNE decompositions in open economies with wedges. This paper takes a different measurement problem as its object: how factor-income, household-expenditure, and firm-demand reallocations shape TFP and household-level incidence. [Section 3](#) shows that these margins generate distinct statistics.

The paper is also related to welfare accounting with heterogeneous agents. [Dávila & Schaab \(2025\)](#) develop an axiomatic decomposition of changes in social welfare using primitives such

as preferences, technologies, and resource constraints. Their framework is deliberately agnostic about equilibrium: allocations need not arise from optimizing agents, and markets need not clear. They separate welfare changes into efficiency and redistribution components, with redistribution reflecting heterogeneity in marginal utilities. By contrast, I study equilibrium changes in aggregate efficiency, measured by TFP, in decentralized distorted economies. Redistribution matters here not because marginal utilities differ, but because income-demand recomposition changes production through a distorted network. [Baqae & Burstein \(2025\)](#) define a heterogeneous-household analogue to [Lucas \(1987\)](#), where aggregate efficiency is the maximal uniform contraction of the consumption-possibility set that keeps every household at least as well off as the status quo. Their first-order decomposition is evaluated around a compensated equilibrium with the first-best allocation. My TFP and PTT decompositions instead operate in decentralized distorted equilibria with preference heterogeneity, where transfers do not neutralize income and expenditure shares.

This paper also contributes to aggregate productivity accounting ([Hulten, 1978](#); [Jorgenson, Gollop, & Fraumeni, 1987](#); [Hall & Diamond, 1990](#); [Basu & Fernald, 2002](#); [Petrin & Levinsohn, 2012](#)) and to the misallocation literature ([Restuccia & Rogerson, 2008](#); [Hsieh & Klenow, 2009](#)). [Baqae & Farhi \(2020\)](#) formalize aggregation in inefficient multisector economies with a representative household, showing that technology shocks and reallocation terms generate first-order changes in TFP. I identify and quantify a distribution-driven source of reallocation: even holding technologies and wedges fixed, changes in income shares, expenditure shares, final demand, intermediate demand, and factor demand can move allocative efficiency. I measure the relevant centralities in U.S. data, decompose the aggregate role of income-demand recomposition, and use the household expenditure ranking M to evaluate redistribution counterfactuals.

Finally, the paper relates to evidence on heterogeneity in household consumption behavior. High-income households have lower MPCs ([Jappelli & Pistaferri, 2014](#)), purchase higher-priced varieties ([Jaimovich et al., 2020](#)), are less price sensitive ([Auer, Burstein, Lein, & Vogel, 2024](#)), and pay higher markups for identical products or baskets with higher average markups ([Sangani, 2024](#)). With heterogeneous MPCs, distributional shocks interact with nominal rigidities—such as downward nominal wage rigidity ([Auclert & Rognlie, 2020](#)) or the zero lower bound ([Mian, Straub, & Sufi, 2021](#))—to affect aggregate demand. In those models, redistribution affects output through spending propensities, unemployment, or monetary constraints. By contrast, the channel in this paper does not rely on nominal rigidities, slack, or capacity constraints. Redistribution affects output because it reallocates expenditure across supply chains with different wedges and network positions, thereby changing TFP through allocative efficiency.

Layout

Section 2 sets up the model. Section 3 derives nonparametric first-order decompositions and sufficient statistics for TFP and PTT, and compares the decentralized economy to a constrained-planner benchmark. Section 4 embeds the accounting framework in a normalized nested-CES general equilibrium model to discipline substitution elasticities and quantify nonlinear responses. Section 5 describes the U.S. implementation and estimates the layered centralities. Section 6 studies budget-neutral transfers targeted by expenditure centrality and evaluates their aggregate TFP and household PTT effects. Section 7 concludes.

2 General Framework

This section sets up the static production-network environment. Firms trade intermediate inputs and face firm-specific markups. Households differ in preferences, factor supply, profit ownership, and transfers, generating heterogeneous income and expenditure shares. The accounting is nonparametric: all objects are evaluated at an equilibrium and do not require functional-form assumptions. The section defines the exposure objects used in Section 3 to derive first-order sufficient statistics for TFP and household real consumption. Let firms be indexed by $i \in \mathcal{N} := \{1, \dots, N\}$ and household types by $h \in \mathcal{H} := \{1, \dots, H\}$.

2.1 Environment

Production. Firm $i \in \mathcal{N}$ produces $y_i = A_i Q_i(L_i, X_i)$, where $L_i = Q_i^\ell(\{\ell_{ih}\}_{h \in \mathcal{H}})$ and $X_i = Q_i^x(\{x_{ij}\}_{j \in \mathcal{N}})$. Here A_i is Hicks-neutral productivity; L_i and X_i are primary-factor and intermediate-input composites; ℓ_{ih} is factor h used by firm i ; and x_{ij} is input j used by firm i .¹ Given $\{p_j\}_{j \in \mathcal{N}}$ and $\{w_h\}_{h \in \mathcal{H}}$, firm i minimizes cost $\bar{c}_i = p_i^\ell L_i + p_i^x X_i = \sum_{h \in \mathcal{H}} w_h \ell_{ih} + \sum_{j \in \mathcal{N}} p_j x_{ij}$, where p_i^ℓ and p_i^x are the unit-cost indices dual to Q_i^ℓ and Q_i^x . Firm i faces an exogenous wedge $mc_i \equiv \frac{\partial \bar{c}_i}{\partial y_i} = \mu_i p_i$, with $\mu_i \in (0, 1]$. When $\mu_i < 1$, a share $1 - \mu_i$ of revenue is profits π_i . Thus, wedges redistribute revenue to profits and alter the equilibrium use of inputs without changing the production possibility set.

Households. Household h chooses $(C_h, L_h, \{C_{hi}\}_{i \in \mathcal{N}})$ to maximize $U_h(C_h, L_h)$, subject to $C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}})$, where C_{hi} is consumption from firm i , and p_h^c denotes the unit expenditure index dual to Q_h^c .² Because U_h is increasing in consumption, the budget constraint binds:

$$E_h \equiv p_h^c C_h = \sum_{i \in \mathcal{N}} p_i C_{hi} = J_h + \Pi_h + T_h, \quad (1)$$

where $J_h = w_h L_h$ is income from the primary factor supplied by household type h , $\Pi_h =$

¹Assume Q_i , Q_i^ℓ , and Q_i^x are C^2 , increasing, concave, constant returns to scale, strictly quasi-concave on the interior, and satisfy Inada conditions on used inputs.

²Assume U_h is increasing in C_h , decreasing in L_h , strictly concave, and satisfies Inada conditions; Q_h^c is positive, degree-one homogeneous, and satisfies Inada conditions on consumed goods.

$\sum_{i \in \mathcal{N}} \kappa_{ih} \pi_i$ is profit income, and T_h is a lump-sum transfer. Ownership shares satisfy $\kappa_{ih} \geq 0$ and $\sum_{h \in \mathcal{H}} \kappa_{ih} = 1$ for each i .

Households differ in preferences, factor supply, and profit ownership. Heterogeneity in Q_h^c generates expenditure-share heterogeneity at common goods prices, while J_h , κ_{ih} , and T_h determine type-specific income exposure. Markets are incomplete: there are no state-contingent claims spanning idiosyncratic household or firm shocks. Households cannot fully insure factor income or dividend risk, so income exposure to shocks differs across types.

Market Clearing. Goods and factor markets clear, and transfers are budget neutral: $y_i = \sum_{h \in \mathcal{H}} C_{hi} + \sum_{j \in \mathcal{N}} x_{ji} \forall i \in \mathcal{N}$, $L_h = \sum_{i \in \mathcal{N}} \ell_{ih} \forall h \in \mathcal{H}$, and $\sum_{h \in \mathcal{H}} T_h = 0$.

Equilibrium. Let the aggregate state be $e \equiv (A, \mu, \kappa, T) \in \mathcal{E}$. $A \equiv (A_i)_{i \in \mathcal{N}}$, $\mu \equiv (\mu_i)_{i \in \mathcal{N}}$, and $T \equiv (T_h)_{h \in \mathcal{H}}$ collect productivities, inverse markups, and transfers. The ownership matrix $\kappa \equiv (\kappa_{ih})_{i \in \mathcal{N}, h \in \mathcal{H}}$ records household h 's share of firm i , with $\sum_{h \in \mathcal{H}} \kappa_{ih} = 1$ for each i . Fix e and suppress its dependence when unambiguous (e.g., $w_h(e) \rightarrow w_h$).

For any $e \in \mathcal{E}$, an equilibrium is an allocation $(\{y_i\}, \{x_{ij}\}, \{\ell_{ih}\}, \{C_{hi}\}, \{L_h\})$ and price system $(\{p_i\}, \{w_h\})$ such that: (i) given $\{p_i\}_{i \in \mathcal{N}}$ and $\{w_h\}_{h \in \mathcal{H}}$, each firm i minimizes cost by choosing $\{\ell_{ih}\}_{h \in \mathcal{H}}$ and $\{x_{ij}\}_{j \in \mathcal{N}}$ subject to its technology and satisfies $mc_i = \mu_i p_i$; (ii) given $\{p_i\}_{i \in \mathcal{N}}$ and $\{w_h, \Pi_h, T_h\}_{h \in \mathcal{H}}$, each household h chooses $(L_h, \{C_{hi}\}_{i \in \mathcal{N}})$ to maximize $U_h(C_h, L_h)$ subject to $C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}})$ and (1); and (iii) profits are distributed according to κ , markets clear, and transfers are budget neutral.

Firm-level wedges do not alter the production possibility set, but they change how sales are split between input payments and rents. Because wages, profits, and transfers enter heterogeneous household budgets, shocks to A , μ , κ , or T change income shares, expenditure shares, and factor supplies. These distributional shifts feed back into firm sales, input demand, wages, and profits through the network, so wedges can make household income and expenditure composition relevant for allocative efficiency. The Online Appendix gives sufficient conditions for equilibrium. In particular, an essential primary factor used directly or indirectly in producing every good rules out disconnected production components and pins down a common relative-price system.

2.2 Measures of Centrality

This subsection defines the direct and network-adjusted exposure objects used in the decompositions. I distinguish cost-based exposures from revenue-based exposures because firm wedges drive a gap between the cost of producing through a supply chain and the revenue passed upstream. Cost-based objects describe downstream marginal-cost and price pass-through from suppliers to buyers. Revenue-based objects describe upstream payment pass-through from

household and firm expenditure to suppliers, factors, and profits. Direct exposures are observed cost, expenditure, and income shares; network-adjusted exposures add all indirect input-output paths through the corresponding Leontief inverse. **Table 1** summarizes direct exposures; **Table 2** reports network-adjusted exposures and the sufficient statistics built from them.

2.2.1 Direct Exposure Shares

The direct cost exposures of firm i are its factor- and intermediate-composite cost shares. Define $\omega_i^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^\ell}$ and $\omega_i^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^x}$. By Shephard's lemma, $\omega_i^\ell = \frac{p_i^\ell L_i}{\bar{c}_i}$ and $\omega_i^x = \frac{p_i^x X_i}{\bar{c}_i}$; collect them in $\omega_\ell \equiv (\omega_i^\ell)_{i \in \mathcal{N}}$ and $\omega_x \equiv (\omega_i^x)_{i \in \mathcal{N}}$. To disaggregate these shares, define $\tilde{\Omega}_\ell \equiv (\tilde{\Omega}_{ih}^\ell)$ and $\tilde{\Omega}_x \equiv (\tilde{\Omega}_{ij}^x)$, with $\tilde{\Omega}_{ih}^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log w_h}$ and $\tilde{\Omega}_{ij}^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_j}$. Again by Shephard's lemma, $\tilde{\Omega}_{ih}^\ell = \frac{w_h \ell_{ih}}{\bar{c}_i}$ and $\tilde{\Omega}_{ij}^x = \frac{p_j x_{ij}}{\bar{c}_i}$. Homogeneity gives $\sum_{h \in \mathcal{H}} \tilde{\Omega}_{ih}^\ell + \sum_{j \in \mathcal{N}} \tilde{\Omega}_{ij}^x = 1$, with $\omega_i^\ell = \sum_{h \in \mathcal{H}} \tilde{\Omega}_{ih}^\ell$ and $\omega_i^x = \sum_{j \in \mathcal{N}} \tilde{\Omega}_{ij}^x$.

Table 1: Direct Exposure Shares

Object	Definition	Properties
Cost-Based Shares		
ω_ℓ	$\omega_i^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^\ell} = \frac{p_i^\ell L_i}{\bar{c}_i}$	$\omega_i^\ell + \omega_i^x = 1$
ω_x	$\omega_i^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_i^x} = \frac{p_i^x X_i}{\bar{c}_i}$	
$\tilde{\Omega}_\ell$	$\tilde{\Omega}_{ih}^\ell \equiv \frac{\partial \log \bar{c}_i}{\partial \log w_h} = \frac{w_h \ell_{ih}}{\bar{c}_i}$	$\sum_{h \in \mathcal{H}} \tilde{\Omega}_{ih}^\ell + \sum_{j \in \mathcal{N}} \tilde{\Omega}_{ij}^x = 1$
$\tilde{\Omega}_x$	$\tilde{\Omega}_{ij}^x \equiv \frac{\partial \log \bar{c}_i}{\partial \log p_j} = \frac{p_j x_{ij}}{\bar{c}_i}$	
α	$\alpha_{ih} \equiv \frac{\partial \log p_i^\ell L_i}{\partial \log w_h} = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$	$\sum_{h \in \mathcal{H}} \alpha_{ih} = 1$
$\mathcal{W}$	$\mathcal{W}_{ij} \equiv \frac{\partial \log p_i^x X_i}{\partial \log p_j} = \frac{p_j x_{ij}}{p_i^x X_i}$	$\sum_{j \in \mathcal{N}} \mathcal{W}_{ij} = 1$
Revenue-Based Shares		
Ω_ℓ	$\Omega_{ih}^\ell \equiv \frac{w_h \ell_{ih}}{S_i} = \mu_i \tilde{\Omega}_{ih}^\ell$	$\sum_{h \in \mathcal{H}} \Omega_{ih}^\ell + \sum_{j \in \mathcal{N}} \Omega_{ij}^x = \mu_i$
Ω_x	$\Omega_{ij}^x \equiv \frac{p_j x_{ij}}{S_i} = \mu_i \tilde{\Omega}_{ij}^x$	
Ω_π	$\Omega_{ih}^\pi \equiv \frac{\kappa_{ih} \pi_i}{S_i} = (1 - \mu_i) \kappa_{ih}$	$\sum_{h \in \mathcal{H}} (\Omega_{ih}^\ell + \Omega_{ih}^\pi) + \sum_{j \in \mathcal{N}} \Omega_{ij}^x = 1$
Household Shares		
β	$\beta_{hi} \equiv \frac{\partial \log E_h}{\partial \log p_i} = \frac{p_i C_{hi}}{E_h}$	$\sum_{i \in \mathcal{N}} \beta_{hi} = 1$
κ	$\kappa_{ih} \equiv \frac{\partial \Pi_h}{\partial \pi_i}$	$\sum_{h \in \mathcal{H}} \kappa_{ih} = 1$
τ	$\tau_h \equiv \frac{T_h}{GDP}$	$\sum_{h \in \mathcal{H}} \tau_h = 0$

Normalize these disaggregated cost shares within each node. For firms with positive composite shares, define $\alpha \equiv \text{diag}(\omega_\ell)^{-1} \tilde{\Omega}_\ell$ and $\mathcal{W} \equiv \text{diag}(\omega_x)^{-1} \tilde{\Omega}_x$. Thus α_{ih} is factor h 's share in firm i 's primary-factor composite, and $\mathcal{W}_{ij}$ is input j 's share in firm i 's intermediate-input composite. Equivalently, by Shephard's lemma applied to the subcost functions, $\alpha_{ih} = \frac{\partial \log p_i^\ell}{\partial \log w_h} = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$ and $\mathcal{W}_{ij} = \frac{\partial \log p_i^x}{\partial \log p_j} = \frac{p_j x_{ij}}{p_i^x X_i}$. Homogeneity implies $\sum_{h \in \mathcal{H}} \alpha_{ih} = 1$ and $\sum_{j \in \mathcal{N}} \mathcal{W}_{ij} = 1$.

Revenue-based upstream shares measure how firm i 's sales are split between payments to primary factors, intermediate suppliers, and profit owners. Under constant returns and $mc_i = \mu_i p_i$, total cost satisfies $\bar{c}_i = \mu_i S_i$. Hence define $\Omega_\ell \equiv \text{diag}(\mu) \tilde{\Omega}_\ell$ and $\Omega_x \equiv \text{diag}(\mu) \tilde{\Omega}_x$.

Elementwise, $\Omega_{ih}^\ell = \frac{w_h \ell_{ih}}{S_i}$ and $\Omega_{ij}^x = \frac{p_j x_{ij}}{S_i}$. When $\mu_i < 1$, only a share μ_i of sales is paid to inputs; the residual $1 - \mu_i$ is profits. Define the profit-share matrix $\Omega_\pi \equiv \text{diag}(\mathbb{1}_N - \mu)\kappa$ and $\Omega_{ih}^\pi = \frac{\kappa_{ih} \pi_i}{S_i}$. Revenue adds up: $\sum_{h \in \mathcal{H}} (\Omega_{ih}^\ell + \Omega_{ih}^\pi) + \sum_{j \in \mathcal{N}} \Omega_{ij}^x = 1$.

For households, let $\beta \equiv (\beta_{hi})_{h \in \mathcal{H}, i \in \mathcal{N}}$ denote the final-expenditure share matrix, with row $\beta_h \equiv (\beta_{hi})_{i \in \mathcal{N}}$. Define the elasticity $\beta_{hi} \equiv \frac{\partial \log E_h}{\partial \log p_i}$. By Shephard's lemma, $\beta_{hi} = \frac{p_i C_{hi}}{E_h}$, and hence $\sum_{i \in \mathcal{N}} \beta_{hi} = 1$. Let $\tau_h \equiv \frac{T_h}{GDP}$. Budget neutrality implies $\sum_{h \in \mathcal{H}} \tau_h = 0$.

2.2.2 Network-Adjusted Centralities

The cost-based firm-to-firm downstream Leontief inverse is $\tilde{\Psi}_x \equiv (I - \tilde{\Omega}_x)^{-1}$. Its element $\tilde{\psi}_{ij}^x$ is firm i 's total cost exposure to firm j , summing all direct and indirect input-output paths from j to i . The revenue-based firm-to-firm upstream Leontief inverse is $\Psi_x \equiv (I - \Omega_x)^{-1}$. Its element ψ_{ij}^x is the network-adjusted share of firm i 's sales reaching firm j , equivalently the marginal revenue generated at j by one dollar of sales at i . With wedges, revenue pass-through differs from cost exposure because part of the revenue is retained as rents along upstream paths.

The firm-to-household downstream exposure matrix is $\tilde{\mathcal{B}} \equiv \beta \tilde{\Psi}_x$. Its element $\tilde{\mathcal{B}}_{hi}$ is household h 's total cost exposure to firm i , summarizing the pass-through from p_i to p_h^c . Aggregating across households with expenditure weights $\chi_h \equiv \frac{E_h}{GDP}$ gives the cost-based Domar weight $\tilde{\lambda}_i \equiv \sum_{h \in \mathcal{H}} \chi_h \tilde{\mathcal{B}}_{hi}$, which measures the share of aggregate value added passing through firm i . The household-to-firm upstream exposure matrix is $\mathcal{B} \equiv \beta \Psi_x$. Its element $\mathcal{B}_{hi}$ is the network-adjusted share of household h 's expenditure reaching firm i . Aggregating across households yields the revenue Domar weight $\lambda_i \equiv \sum_{h \in \mathcal{H}} \chi_h \mathcal{B}_{hi} = \frac{S_i}{GDP}$. The gap $\tilde{\lambda}_i - \lambda_i$ reflects rents retained along downstream paths from firm i to final expenditure.

The factor-to-firm downstream exposure matrix is $\tilde{\Psi}_\ell \equiv \tilde{\Psi}_x \tilde{\Omega}_\ell$. Its element $\tilde{\psi}_{ih}^\ell$ is firm i 's total cost exposure to factor h , or equivalently the share of firm i 's cost attributable to factor h 's. Since $\sum_{h \in \mathcal{H}} \tilde{\psi}_{ih}^\ell = 1$, each firm's production cost can be fully traced back to primary-factor payments. The firm-to-factor upstream exposure matrix is $\Psi_\ell \equiv \Psi_x \Omega_\ell$. Its element ψ_{ih}^ℓ is the network-adjusted share of firm i 's sales reaching factor h . With wedges, $\sum_{h \in \mathcal{H}} \psi_{ih}^\ell \leq 1$, because part of sales is retained as rents before reaching primary factors.

The factor-to-household downstream exposure matrix is $\tilde{\mathcal{C}} \equiv \beta \tilde{\Psi}_\ell$. Its element $\tilde{\mathcal{C}}_{hb}$ is the share of household h 's consumption cost ultimately attributable to factor b 's value added. Since $\sum_{b \in \mathcal{B}} \tilde{\mathcal{C}}_{hb} = 1$, each household's consumption cost can be fully traced back to primary-factor payments. Aggregating across households gives the cost-based factor weight $\tilde{\Lambda}_b \equiv \sum_{h \in \mathcal{H}} \chi_h \tilde{\mathcal{C}}_{hb}$, which measures factor b 's share of aggregate value added; hence $\sum_{b \in \mathcal{B}} \tilde{\Lambda}_b = 1$. The household-to-factor upstream exposure matrix is $\mathcal{C} \equiv \beta \Psi_\ell$. Its element $\mathcal{C}_{hb}$ is the network-adjusted share of household h 's expenditure reaching factor b . Aggregating gives the revenue-based factor

Table 2: Network-Adjusted Exposures and Distributional Centralities

<i>Object</i>	<i>Definition and Interpretation</i>	<i>Properties</i>
Downstream / Cost-Based Exposures		
$\tilde{\Psi}_x = (I - \tilde{\Omega}_x)^{-1}$	$\tilde{\psi}_{ij}^x$: Cost exposure of firm i to firm j	
$\tilde{\mathcal{B}} = \beta \tilde{\Psi}_x$	$\tilde{\mathcal{B}}_{hi}$: Cost exposure of household h to firm i	
$\tilde{\Psi}_\ell = \tilde{\Psi}_x \tilde{\Omega}_\ell$	$\tilde{\psi}_{ih}^\ell$: factor h 's value-added share in $\bar{c}_i$	$\sum_{h \in \mathcal{H}} \tilde{\psi}_{ih}^\ell = 1$
$\tilde{\mathcal{C}} = \beta \tilde{\Psi}_\ell$	$\tilde{\mathcal{C}}_{hb}$: factor b 's value-added share in E_h	$\sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{hb} = 1$
$\tilde{\lambda} = \tilde{\mathcal{B}}' \chi$	$\tilde{\lambda}_i$: Value added that passes through firm i	$\sum_{i \in \mathcal{N}} \omega_i^\ell \tilde{\lambda}_i = 1$
$\tilde{\Lambda} = \tilde{\mathcal{C}}' \chi$	$\tilde{\Lambda}_h$: Value added generated by factor h	$\sum_{h \in \mathcal{H}} \tilde{\Lambda}_h = 1$
Upstream / Revenue-Based Exposures		
$\Psi_x = (I - \Omega_x)^{-1}$	ψ_{ij}^x : Share of S_i reaching to firm j	
$\mathcal{B} = \beta \Psi_x$	$\mathcal{B}_{hi}$: Share of E_h reaching to firm i	
$\Psi_\ell = \Psi_x \Omega_\ell$	ψ_{ih}^ℓ : Share of S_i reaching to factor h	
$\mathcal{C} = \beta \Psi_\ell$	$\mathcal{C}_{hb}$: Share of E_h reaching to factor b	
$\lambda = \mathcal{B}' \chi$	λ_i : Aggregate sales share S_i / GDP	$\sum_{i \in \mathcal{N}} \lambda_i \geq 1$
$\Lambda = \mathcal{C}' \chi$	Λ_h : Factor income share J_h / GDP	$\sum_{h \in \mathcal{H}} \Lambda_h \leq 1$
$\chi = (\Omega_\ell + \Omega_\pi)' \lambda + \tau$	χ_h : Consumption expenditure share E_h / GDP	$\sum_{h \in \mathcal{H}} \chi_h = 1$
Distributional Centralities		
$\delta = \text{diag}(\Lambda)^{-1} \tilde{\Lambda}$	δ_h : Distortion centrality for factor h	$\delta_h = \tilde{\Lambda}_h / \Lambda_h \geq 1$
$M = \mathcal{C} \delta$	M_h : Expenditure centrality for household h	$M_h = \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} \delta_b$
$F = \Psi_\ell \delta$	F_i : Revenue centrality for firm i	$F_i = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell \delta_h$

weight $\Lambda_b \equiv \sum_{h \in \mathcal{H}} \chi_h \mathcal{C}_{hb} = \frac{J_b}{\text{GDP}}$, factor b 's income share. Using the one-to-one household-factor mapping, the household expenditure shares satisfy $\chi_h = \Lambda_h + \sum_{i \in \mathcal{N}} \Omega_{ih}^\pi \lambda_i + \tau_h$.

Because firm-level wedges retain part of sales as profits before revenue reaches upstream input suppliers and factors, cost-based exposures weakly dominate their revenue-based counterparts: $\tilde{\Psi}_x \succeq \Psi_x$, $\tilde{\mathcal{B}} \succeq \mathcal{B}$, $\tilde{\Psi}_\ell \succeq \Psi_\ell$, $\tilde{\mathcal{C}} \succeq \mathcal{C}$, $\tilde{\lambda} \succeq \lambda$, $\tilde{\Lambda} \succeq \Lambda$. For $\Lambda_h > 0$, define factor h 's *distortion centrality* as $\delta_h \equiv \frac{\tilde{\Lambda}_h}{\Lambda_h} \geq 1$. This object compares factor h 's cost-based value-added share with its revenue-based income share. A high δ_h means that factor h accounts for a large share of embodied value added but receives a small share of factor income, because part of the revenue generated downstream is retained as profits before reaching factors.

This use of the term *distortion centrality* is related to, but distinct from, Liu (2019). Liu's distortion centrality is a sector-level statistic: it is the ratio of sectoral influence to the sector's Domar weight and ranks sectors by the marginal social value of industrial-policy subsidies. His object delivers a weighted-Hulten policy result: the first-order aggregate effect of sectoral interventions is summarized by the covariance between distortion centrality and subsidy spending. The underlying wedge accounting is different. In Liu's environment, market-imperfection payments are quasi-rents that are wasted as deadweight losses; they raise production costs but do not become household income. Here, firm-level wedges are rebated: they split sales between input payments and profits, and profits are distributed to households through κ . Hence wedge revenues affect both allocative efficiency and the distribution of income and expenditure. The

object δ_h is therefore not a sectoral subsidy statistic. It is a factor-level incidence and reallocation statistic: it compares the value added embodied in factor h with the factor income reaching h , while the residual wedge revenue is tracked as profit income to households.

Define household h 's *expenditure centrality* and firm i 's *revenue centrality* by $M_h \equiv \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} \delta_b$ and $F_i \equiv \sum_{b \in \mathcal{H}} \psi_{ib}^\ell \delta_b$. M_h is the distortion-centrality exposure embedded in household h 's expenditure: it is high when spending by h reaches high- δ factors. F_i is the corresponding exposure embedded in firm i 's sales: it is high when sales by i reach high- δ factors. Both indices combine the factor-level gap δ_b with the strength of the revenue path to that factor, captured by $\mathcal{C}_{hb}$ or ψ_{ib}^ℓ . Thus, M ranks households for expenditure reallocation, while F ranks firms for demand recomposition.

2.2.3 Layered Factorization and Measurement

The layered objects above can be embedded in the HA-IO representation of [Baqae & Farhi \(2019b, 2024\)](#). When all layers are observed, the distinction is not algebraic: the layered matrices are blocks of the corresponding stacked HA-IO Leontief inverse. For example, the cost-based stacked operator can be written as $\tilde{\Omega}^{HA} = \begin{pmatrix} \tilde{\Omega}_x & 0 & \tilde{\Omega}_\ell \\ \tilde{\beta} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $(I - \tilde{\Omega}^{HA})^{-1} = \begin{pmatrix} \tilde{\Psi}_x & 0 & \tilde{\Psi}_\ell \\ \tilde{\mathcal{B}} & I & \tilde{\mathcal{C}} \\ 0 & 0 & I \end{pmatrix}$, and the revenue-based operator is $\Omega^{HA} = \begin{pmatrix} \Omega_x & 0 & \Omega_\ell \\ \beta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $(I - \Omega^{HA})^{-1} = \begin{pmatrix} \Psi_x & 0 & \Psi_\ell \\ \mathcal{B} & I & \mathcal{C} \\ 0 & 0 & I \end{pmatrix}$.

The difference is one of measurement. First, the layered representation separates the firm-production inverse from the household-expenditure and factor-income mappings. A single inverse records total node-to-node exposure; the layered system identifies which exposure is relevant for each policy margin. Second, the factorization delivers distinct empirical measures for those margins. The cost- and revenue-based blocks imply $\tilde{\Lambda} = \tilde{\mathcal{C}}' \chi$, $\Lambda = \mathcal{C}' \chi$, $\delta = \text{diag}(\Lambda)^{-1} \tilde{\Lambda}$, and the contractions $M = \mathcal{C} \delta$, and $F = \Psi_\ell \delta$. [Section 3](#) shows that these objects are the sufficient statistics for factor-income shifts, household-expenditure redistribution, and firm-demand recomposition.

The factorization also has practical advantages. Instead of inverting the full $(N + H + F) \times (N + H + F)$ HA-IO matrix, or $(N + 2H) \times (N + 2H)$ under the one-factor-per-household normalization, the only nontrivial inversions are the $N \times N$ firm blocks $(I - \tilde{\Omega}_x)^{-1}$ and $(I - \Omega_x)^{-1}$. The remaining objects are obtained by matrix multiplication. The representation is modular: firm-to-firm exposures require only the production layer; household-to-firm exposures add β ; firm-to-factor exposures add Ω_ℓ ; household-to-factor exposures use all three. Profit ownership κ and transfers τ are needed to close household expenditure shares and incidence, but not to compute production or factor pass-through. Missing layers therefore restrict which statistics can be measured without forcing the researcher to impute a full HA-IO matrix.

3 Aggregate and Distributional Accounting

This section derives the accounting identities that turn the network exposure and centrality objects in [Section 2](#) into sufficient statistics for aggregate TFP and household real consumption. [Proposition 1](#) gives cost-based price pass-through. [Theorem 1](#) decomposes changes in Domar weights and factor-income shares into expenditure-share shifts and demand, input, and factor recomposition. [Theorem 2](#) decomposes TFP into technology, competitiveness, and distributional reallocation, and shows how δ , M , and F rank factor-income, household-expenditure, and firm-demand shifts. [Theorem 3](#) compares the decentralized economy with a constrained-planner benchmark. [Theorem 4](#) introduces positional terms of trade (PTT), a household-level incidence object whose expenditure-weighted average equals aggregate TFP growth.

3.1 Price Pass-Through

[Proposition 1](#) derives cost-based price pass-through. Given factor-price changes, firm prices respond to productivity and markup shocks through downstream input-output linkages. Household cost-of-living indices aggregate the same firm-level pass-through using household expenditure shares, while the aggregate price index uses expenditure-weighted cost-based Domar weights. These elasticities summarize price incidence; income incidence is characterized below.

Proposition 1. Price Variation Across Firms and Households. The first-order variation in firm price p_i , household consumption price index p_h^c , and the aggregate price index p_Y satisfy

$$\begin{aligned} d \log p_i &= - \sum_{j \in \mathcal{N}} \tilde{\psi}_{ij}^x (d \log A_j + d \log \mu_j) + \sum_{h \in \mathcal{H}} \tilde{\psi}_{ih}^\ell d \log w_h, \\ d \log p_h^c &= - \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} (d \log A_i + d \log \mu_i) + \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{hb} d \log w_b, \\ d \log p_Y &= - \sum_{i \in \mathcal{N}} \tilde{\lambda}_i (d \log A_i + d \log \mu_i) + \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h d \log w_h. \end{aligned}$$

These equations imply three price-pass-through results. First, productivity and inverse-markup shocks enter symmetrically: both lower prices through the cost-based firm-to-firm and firm-to-household exposure matrices, $\tilde{\Psi}_x$ and $\tilde{\mathcal{B}}$. Second, factor-price shocks pass through according to factor-to-firm and factor-to-household exposures, $\tilde{\Psi}_\ell$ and $\tilde{\mathcal{C}}$, while aggregate price exposure is summarized by $\tilde{\lambda}$ and $\tilde{\Lambda}$. Third, household price incidence is heterogeneous because expenditure baskets differ: variation in β_h changes households' exposure to firm and factor prices. Income sources—factor income, profit ownership κ , and transfers τ —do not directly affect price pass-through; they enter household real-consumption incidence below.

3.2 Revenue and Factor-Income Shares

Theorem 1 differentiates the revenue-share system $\lambda = \mathcal{B}'\chi$ and the factor-income system $\Lambda = \mathcal{C}'\chi = \Omega_\ell'\lambda$, decomposing realized changes in firm sales and factor-income shares into expenditure reallocation, final-demand recomposition, intermediate-input recomposition, and markup channels, with the factor-income decomposition additionally isolating direct factor-demand recomposition. These are nonparametric decompositions in realized share changes; they do not require a parametric propagation system.

Theorem 1. Revenue and Factor-Income Shares. The first-order changes in λ_i and Λ_h satisfy

$$d\lambda_i = \underbrace{\sum_{b \in \mathcal{H}} \mathcal{B}_{bi} d\chi_b}_{\text{Expenditure Reallocation}_i} + \underbrace{\sum_{b \in \mathcal{H}} \chi_b d\mathcal{B}_{bi}}_{\text{Sales-Exposure Recomposition}_i}, \quad (2a) \quad d\Lambda_h = \underbrace{\sum_{b \in \mathcal{H}} \mathcal{C}_{bh} d\chi_b}_{\text{Expenditure Reallocation}_h} + \underbrace{\sum_{b \in \mathcal{H}} \chi_b d\mathcal{C}_{bh}}_{\text{Income-Exposure Recomposition}_h}. \quad (2b)$$

The sales-exposure and income-exposure recomposition terms are

$$\begin{aligned} \text{Sales-Exposure Recomposition}_i &= \underbrace{\sum_{j \in \mathcal{N}} \psi_{ji}^x \sum_{b \in \mathcal{H}} \chi_b d\beta_{bj}}_{\text{Final-Demand Recomposition}_i} + \underbrace{\sum_{j \in \mathcal{N}} \psi_{ji}^x \sum_{m \in \mathcal{N}} \mu_m \lambda_m d\tilde{\Omega}_{mj}^x}_{\text{Intermediate-Input Recomposition}_i} + \underbrace{\sum_{j \in \mathcal{N}} (\psi_{ji}^x - \mathbb{1}_{\{j=i\}}) \lambda_j d \log \mu_j}_{\text{Markup Revenue}_i}, \\ \text{Income-Exposure Recomposition}_h &= \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \sum_{b \in \mathcal{H}} \chi_b d\beta_{bi}}_{\text{Final-Demand Recomposition}_h} + \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \sum_{j \in \mathcal{N}} \mu_j \lambda_j d\tilde{\Omega}_{ji}^x}_{\text{Intermediate-Input Recomposition}_h} + \underbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i d\tilde{\Omega}_{ih}^\ell}_{\text{Factor-Demand Recomposition}_h} + \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \lambda_i d \log \mu_i}_{\text{Markup Income}_h}. \end{aligned}$$

Equation (2a) decomposes $d\lambda_i$ into two margins. The *expenditure-reallocation* term holds household-to-firm revenue exposures $\mathcal{B}$ fixed and reallocates expenditure shares χ . Thus, shifting expenditure share from household q to household b raises firm i 's revenue Domar weight whenever $\mathcal{B}_{bi} > \mathcal{B}_{qi}$. The *sales-exposure recomposition* term holds χ fixed and changes the exposure of household expenditure to firm i , raising λ_i when $\sum_{b \in \mathcal{H}} \chi_b d\mathcal{B}_{bi} > 0$. This recomposition has three components. Final-demand recomposition, $d\beta$, raises λ_i when final demand shifts toward firms whose revenue reaches i more strongly; e.g., shifting demand from firm k to firm j raises λ_i if $\psi_{ji}^x > \psi_{ki}^x$. Intermediate-input recomposition, $d\tilde{\Omega}_x$, captures changes in firms' input sourcing, weighted by firm size and revenue pass-through, $\mu_m \lambda_m$. Finally, the *markup-revenue* term captures how changes in μ_j alter upstream revenue pass-through to firm i .

Equation (2b) maps λ movements into factor-income shares. The *expenditure-reallocation* term now holds household-to-factor revenue exposures $\mathcal{C}$ fixed and reallocates expenditure shares χ . Thus, shifting expenditure share from household q to household b raises factor h 's income share whenever $\mathcal{C}_{bh} > \mathcal{C}_{qh}$. The *income-exposure recomposition* term holds χ fixed and changes the exposure of household expenditure to factor h , raising Λ_h when $\sum_{b \in \mathcal{H}} \chi_b d\mathcal{C}_{bh} > 0$. Its final-demand and intermediate-input components raise Λ_h when demand shifts toward firms whose revenue reaches factor h more strongly. Factor-demand recomposition, $d\tilde{\Omega}_\ell$, captures

changes in firms' direct factor intensities, weighted by $\mu_i \lambda_i$. The *markup-income* term captures how changes in μ_i shift revenue toward factor h , with weights $\lambda_i \psi_{ih}^\ell$.

These decompositions are closest to the propagation accounting in [Baqee & Rubbo \(2023\)](#). Their representative-household, single-factor benchmark is parametric: share adjustment is disciplined by substitution elasticities and price variation. [Theorem 1](#) is instead nonparametric. It decomposes realized movements in revenue Domar weights and factor-income shares in a heterogeneous-household, multi-factor economy. The revenue-share decomposition separates expenditure reallocation from sales-exposure recomposition, with the latter split into final-demand, intermediate-input, and markup-revenue components. The factor-income decomposition maps these revenue movements into factor income and adds direct factor-demand and markup-income margins. The decompositions are therefore directly implementable with repeated input-output, expenditure, and factor-income data.³

3.3 Aggregate Accounting with Heterogeneous Households

This subsection derives a first-order TFP decomposition for the decentralized heterogeneous-household economy. Real GDP can be written either from the expenditure side, as an aggregator of household real consumption, or from the production side, as TFP times a value-added aggregator over primary factors. [Theorem 2](#) uses this equivalence to decompose $d \log TFP$ into technology, competitiveness, and distributional reallocation. The four representations for distributional reallocation, characterize how δ , M , and F are sufficient statistics for factor-income reallocation, household-expenditure redistribution, and final-demand, intermediate-input, and factor-demand recomposition.

Theorem 2. Aggregate TFP with Heterogeneous Households. In equilibrium, real GDP admits the local expenditure- and production-side representations

$$Y = Q_Y(\{C_h\}_{h \in \mathcal{H}}) = TFP \mathcal{F}(\{L_h\}_{h \in \mathcal{H}}), \quad (3)$$

where Q_Y and $\mathcal{F}$ satisfy $\frac{\partial \log Q_Y}{\partial \log C_h} = \chi_h$ and $\frac{\partial \log \mathcal{F}}{\partial \log L_h} = \tilde{\Lambda}_h$.

To first order, Y and TFP satisfy

$$d \log Y = d \log TFP + \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h d \log L_h, \quad (4)$$

$$d \log TFP = \underbrace{\sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log A_i}_{\text{Technology}} + \underbrace{\sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log \mu_i}_{\text{Competitiveness}} + \mathcal{R}, \quad (5)$$

³[Bigio & La'O \(2020\)](#) show that, near efficiency in a representative-household, single-factor economy, sectoral distortions affect the labor wedge at first order but not TFP. Here the objects are firm sales-share and factor-income-share movements at an arbitrary distorted equilibrium with heterogeneous households and multiple factors.

where $\mathcal{R}$ denotes distributional reallocation. The term $\mathcal{R}$ has four equivalent representations:

$$\mathcal{R} = - \overbrace{\sum_{h \in \mathcal{H}} \delta_h d\Lambda_h}^{\text{Entropic TT}} \quad (\text{R1})$$

$$= - \overbrace{\sum_{h \in \mathcal{H}} (\delta_h - 1) \Lambda_h d \log J_h}^{\text{Factor Terms of Trade (TT)}} + \overbrace{\sum_{i \in \mathcal{N}} \lambda_i ((1 - \mu_i) d \log S_i - d\mu_i)}^{\text{Corporate Income}} \quad (\text{R2})$$

$$= - \overbrace{\sum_{h \in \mathcal{H}} M_h d\chi_h}^{\text{Distributive TT}} - \overbrace{\sum_{h \in \mathcal{H}} \chi_h \sum_{b \in \mathcal{H}} \delta_b d\mathcal{C}_{hb}}^{\text{Centrality TT}} \quad (\text{R3})$$

$$= - \overbrace{\sum_{h \in \mathcal{H}} M_h d\chi_h}^{\text{Distributive TT}} - \overbrace{\sum_{h \in \mathcal{H}} \chi_h \sum_{i \in \mathcal{N}} F_i d\beta_{hi}}^{\text{Final-Demand TT}} - \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{j \in \mathcal{N}} F_j d\tilde{\Omega}_{ij}^x}^{\text{Intermediate-Input TT}} \\ - \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{h \in \mathcal{H}} \delta_h d\tilde{\Omega}_{ih}^\ell}^{\text{Factor-Demand TT}} - \overbrace{\sum_{i \in \mathcal{N}} \lambda_i F_i d \log \mu_i}^{\text{Competitive TT}}. \quad (\text{R4})$$

From [equation \(3\)](#), real GDP has two equilibrium representations. On the expenditure side, Q_Y aggregates household real consumption with elasticities equal to expenditure shares, χ_h . On the production side, real GDP equals TFP times a value-added aggregator $\mathcal{F}$ over primary factors, with elasticities equal to cost-based factor value-added shares, $\tilde{\Lambda}_h$.

[Equation \(4\)](#) decomposes $d \log Y$ into TFP growth and primary-factor growth. [Equation \(5\)](#) decomposes $d \log TFP$ into technology, competitiveness, and distributional reallocation. *Technology* is the Hulten term: productivity changes evaluated at current cost-based Domar weights, holding the allocation fixed. *Competitiveness* is the direct effect of changes in markups, also weighted by cost-based Domar weights. Hence, absent distributional reallocation, shocks to A_i and μ_i affect TFP in proportion to $\tilde{\lambda}_i$. *Distributional reallocation* captures the allocative-efficiency effect of changes in factor-income shares and demand composition. Together, *competitiveness* and *distributional reallocation* are allocative efficiency: technology shifts the feasible frontier, while allocative efficiency moves the economy along that frontier. In the first best, Hulten's theorem implies $d \log TFP = \lambda' d \log A$, and local allocative neutrality holds: *competitiveness* + *distributional reallocation* = 0.

[Theorem 2](#) gives four equivalent representations of the *distributional reallocation* channel. They express the same second-best force: with wedges, cost-based value-added exposure and revenue-based income exposure need not coincide. Rents retained reduce the revenue reaching some upstream factors, so a factor can embody a large share of value added while receiving a small share of income. Distortion centrality, δ_h , summarizes this gap. The remaining representations project the same gap onto policy-relevant margins. Increasing the income share of high- δ factors, the expenditure share of high- M households, or demand toward high- F firms lowers TFP at the margin; reallocations in the opposite direction raise TFP. Thus, δ , M , and F

translate factor-income, household-expenditure, and firm-demand recomposition into marginal TFP effects.

In representation (R1), the *entropic terms of trade* summarize how changes in factor-income shares affect the distance between the cost-based value-added and the revenue-based factor-income distributions.⁴ This representation is the same network accounting as in [Baqae & Farhi \(2020\)](#), written in terms of the factor-income shares of the decentralized heterogeneous-household economy. It is therefore the accounting starting point, not the main novelty. Its role is to show that δ_h ranks factors by the marginal TFP effect of factor-income reallocation. Since $\mathcal{R} = -\sum_h \delta_h d\Lambda_h$, shifting income from factor q to factor b raises TFP if $\delta_q > \delta_b$. Hence TFP rises when factor income moves away from high- δ factors and toward low- δ factors, or equivalently, when the distance between Λ and $\tilde{\Lambda}$ increases.

Representation (R2) rewrites *distributional reallocation* in terms of nominal income components. The *factor terms of trade* term captures how changes in factor compensation affect TFP. Since the weight is $\delta_h - 1$, an increase in nominal income for a high- δ factor lowers the reallocation component, while a decline raises it. Intuitively, high- δ factors embody more value added than they receive as income; increasing their compensation accentuates misallocation. The *corporate income* term accounts for retained rents generated by firm-level wedges. Changes in firm sales and markups change the profit share $(1 - \mu_i)S_i/GDP$, and therefore change the wedge between factor income and embodied value added. This does not mean that higher profits mechanically raise total TFP. The statement is conditional: it applies to the *distributional reallocation* component, holding the factor-income distribution fixed. For example, a 1% increase in μ_i , holding factor-income and sales distributions fixed, lowers distributional reallocation by $\lambda_i \mu_i$ but raises the competitiveness term by $\tilde{\lambda}_i$. Hence total allocative efficiency changes by $\tilde{\lambda}_i - \lambda_i \mu_i \geq 0$, since $\tilde{\lambda}_i \geq \lambda_i$ and $\mu_i \leq 1$.

The final two representations in [Theorem 2](#) are the main measurement contribution. They build on the factor-income-share decomposition in [Theorem 1](#) and rewrite the same distributional-reallocation term in policy margins. Representation (R3) separates household-expenditure redistribution, $d\chi$, from changes in household-to-factor revenue exposure, $d\mathcal{C}$. The *distributive terms of trade* imply that increasing household h 's expenditure share lowers TFP in proportion to M_h . Hence a shift of expenditure from household q to household b raises TFP if $M_q > M_b$. Thus, M is the sufficient statistic for the TFP effect of household-expenditure redistribution. The *centrality terms of trade* capture exposure recomposition holding χ fixed: TFP falls when factor income exposure to household expenditure rises, especially for high- δ factors, and for

⁴Holding the current cost-based weights $\tilde{\Lambda}$ fixed, define $D_{KL}(\tilde{\Lambda} \mid \Lambda) \equiv \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h \log \left(\frac{\tilde{\Lambda}_h}{\Lambda_h} \right)$. Then $dD_{KL}(\tilde{\Lambda} \mid \Lambda) = -\sum_{h \in \mathcal{H}} \tilde{\Lambda}_h d \log \Lambda_h = -\sum_{h \in \mathcal{H}} \delta_h d\Lambda_h$, which is representation (R1).

households with large expenditure shares.

Corollary 1. Distributive Neutrality. Variations in the household expenditure distribution have no first-order effect on TFP when expenditure centrality is symmetric, $M_h = M$ for all $h \in \mathcal{H}$. This neutrality holds in several canonical environments: (i) an undistorted economy, $\mu_i = 1$ for all $i \in \mathcal{N}$; (ii) homogeneous consumption bundles, $\beta_{hi} = \beta_i$ for all $h \in \mathcal{H}$; (iii) no intermediate inputs and uniform distortions, $\omega_i^x = 0$ and $\mu_i = \mu$ for all $i \in \mathcal{N}$; and (iv) no intermediate inputs and firm-specific factor supply, $\omega_i^\ell = \alpha_{ii} = 1$ for all $i \in \mathcal{N}$.

Corollary 1 gives conditions under which the expenditure distribution can be ignored for first-order TFP measurement. First-best allocations, identical consumption bundles, and special no-intermediate-input structures all make M symmetric, but outside these cases household expenditure heterogeneity interacts with wedges and network exposure. Thus, heterogeneity in consumption bundles is necessary but not sufficient for expenditure redistribution to affect TFP; the relevant object is heterogeneity in expenditure centrality.

Representation (R4) decomposes the centrality terms of trade into final-demand, intermediate-input, factor-demand, and competitive margins. The final-demand term implies that shifting household demand toward high- F firms lowers TFP; equivalently, reallocating demand from firm i to firm j raises TFP if $F_i > F_j$. The intermediate-input term applies the same logic within production: increasing input demand for high- F suppliers lowers TFP. The factor-demand term ranks factor recomposition by δ : shifting factor demand from h to b raises TFP if $\delta_h > \delta_b$. More generally, reallocating a firm's cost share from factor h to intermediate input j raises TFP if $\delta_h > F_j$. The competitive term captures how changes in markups reallocate revenue through the network. A 1% increase in μ_i , holding other terms fixed, lowers distributional reallocation by $\lambda_i F_i$, while the direct competitiveness term raises TFP by $\tilde{\lambda}_i$. Hence total allocative efficiency changes by $\tilde{\lambda}_i - \lambda_i F_i$, which need not be positive. Overall, M , F , δ , cost shares, and Domar weights are sufficient statistics for the centrality terms of trade. **Corollary 2** gives neutrality conditions under which these cost-recomposition margins are first-order irrelevant for TFP.

Corollary 2. Recomposition Neutrality.

1. At the first best, recomposition is first-order neutral for TFP.
2. Household h 's final-demand recomposition is TFP-neutral if all firms whose expenditure shares change have the same revenue centrality. In particular, neutrality holds if $F_i = F_h$ for all i with $\beta_{hi} > 0$.
3. Firm i 's input-cost recomposition is TFP-neutral if all changing input margins have the same centrality weight: there exists a scalar z_i such that $F_j = z_i$ for all j with $d\tilde{\Omega}_{ij}^x \neq 0$, and

$\delta_h = z_i$ for all h with $d\tilde{\Omega}_{ih}^\ell \neq 0$. Sufficient cases are: (a) firm i uses no direct intermediates and δ_h is constant across employed factors; (b) firm i uses no direct primary factors and F_j is constant across sourced intermediates; or (c) directly used factors and sourced intermediates have aligned centralities, $\delta_h = F_j$, for all h, j with $\tilde{\Omega}_{ih}^\ell > 0$ and $\tilde{\Omega}_{ij}^x > 0$.

3.4 Aggregate Accounting under a Constrained Planner

This subsection introduces a constrained-planner benchmark rather than a first-best planner. Firms continue to cost-minimize under the same technologies, input-output linkages, and firm-level wedges. The planner centralizes the household side of the economy: consumption, factor supplies, and lump-sum transfers. It chooses $\{C_h, L_h, T_h, \{C_{hi}\}_{i \in \mathcal{N}}\}_{h \in \mathcal{H}}$ to maximize $W(Y, L)$, subject to the household implementability constraints implied by (1) and budget neutrality, $\sum_{h \in \mathcal{H}} T_h = 0$. The implementability constraints require each household's allocation to be financeable by its factor income, profit income, and transfer. Because transfers are flexible but budget neutral, the planner can reallocate purchasing power across households; at an interior solution, the transfer FOC equalizes the shadow value of household resources across implementability constraints. As in [Theorem 2](#), aggregate output and factor supply satisfy $Y = Q_Y(\{C_h\}_{h \in \mathcal{H}})$ and $L = \mathcal{F}(\{L_h\}_{h \in \mathcal{H}})$.⁵ [Theorem 3](#) characterizes Y and TFP under this constrained planner with endogenous factor supply.

Theorem 3. Real GDP and TFP under a Constrained Planner.

Let $\Gamma \equiv \sum_{h \in \mathcal{H}} \Lambda_h$. At an interior constrained-planner allocation,

$$\frac{W_L}{W_Y} + \Gamma \frac{Y}{L} = 0, \quad \delta_b = \Gamma^{-1} \quad \forall b \in \mathcal{H}. \quad (6)$$

The first-order variation in real GDP satisfies

$$d \log Y = \overbrace{\sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log A_i + \sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log \mu_i - d \log \Gamma}^{d \log TFP} + \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h d \log L_h. \quad (7)$$

Moreover, the first-order variations in Λ_h and Γ satisfy

$$d\Lambda_h = \sum_{i \in \mathcal{N}} \psi_{ih}^\ell \lambda_i d \log \mu_i + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\tilde{\Omega}_{ih}^\ell + \sum_{j \in \mathcal{N}} \psi_{jh}^\ell \left(d\tilde{\beta}_j + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\tilde{\Omega}_{ij}^x \right), \quad (8)$$

$$d\Gamma = \sum_{i \in \mathcal{N}} \psi_i^\ell \lambda_i d \log \mu_i + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\omega_i^\ell + \sum_{i \in \mathcal{N}} \psi_i^\ell \left(d\tilde{\beta}_i + \sum_{j \in \mathcal{N}} \mu_j \lambda_j d\tilde{\Omega}_{ji}^x \right), \quad (9)$$

where $\tilde{\beta}_i \equiv \sum_{h \in \mathcal{H}} \frac{p_i C_{hi}}{GDP}$ and $\psi_i^\ell \equiv \sum_{h \in \mathcal{H}} \psi_{ih}^\ell$.

[Equation \(6\)](#) is the planner's balancing condition. It equates the planner's marginal rate of substitution between aggregate output and aggregate factor supply to the distorted marginal rate of transformation $-\frac{W_L}{W_Y} = \Gamma \frac{Y}{L}$. At the constrained-planner allocation, Γ equals the aggregate

⁵ W is increasing in Y , decreasing in L , strictly concave, C^2 , and satisfies Inada conditions. Q_Y and $\mathcal{F}$ are neoclassical, CRS, increasing, C^2 , strictly quasi-concave on the interior, and satisfy Inada conditions.

factor-income share and the inverse of the common distortion centrality, $\delta_h = \Gamma^{-1}$ for all factors. Transfers equalize the shadow value of household resources across implementability constraints, while centralized factor-supply choices equalize distortion-adjusted marginal tradeoffs across factors. Hence the cross-factor dispersion in δ_h , which drives factor-income reallocation in the decentralized economy, disappears.

Equation (7) decomposes first-order real GDP growth into TFP growth and primary-factor growth. As in **Theorem 2**, the TFP term contains technology and competitiveness components. Under the constrained planner, however, the decentralized distributional-reallocation channel collapses to $-d \log \Gamma$. Now, the relevant distributional object is the aggregate split between factor payments and residual wedge income, not the full distribution of factor-income.

Equations (8) and (9) are the planner analogues of the factor-income and reallocation decompositions in **Theorem 1** and **Theorem 2**. The key simplification is that the decentralized $d\chi$ margins disappear: there is no expenditure-reallocation term in factor-income shares and no distributive terms-of-trade term in TFP. This neutrality comes from the planner's transfer FOC. After any shock, the constrained planner adjusts lump-sum transfers so that household budget multipliers remain equalized. Hence household expenditure shares may change, but they do not enter as independent distributional wedges. They matter only through the aggregate final-demand vector, $\bar{\beta}_i = \sum_{h \in \mathcal{H}} \chi_h \beta_{hi}$. Equivalently, the planner's consumption choices equalize marginal valuations across household-good pairs, $\frac{p_i}{p_Y} = \frac{\partial Q_Y}{\partial C_h} \frac{\partial Q_h^c}{\partial C_{hi}^c}$, so household identity no longer creates an additional demand wedge for aggregate production. This is why the decentralized distributive term, $\sum_h M_h d\chi_h$, is replaced by aggregate final-demand recomposition, $d\bar{\beta}$. The firm-demand margins also collapse from revenue centralities F_i to aggregate firm-to-factor pass-through, $\psi_i^\ell \equiv \sum_{h \in \mathcal{H}} \psi_{ih}^\ell$. Together with $\bar{\beta}$, input-cost shares, inverse markups, and Domar weights, ψ_i^ℓ characterizes how recomposition affects the aggregate factor-income share Γ and TFP under the constrained planner.

The contrast between **Theorem 2** and **Theorem 3** is therefore substantive, not notational. **Theorem 2** evaluates TFP at the decentralized heterogeneous-household allocation, where household expenditure choices, factor-income shares, and firm-demand composition enter through the sufficient statistics (δ, M, F) . **Theorem 3** evaluates network accounting at the constrained-planner allocation. Centralization changes the allocation, equalizes the relevant marginal tradeoffs, neutralizes the decentralized $d\chi$ wedge, and collapses the sufficient statistics from (δ, M, F) to Γ and aggregate firm-to-factor pass-through, ψ_i^ℓ . Thus, even if representative-household production-network accounting, as in [Baqaee & Farhi \(2020\)](#), and the decentralized heterogeneous-household accounting in **Theorem 2** have similar formal structures, they do not describe the same economic object or imply the same TFP measurement. The relevant sufficient

statistics are allocation-dependent.

3.5 Distributional Accounting

Theorem 4 characterizes household real consumption and introduces *positional terms of trade* (PTT). PTT is the household analogue of TFP: it summarizes how the production network transforms primary factors into household h 's real consumption. Unlike the household cost-of-living index in **Proposition 1**, which captures only price pass-through, PTT combines expenditure exposure with income-side exposure through factor earnings, profit income, ownership shares, and transfers. The theorem decomposes PTT growth into household-specific technology, competitiveness, and distributional-reallocation terms.

Theorem 4. Real Consumption and PTTs in a Heterogeneous-Household Economy.

In equilibrium, household h 's real consumption satisfies

$$C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}}) = PTT_h f_h(\{L_b\}_{b \in \mathcal{H}}), \quad (10)$$

where f_h satisfies $\frac{\partial \log f_h}{\partial \log L_b} = \tilde{\mathcal{C}}_{hb}$.

To first order, C_h and PTT_h satisfy

$$d \log C_h = d \log PTT_h + \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{hb} d \log L_b, \quad (11)$$

$$d \log PTT_h = \underbrace{\sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} d \log A_i}_{\text{Technology}_h} + \underbrace{\sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} d \log \mu_i}_{\text{Competitiveness}_h} + \mathcal{R}_h, \quad (12)$$

where $\mathcal{R}_h$ is household h 's distributional reallocation, which has four equivalent representations:

$$\mathcal{R}_h = d \log \chi_h - \overbrace{\sum_{b \in \mathcal{H}} \delta_{b|h} d \Lambda_b}^{\text{Entropic } TT_h} \quad (R1h)$$

$$= \overbrace{\frac{J_h}{E_h} d \log J_h - \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{hb} d \log J_b}^{-\text{Factor } TT_h} + \overbrace{\frac{dT_h}{E_h}}^{\text{Transfers}_h} + \overbrace{\sum_{i \in \mathcal{N}} \frac{S_i}{E_h} \left((1 - \mu_i) (d\kappa_{ih} + \kappa_{ih} d \log S_i) - \kappa_{ih} d \mu_i \right)}^{\text{Corporate Income}_h} \quad (R2h)$$

$$= d \log \chi_h - \overbrace{\sum_{b \in \mathcal{H}} M_{b|h} d \chi_b}^{\text{Distributive } TT_h} - \overbrace{\sum_{b \in \mathcal{H}} \chi_b \sum_{q \in \mathcal{H}} \delta_{q|h} d \mathcal{C}_{bq}}^{\text{Centrality } TT_h} \quad (R3h)$$

$$= d \log \chi_h - \overbrace{\sum_{b \in \mathcal{H}} M_{b|h} d \chi_b}^{\text{Distributive } TT_h} - \overbrace{\sum_{b \in \mathcal{H}} \chi_b \sum_{i \in \mathcal{N}} F_{i|h} d \beta_{bi}}^{\text{Final-Demand } TT_h} - \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{j \in \mathcal{N}} F_{j|h} d \tilde{\Omega}_{ij}^x}^{\text{Intermediate-Input } TT_h} \\ - \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{b \in \mathcal{H}} \delta_{b|h} d \tilde{\Omega}_{ib}^\ell}^{\text{Factor-Demand } TT_h} - \overbrace{\sum_{i \in \mathcal{N}} \lambda_i F_{i|h} d \log \mu_i}^{\text{Competitive } TT_h}. \quad (R4h)$$

The household-specific centralities are $\delta_{b|h} \equiv \tilde{\mathcal{C}}_{hb}^{\ell}$, $M_{b|h} \equiv \sum_{q \in \mathcal{H}} \mathcal{C}_{bq} \delta_{q|h}$, and $F_{i|h} \equiv \sum_{q \in \mathcal{H}} \psi_{iq}^\ell \delta_{q|h}$.

Household expenditure shares satisfy

$$d\chi_h = d\Lambda_h + \sum_{i \in \mathcal{N}} [(1 - \mu_i) (\lambda_i d\kappa_{ih} + \kappa_{ih} d\lambda_i) - \kappa_{ih} \lambda_i d\mu_i] + d\tau_h. \quad (13)$$

Finally, aggregate TFP growth is the expenditure-weighted average of household PTT growth:

$$d \log TFP = \sum_{h \in \mathcal{H}} \chi_h d \log PTT_h. \quad (14)$$

Equation (10) gives the household analogue of the aggregate representation in **equation (3)**. On the expenditure side, Q_h^c aggregates household h 's final-good consumption. On the factor side, real consumption equals PTT_h times a primary-factor aggregator f_h . The elasticity $\tilde{\mathcal{E}}_{hb}$ is the share of value added embodied in household h 's consumption bundle that is attributable to factor b . Thus, PTT_h plays for household real consumption the same role that TFP plays for aggregate output. **Equation (11)** decomposes household real-consumption growth into PTT growth and factor-supply growth. **Equation (12)** decomposes $d \log PTT_h$ into household-specific technology, competitiveness, and distributional reallocation. Absent distributional reallocation, household h 's exposure to firm i 's productivity and inverse-markup shocks is $\tilde{\mathcal{B}}_{hi}$, the same downstream pass-through object that appears in aggregate form in **Theorem 2**.

The four representations of $\mathcal{R}_h$ parallel the aggregate decomposition but shift the object from aggregate TFP to household h 's incidence. Representation (R1h) expresses incidence through factor-income reallocation. Representation (R2h) rewrites the same term in nominal income components: factor earnings, transfers, and corporate income. Representation (R3h) projects factor-income reallocation onto household-expenditure redistribution and household-to-factor exposure recomposition. Representation (R4h) decomposes that exposure recomposition into final-demand, intermediate-input, factor-demand, and competitive margins.

Representation (R1h) writes household h 's distributional reallocation as an expenditure-share term, $d \log \chi_h$, minus an *entropic terms-of-trade* term. The household-specific distortion centrality $\delta_{b|h} \equiv \frac{\tilde{\mathcal{E}}_{hb}}{\Lambda_b}$ compares the value-added share of factor b embodied in household h 's consumption bundle with factor b 's aggregate income share. A high $\delta_{b|h}$ means that factor b is important for household h 's consumption costs relative to the income paid to that factor. Holding χ_h fixed, reallocating factor income from q to b raises household h 's PTT if $\delta_{q|h} > \delta_{b|h}$. Thus, $\delta_{\cdot|h}$ ranks factor-income reallocations by their marginal incidence effect on household h .

Representation (R2h) rewrites household h 's distributional reallocation in nominal-income components: factor terms of trade, transfers, and corporate income. The *factor terms of trade* compare the growth of household h 's own factor income with the growth of the nominal factor payments embodied in its consumption bundle. The term $(J_h/E_h)d \log J_h$ is household h 's factor-income growth scaled by its expenditure, while $\sum_b \tilde{\mathcal{E}}_{hb} d \log J_b$ is the factor cost exposure, weighting each factor's income growth by its value-added share in household h 's bundle. Hence

PTT_h rises when household h 's factor income grows relative to the factor costs embodied in what it consumes. The *transfer* term, dT_h/E_h , captures lump-sum income changes relative to household expenditure. The *corporate-income* term captures changes in profit receipts. Household h 's profit income from firm i equals $\kappa_{ih}(1 - \mu_i)S_i$, so it rises with ownership gains $d\kappa_{ih}$, firm-sales growth $d \log S_i$, and higher markups. Thus, (R2h) shows how functional income, transfers, and profit claims enter household real-consumption incidence.

Representation (R3h) separates household h 's distributional reallocation into expenditure-share changes, $d\chi$, and household-to-factor exposure recomposition, $d\mathcal{C}$. The direct term $d \log \chi_h$ captures the effect of changing household h 's own expenditure share. The *distributive terms of trade* capture how shifts in the expenditure distribution across all households affect PTT_h through the centrality vector $M_{\cdot|h}$. A high $M_{b|h}$ means that household b 's expenditure reaches factors with high incidence weight for household h , summarized by $\delta_{\cdot|h}$. Holding household h 's own expenditure share fixed, shifting expenditure from household b to household q raises PTT_h if $M_{b|h} > M_{q|h}$. Thus, $M_{\cdot|h}$ is the sufficient statistic for how household-expenditure redistribution affects household h 's PTT. The *centrality terms of trade* hold expenditure shares fixed and capture changes in household-to-factor exposure. An increase in $\mathcal{C}_{bq}$ lowers PTT_h in proportion to $\chi_b \delta_{q|h}$: the effect is larger when household b has a larger expenditure share and factor q has higher distortion centrality from household h 's perspective. This term isolates exposure recomposition holding the expenditure distribution fixed.

Representation (R4h) decomposes the centrality terms of trade into final-demand, factor-demand, intermediate-input, and competitive margins. The final-demand and intermediate-input terms imply that PTT_h falls when demand shifts toward firms with high household-specific revenue centrality, $F_{i|h}$. A high $F_{i|h}$ means that revenue reaching firm i is routed toward factors with high incidence weight for household h . Hence reallocating final or intermediate demand from firm i to firm j raises PTT_h if $F_{i|h} > F_{j|h}$. The factor-demand term applies to primary factors. Shifting factor demand from b to m raises PTT_h if $\delta_{b|h} > \delta_{m|h}$. More generally, reallocating firm i 's cost share from factor b to intermediate input j raises PTT_h if $\delta_{b|h} > F_{j|h}$. Finally, the competitive term captures the household-specific incidence of markup changes.

Equation (13) decomposes the household expenditure-share margin that appears in (R1h), (R3h), and (R4h). Changes in χ_h come from variations in factor-income, profit-ownership, firm sales-share, markups, and transfers. Finally, **Equation (14)** links household incidence back to aggregate efficiency: $d \log TFP = \sum_{h \in \mathcal{H}} \chi_h d \log PTT_h$. Aggregate TFP growth is therefore the expenditure-weighted average of household PTT growth, even though individual PTTs can differ because households vary in expenditure baskets, factor-income exposure, profit ownership, and transfers. Under the first best, the expenditure-weighted sum of household

allocative-efficiency terms is zero, but household-level PTT incidence can remain heterogeneous.

Representations (R1h) and (R2h) are closest to the decomposition in [Baqae & Farhi \(2024\)](#): they use analogous factorial-terms-of-trade accounting, comparing household income exposure with the factor content embodied in consumption costs, but here the income side is decomposed into factor income, ownership, firm-sales, markup, and transfer margins.⁶ The remaining representations are novel and specific to the incidence problem: $\delta_{\cdot|h}$, $M_{\cdot|h}$, and $F_{\cdot|h}$ rank factor-income, household-expenditure, and firm-demand reallocations by their marginal effect on household h 's PTT.

3.6 What the Decompositions Add

The decompositions above use the cost- and revenue-based network-accounting developed in the production-network literature. The contribution is to identify the margins inside distributional reallocation as distinct first-order channels and derive their sufficient statistics. The layered representation separates objects that a stacked HA-IO operator records jointly: household expenditure, firm sales, factor income, profit ownership, transfers, and input-cost recomposition. This separation matters because the same aggregate reallocation term has different coefficients depending on whether the perturbation is a factor-income shift, a household-expenditure shift, a firm-demand shift, or an ownership/transfer shift.

Theorem 1 provides the first measurement layer. It decomposes realized changes in revenue Domar weights and factor-income shares into observed share changes: expenditure reallocation, final-demand recomposition, intermediate-input recomposition, factor-demand recomposition, and markup channels. This differs from a parametric propagation exercise in which share changes are predicted from substitution elasticities and price changes. Here, the decomposition takes realized changes in χ , β , $\tilde{\Omega}_x$, $\tilde{\Omega}_\ell$, and μ as inputs. This is why the theorem is directly implementable with repeated input-output, expenditure, and factor-income data.

Theorem 2 and **Theorem 4** use existing network-accounting identities as starting points but project them onto different margins. In **Theorem 2**, representation (R1) is the closest point of contact with [Baqae & Farhi \(2020\)](#): it is the distorted-network reallocation identity written in factor-income shares. In **Theorem 4**, representations (R1h) and (R2h) are closest to the welfare decomposition in [Baqae & Farhi \(2024\)](#): they use analogous factorial-terms-of-trade accounting,

⁶Using the this paper's notation, their country-level welfare decomposition is $d \log W_h = \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} d \log A_i + \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{hb} d \log L_b + \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} d \log \mu_i + \sum_{f \in \mathcal{H}^*} \left(\Lambda_{f*}^h - \tilde{\mathcal{C}}_{hf} \right) d \log \Lambda_f + \frac{dT_h}{E_h}$. The common statistics are $\tilde{\mathcal{B}}_{hi}$ and $\tilde{\mathcal{C}}_{hb}$. The object not used here is Λ_{f*}^h , income exposure over true and fictitious factors $\mathcal{H}^*$, where fictitious factors collect profits. In (R2h), I instead split this term into true factor income, $\frac{I_h}{E_h} d \log J_h - \sum_b \tilde{\mathcal{C}}_{hb} d \log J_b$, and corporate income, $\sum_i \frac{S_i}{E_h} ((1 - \mu_i)(d\kappa_{ih} + \kappa_{ih} d \log S_i) - \kappa_{ih} d \mu_i)$. Equation (13) and **Theorem 1** then decompose these income margins further.

comparing household income exposure with the factor content embodied in consumption costs. These initial identities are the accounting backbone. The contribution is the projection of the same reallocation and incidence terms onto policy margins. Representations (R3)–(R4) show that δ , M , and F are the coefficients on factor-income reallocation, household-expenditure redistribution, and firm-demand recomposition. Representations (R3h)–(R4h) provide the household-incidence analogues, $\delta_{\cdot|h}$, $M_{\cdot|h}$, and $F_{\cdot|h}$, which rank the same margins by their effect on household h 's PTT.

Theorem 3 shows that these margins are not merely notational. Under the constrained planner, budget-neutral transfers equalize the shadow value of household resources and centralized factor-supply choices equalize distortion-adjusted factor tradeoffs. The decentralized $d\chi$ margins and cross-factor dispersion in δ disappear, and the sufficient statistics collapse from (δ, M, F) to the aggregate wedge Γ and aggregate firm-to-factor pass-through ψ_i^ℓ . Thus, centralization changes the allocation at which the accounting identity is evaluated; similar first-order identities need not imply the same economic mechanism or the same TFP measurement.

A final distinction concerns numéraire normalization. [Baqee & Farhi \(2020, 2024\)](#) fix nominal GDP as the numéraire. The decompositions here do not require such a normalization. They are written in shares, share-weighted differentials, and budget-share identities, so a common rescaling of nominal prices cancels by construction. Nominal-income representations such as (R2) and (R2h) use factor income, sales, profits, and transfers, but their common nominal component cancels through the relevant expenditure-share and budget-share add-up conditions. Thus, the sufficient statistics δ , M , F , PTT , and their household-specific analogues are not artifacts of a GDP numéraire. A numéraire is needed only in the parametric model, where nominal prices and income responses must be pinned down to solve for endogenous substitution and income effects.

4 Parametric General Equilibrium

While **Theorems 1–4** deliver nonparametric decompositions of realized changes in real outcomes, counterfactual analysis requires a parametric structure to solve for endogenous prices, expenditure shares, and cost shares. I therefore embed the accounting system in a normalized nested-CES environment. The normalization preserves the interpretation of the sufficient statistics from [Section 2](#): the CES weights are calibrated to observed equilibrium cost and expenditure shares, while the elasticities discipline how those shares adjust away from the equilibrium.

4.1 Normalized CES Environment

Let $\hat{z} \equiv z/\bar{z}$ denote a variable normalized by its reference-equilibrium value, and normalize $A_i = 1$ at the reference equilibrium. Firm $i \in \mathcal{N}$ produces according to $\hat{y}_i = A_i(\omega_i^\ell \hat{L}_i^{\frac{\theta-1}{\theta}} + \omega_i^x \hat{X}_i^{\frac{\theta-1}{\theta}})^{\frac{\theta}{\theta-1}}$, with primary-factor and intermediate-input nests $\hat{L}_i = (\sum_{h \in \mathcal{H}} \alpha_{ih} \hat{\ell}_{ih}^{\frac{\theta^\ell-1}{\theta^\ell}})^{\frac{\theta^\ell}{\theta^\ell-1}}$, $\hat{X}_i = (\sum_{j \in \mathcal{N}} \mathcal{W}_{ij} \hat{x}_{ij}^{\frac{\theta^x-1}{\theta^x}})^{\frac{\theta^x}{\theta^x-1}}$. Here, θ is the elasticity of substitution between the primary-factor and intermediate-input composites, θ^ℓ is the elasticity within the primary-factor nest, and θ^x is the elasticity within the intermediate-input nest. Household h 's final consumption aggregator is $\hat{C}_h = (\sum_{i \in \mathcal{N}} \beta_{hi} \hat{C}_{hi}^{\frac{\varrho-1}{\varrho}})^{\frac{\varrho}{\varrho-1}}$, where ϱ is the elasticity of substitution across final goods. The normalized-CES specification follows the calibration logic in [de La Grandville \(1989\)](#); [Klump & de La Grandville \(2000\)](#) and is standard in production-network counterfactuals ([Baqae & Farhi, 2019a, 2020, 2024](#)). Its advantage is that ω_i^ℓ , ω_i^x , α_{ih} , $\mathcal{W}_{ij}$, and β_{hi} retain the equilibrium-share interpretation used in the nonparametric decompositions. The Online Appendix derives the linear general-equilibrium system for the normalized nested-CES economy. It solves for endogenous prices, household expenditure, firm sales, and cost shares under shocks to productivity, inverse markups, ownership, transfers, preferences, and production share shifters. These responses provide the counterfactual inputs for the decompositions above.

Theorems 1–4 take changes in share objects—for example $d\beta_{hi}$, $d\omega_i^\ell$, $d\mathcal{W}_{ij}$, and $d\alpha_{ih}$ —as inputs. In the normalized CES model, the corresponding CES weights are calibrated to equilibrium expenditure and cost shares, while the realized shares adjust endogenously with relative prices according to the substitution elasticities. Thus, share variation can reflect either exogenous preference or technology shifters, or endogenous general-equilibrium adjustment under fixed CES weights. The nonparametric decompositions remain valid in either case.

Nonlinearities. The sufficient-statistic decompositions in **Theorems 1–4** are local. For finite counterfactuals, I use the differentiated hat-algebra approach of [Baqae & Farhi \(2024\)](#); [Baqae & Burstein \(2025\)](#). The procedure divides a shock into small increments and repeatedly solves the log-linear general-equilibrium system for $d \log E_h$, $d \log S_i$, and $d \log w_h$. At each step, **Proposition 1** updates prices, the CES share equations update β , $\tilde{\Omega}_\ell$, and $\tilde{\Omega}_x$, and the updated economy becomes the new baseline for the next increment. This captures nonlinear general-equilibrium effects through repeated local approximations, rather than by solving the full nonlinear equilibrium system as in exact hat algebra ([Dekle et al., 2008](#); [Costinot & Rodríguez-Clare, 2014](#)). Implementation details are in the Online Appendix.

4.2 Example 1: Redistribution and Aggregate Efficiency

To isolate the effect of redistribution on aggregate efficiency, consider a two-sector, two-household Cobb-Douglas economy. Firms $i \in \{m, a\}$, manufacturing and agriculture, produce with equal factor shares, $y_i = (\ell_{ih}\ell_{il})^{1/2}$, where h and l denote high- and low-skill labor. Households supply labor inelastically, hold equal profit claims, and consume final goods with Cobb-Douglas preferences, $C_b = C_{bm}^{\beta_b} C_{ba}^{1-\beta_b}$ for $b \in \{h, l\}$. The government implements a budget-neutral lump-sum transfer $d\tau_h$ to high-skill households financed by $d\tau_l = -d\tau_h$. Preference heterogeneity, $\beta_h \neq \beta_l$, makes redistribution change the composition of final demand. Firm-specific inverse markups, $\mu_m \neq \mu_a$, make this demand recomposition relevant for aggregate efficiency. I study redistribution around the no-transfer baseline, $\tau_h = \tau_l = 0$.

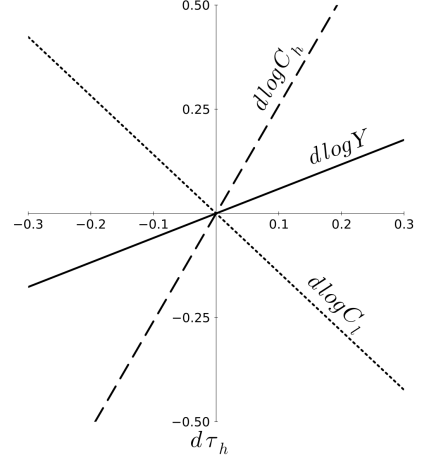
This configuration is analytically tractable and shuts down all channels except redistribution-induced final-demand recomposition. Labor supply is inelastic, there are no intermediate inputs, Cobb-Douglas preferences fix household expenditure shares within each basket, and Cobb-Douglas production fixes firms' factor-cost shares. Hence there is no endogenous labor-supply response, no price-induced substitution in consumption, no intermediate-input propagation, and no cost-share recomposition. Around the no-transfer baseline, the only first-order effect of $d\tau_h$ is to shift purchasing power across households with different consumption weights. This recomposes aggregate final demand across firms, reallocates labor across sectors, and changes TFP when $\mu_m \neq \mu_a$. The example therefore isolates the core mechanism: redistribution can affect aggregate efficiency even when productivity, wedges, preferences, and technologies are held fixed.

At the no-transfer baseline, $\tau_h = \tau_l = 0$, both household types have the same expenditure share, $\chi_h = \chi_l = 1/2$, and the same cost-based value-added share, $\tilde{\Lambda}_h = \tilde{\Lambda}_l = 1/2$. The manufacturing and agriculture sales shares are $\lambda_m = \tilde{\beta} = \frac{\beta_h + \beta_l}{2}$ and $\lambda_a = 1 - \tilde{\beta}$. Each worker type earns the same factor-income share, $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + \frac{1}{2}\tilde{\beta}(\mu_m - \mu_a)$. The expenditure centrality of household $b \in \{h, l\}$ is $M_b = \frac{\mu_a + \beta_b(\mu_m - \mu_a)}{\mu_a + \tilde{\beta}(\mu_m - \mu_a)}$. Consider an infinitesimal transfer $d\tau_h$, measured as a share of GDP, from low-skill to high-skill households, so $d\chi_h = d\tau_h$ and $d\chi_l = -d\tau_h$. Since household demands are Cobb-Douglas, the transfer changes the manufacturing sales share by $d\lambda_m = (\beta_h - \beta_l)d\tau_h$ and $d\lambda_a = -d\lambda_m$. Because both sectors use high- and low-skill labor with equal cost shares, the induced change in each factor-income share is $d\Lambda_h = d\Lambda_l = \frac{1}{2}(\beta_h - \beta_l)(\mu_m - \mu_a)d\tau_h$. Factor supplies are fixed, so $d\log Y = d\log TFP$. Applying the entropic and distributive representations in [Theorem 2](#),

$$d\log Y = - \sum_{b \in \{h, l\}} \frac{\tilde{\Lambda}_b}{\Lambda_b} d\Lambda_b = -(M_h - M_l)d\chi_h = \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)} d\tau_h.$$

This expression shows that redistribution affects aggregate efficiency only through final-demand recomposition. A transfer to high-skill households raises TFP iff it shifts demand toward the more distorted sector: $(\beta_h - \beta_l)(\mu_a - \mu_m) > 0$. For example, if high-skill households spend relatively more on manufacturing ($\beta_h > \beta_l$), then the transfer raises TFP when manufacturing is more distorted than agriculture ($\mu_m < \mu_a$). Conversely, if the transfer shifts demand toward the less distorted sector, aggregate output falls. Thus, redistribution can have first-order TFP effects even though productivity, wedges, preferences, and technologies are held fixed.

Figure 1: Output and Consumption



From [Theorem 4](#), household real-consumption responses are $d \log C_h = 2 d \tau_h + d \log Y$ and $d \log C_l = -2 d \tau_h + d \log Y$. The common term $d \log Y$ is the aggregate-efficiency effect shared by both households, while the terms $\pm 2 d \tau_h$ capture the direct redistribution of purchasing power. A transfer from low-skill to high-skill households, $d \tau_h > 0$, therefore raises C_h and lowers C_l whenever the transfer effect dominates the common efficiency gain for the taxed household. [Figure 1](#) plots the local responses at $(\beta_h, \beta_l, \mu_m, \mu_a) = (0.8, 0.3, 0.2, 0.7)$. At these parameters, the transfer raises output because high-skill households spend more on the more distorted manufacturing firm. The high-skill household gains from both the transfer and the efficiency effect, while the low-skill household loses from the tax but partially recovers through the shared increase in aggregate output. The example illustrates that redistribution-driven efficiency gains need not be zero-sum, even though incidence remains heterogeneous.

4.3 Example 2: Redistribution by the Constrained Social Planner

Return to the economy in [Example 1](#). Normalize baseline productivities to $A_m = A_a = 1$ and fix factor supplies at $L_h = L_l = 1$. At the first best, labor is allocated according to final-demand weights: $y_m = \ell_{mh} = \ell_{ml} = \lambda_m = \bar{\beta}$ and $y_a = \ell_{ah} = \ell_{al} = \lambda_a = 1 - \bar{\beta}$. For productivity shocks around the first best, Hulten's theorem gives $d \log Y = \bar{\beta} d \log A_m + (1 - \bar{\beta}) d \log A_a$.

With firm-level wedges, the decentralized no-transfer equilibrium is inefficient. Labor and output are allocated as $y_m = \frac{\bar{\beta} \mu_m}{\bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a}$ and $y_a = \frac{(1 - \bar{\beta}) \mu_a}{\bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a}$, while sales shares remain $\lambda_m = \bar{\beta}$ and $\lambda_a = 1 - \bar{\beta}$. Without intermediate inputs, misallocation arises only from relative wedges across final-good producers, $\mu_m \neq \mu_a$. Let $\Gamma \equiv \bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a$ denote the aggregate factor-income share. With real GDP aggregator $Y = (C_h C_l)^{1/2}$, inefficient output relative to

first-best output is $\frac{Y}{Y_{\text{eff}}} = \frac{\mu_m^\beta \mu_a^{1-\beta}}{\bar{\beta} \mu_m + (1-\bar{\beta}) \mu_a} \leq 1$. By **Theorem 2**, first-order output growth satisfies

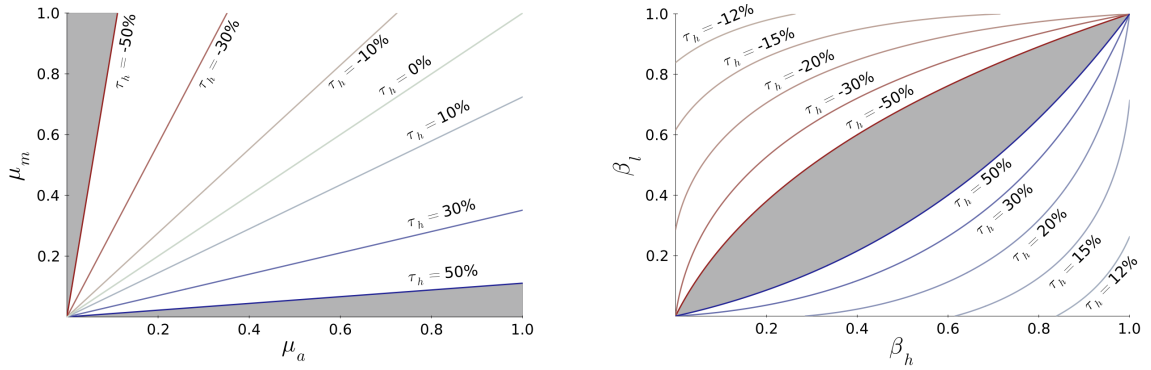
$$d \log Y = \bar{\beta} d \log A_m + (1 - \bar{\beta}) d \log A_a + \frac{\mu_a - \mu_m}{\Gamma} \left(\bar{\beta}(1 - \bar{\beta}) d \log \left(\frac{\mu_m}{\mu_a} \right) + (\beta_h - \beta_l) d \chi_h \right).$$

This expression isolates three mechanisms. First, productivity shocks affect output with Domar weights as elasticities. Second, lower relative wedge dispersion raises output: if manufacturing is more distorted ($\mu_m < \mu_a$), increasing μ_m / μ_a improves allocative efficiency. Third, redistribution affects output by changing final-demand composition. A $d\tau_h > 0$ raises output when high-skill households spend relatively more on the more distorted sector, i.e., $(\beta_h - \beta_l)(\mu_a - \mu_m) > 0$.

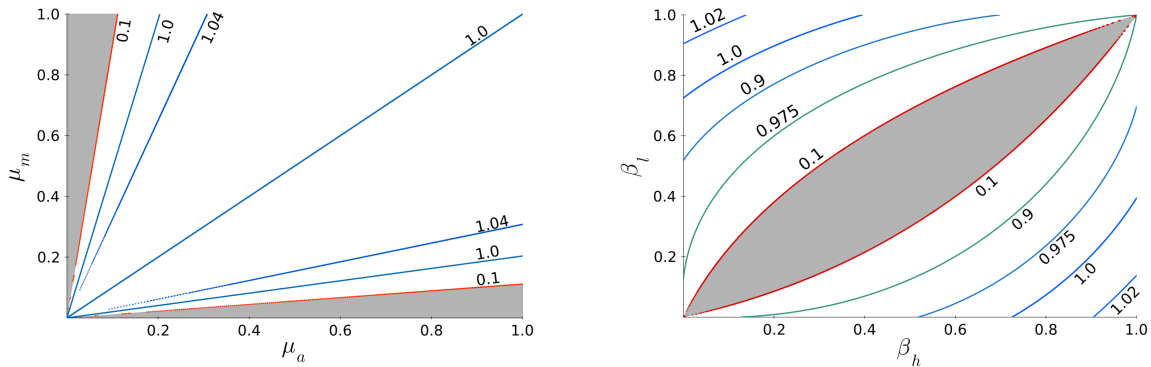
The constrained planner in **Theorem 3** can alter the expenditure distribution through lump-sum transfers, since $\chi_h = 1/2 + \tau_h$ and $\chi_l = 1/2 - \tau_h$. One feasible target is to restore the efficient sectoral labor allocation, $y_m = \bar{\beta}$ and $y_a = 1 - \bar{\beta}$. This requires setting the high-skill transfer to $\tilde{\tau}_h = \frac{\bar{\beta}(1-\bar{\beta})}{\bar{\beta}\mu_a + (1-\bar{\beta})\mu_m} \left(\frac{\mu_a - \mu_m}{\beta_h - \beta_l} \right)$, provided $|\tilde{\tau}_h| \leq 1/2$.

Figure 2: Transfers for Efficient Worker Allocation: Schedule and Output Effects

A. $\tilde{\tau}_h$ Schedule as a Function of (μ_a, μ_m) and (β_h, β_l)



B. Output Ratio Relative to the No-Transfer Baseline



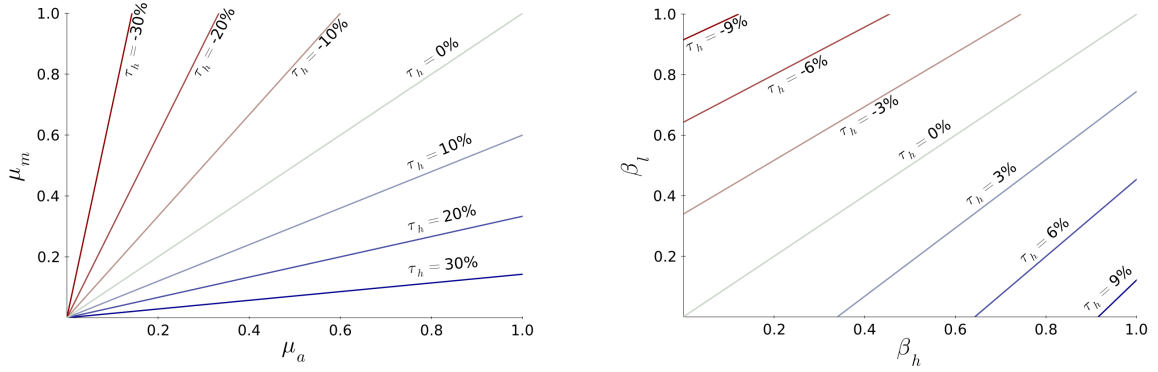
Note: Panel A: $\tilde{\tau}_h(\mu_a, \mu_m)$ at $(\beta_h, \beta_l) = (0.9, 0.1)$ (left) and $\tilde{\tau}_h(\beta_h, \beta_l)$ at $(\mu_m, \mu_a) = (0.4, 0.6)$. Panel B reports $Y(\tilde{\tau}_h)/Y(0)$ for the same calibrations. Shaded gray regions indicate efficient worker allocation infeasible ($|\tilde{\tau}_h| > 1/2$). Positive $\tilde{\tau}_h$ denotes transfers from low- to high-skill households.

In a representative-household economy, restoring the efficient input allocation is sufficient to restore first-best output. With heterogeneous households, this is no longer true because real

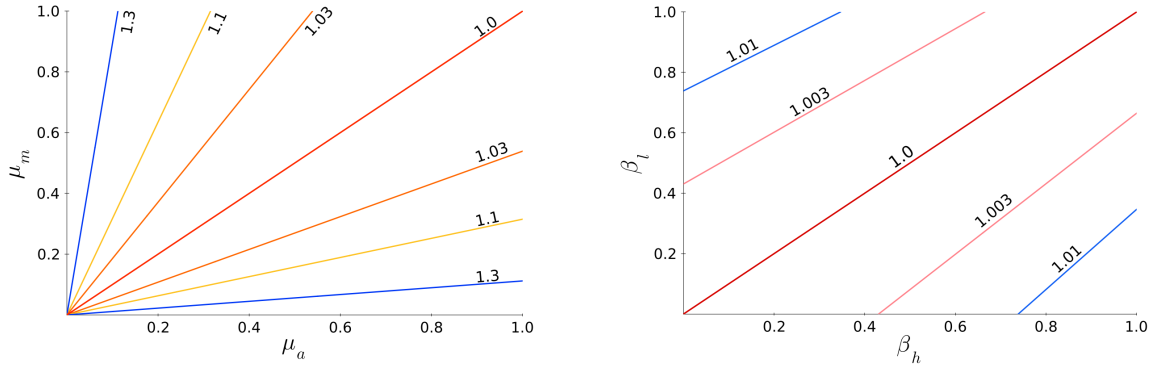
GDP aggregates household real consumption. A transfer that corrects production misallocation can also increase consumption inequality, which lowers $Y = \sqrt{C_h C_l}$. **Figure 2** illustrates this tradeoff. Panel A reports the transfer $\tilde{\tau}_h$ required to restore the efficient worker allocation; gray regions indicate infeasibility, $|\tilde{\tau}_h| > 1/2$, where the required expenditure shift exceeds the feasible transfer range. Panel B reports $Y(\tilde{\tau}_h)/Y(0)$. Values above one indicate that the efficiency gain dominates the induced consumption inequality, while values below one indicate that the inequality cost dominates. The non-monotonicity reflects the central lesson: correcting production distortions and maximizing aggregate real consumption need not coincide once households are heterogeneous.⁷

Figure 3: Optimal Transfer: Schedule and Output Effects

A. τ_h^* Schedule as a Function of (μ_a, μ_m) and (β_h, β_l)



B. Output Ratio Relative to the No-Transfer Baseline



Note: Panel A: $\tau_h^*(\mu_a, \mu_m)$ at $(\beta_h, \beta_l) = (0.9, 0.1)$ (left) and $\tau_h^*(\beta_h, \beta_l)$ at $(\mu_m, \mu_a) = (0.4, 0.6)$ (right). Panel B reports $Y(\tau_h^*)/Y(0)$ for the same calibrations.

The optimal transfer does not generally replicate the first-best worker allocation. Instead, it trades off the production-efficiency gain from shifting expenditure toward the more distorted sector against the real-consumption inequality created by the transfer. In this example, the unconstrained optimal transfer is $\tau_h^* = \frac{1}{4} \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\beta \mu_m + (1 - \beta)\mu_a}$. **Figure 3** plots τ_h^* and the implied output ratio $Y(\tau_h^*)/Y(0)$ relative to the no-transfer equilibrium. Panel A reports τ_h^* as a function of

⁷Small $|\tilde{\tau}_h|$ reallocate labor toward the more distorted firm and raise Y , while larger $|\tilde{\tau}_h|$ increases real consumption inequality, which the aggregator penalizes.

(μ_a, μ_m) , holding (β_h, β_l) fixed, and as a function of (β_h, β_l) , holding (μ_m, μ_a) fixed. Panel B reports $Y(\tau_h^*)/Y(0)$. Values above one indicate that the optimal transfer raises output relative to the decentralized no-transfer equilibrium. The output gain reflects the planner's balance between correcting production misallocation and limiting the consumption-inequality cost generated by redistribution.

At τ_h^* , the manufacturing sales share and aggregate factor-income share are $\lambda_m^* = \bar{\beta} + (\beta_h - \beta_l)\tau_h^*$ and $\Gamma^* = \bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h^*$. Relative to the no-transfer equilibrium, expenditure shifts toward the sector with the higher markup, and the aggregate factor-income share falls. Evaluated at the optimal-transfer allocation, [Theorem 3](#) gives

$$d \log Y = \bar{\beta} d \log A_m + (1 - \bar{\beta}) d \log A_a + (\beta_h - \beta_l)\tau_h^* d \log \left(\frac{A_m}{A_a} \right) + (\mu_a - \mu_m) \frac{\lambda_m^*(1 - \lambda_m^*)}{\Gamma^*} d \log \left(\frac{\mu_m}{\mu_a} \right).$$

This expression illustrates why the constrained-planner allocation changes measurement. The sufficient statistics are allocation-dependent: the first-best, the decentralized no-transfer equilibrium, and the optimal-transfer allocation attach different weights to the same shocks. At the optimum, marginal expenditure reallocation is neutral, but the transfer changes Do-mar weights and the aggregate factor-income share. Relative technology and markup shocks are therefore measured at the planner-induced allocation, not at the decentralized one. The example reinforces the main implication of [Theorem 3](#): decentralized choices can generate aggregate-efficiency externalities, and centralization changes both the allocation and the TFP decomposition.

5 Quantitative Implementation

This section takes the sufficient statistics to U.S. data. I first construct a representative-household benchmark to assess aggregate fit and select the baseline production-network specification. I then introduce occupational heterogeneity by combining occupation-industry earnings exposure with income-conditioned consumption bundles.

5.1 Representative Household Model

I combine three BEA/BLS data sources. First, BEA's Industry-Level Production Accounts (ILPA) with Extended Capital Detail provide sectoral TFP and capital compensation by nine asset classes. I harmonize ILPA to the Production Accounts classification and construct a balanced industry-asset-year panel for 1997–2021. Second, BEA Use and Make tables provide input-output linkages for 66 private industries after dropping government and scrap/used categories. Row-normalizing input expenditures yields the industry input-output matrix $\mathcal{W}$.

For each industry-year, I construct sales, value added, and input costs, where costs include labor, intermediates, and capital by asset. I measure inverse markups as $\mu_{i,t} = \text{Cost}_{i,t} / \text{Sales}_{i,t}$. When measured costs exceed sales, $\mu_{i,t} > 1$; I interpret these cases as operating-loss industry-years. The baseline model uses measured $\mu_{i,t}$ directly, including observations above one, and therefore does not impose $\mu_{i,t} \leq 1$. Online Appendix F.1 reports the frequency of these observations and evaluates robustness to capping, interpolation, and winsorization. The results show that centrality rankings and transfer counterfactuals are robust to alternative treatments of $\mu_{i,t} > 1$, while the historical relevance from the distributional-reallocation share is sensitive to broader tail variation in measured μ . Third, I construct productivity growth as $d \log A_{i,t} = \log(\text{TFP}_{i,t} / \text{TFP}_{i,t-1})$, mapped from ILPA to the 66-industry classification. Summary statistics for μ and $d \log A$ are reported in the Online Appendix.

The representative-household model spans 66 sectors. The household receives income from labor, nine capital assets, and profits. I estimate eight specifications that vary along three dimensions: labor only versus labor plus capital; no intermediates versus input-output networks; and input-output final-demand shares β_{IO} versus PCE consumption bundles β_{PCE} . Each specification produces a $d \log \text{TFP}$ series from [Theorem 2](#), using industry-level variation in productivity, markups, and factor-income shares. I compare these series to BLS aggregate TFP growth. The BLS series is a benchmark, not a gold standard: the model-implied $d \log \text{TFP}$ is not a residual, but a constructed accounting object. [Table 3](#) compares the eight specifications using four fit statistics: RMSE, bias, correlation, and the index of agreement. The best-fitting specification is column (4), which includes labor and capital, input-output networks, and β_{IO} ; it attains $\text{IA} = 0.923$, $\text{correlation} = 0.875$, and the lowest RMSE. I use this specification as the baseline.

Table 3: Comprehensive Model Comparison Metrics

Metric	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MAE	$6.2e^{-3}$	$1.5e^{-2}$	$6.2e^{-3}$	$4.7e^{-3}$	$7.7e^{-3}$	$1.8e^{-2}$	$7.6e^{-3}$	$7.4e^{-3}$
RMSE	$7.4e^{-3}$	$1.9e^{-2}$	$7.3e^{-3}$	$5.8e^{-3}$	$9.5e^{-3}$	$2.4e^{-2}$	$9.5e^{-3}$	$9.5e^{-3}$
Bias	$5.4e^{-3}$	$-8.9e^{-3}$	$5.3e^{-3}$	$2.1e^{-4}$	$6.9e^{-3}$	$-8.5e^{-3}$	$6.9e^{-3}$	$1.3e^{-3}$
Corr	0.871	0.702	0.873	0.875	0.748	0.597	0.748	0.767
IA	0.815	0.646	0.819	0.923	0.760	0.562	0.761	0.839

Notes: Columns (1)-(8): labor (L) vs. labor+capital (LK) $\times$ no-intermediates (NN) vs. networks (N) $\times$ β_{IO} vs. β_{PCE} : (1) L,NN, β_{IO} ; (2) L,N, β_{IO} ; (3) LK,NN, β_{IO} ; (4) LK,N, β_{IO} ; (5) L,NN, β_{PCE} ; (6) L,N, β_{PCE} ; (7) LK,NN, β_{PCE} ; (8) LK,N, β_{PCE} . Closer is better for MAE, RMSE, and Bias; higher is better for corr, NSE, and IA.

5.2 Occupational Heterogeneity

I introduce household heterogeneity using occupations. For each year $t \in [1997, 2021]$, I use BLS Occupational Employment and Wage Statistics (OEWS) to construct occupation o 's labor-cost share in industry i : $\alpha_{io,t} = \frac{\text{Labor Income}_{io,t}}{\sum_{o'} \text{Labor Income}_{io',t}}$. The household set contains 843 types: 833

occupational labor-income households, nine capital-asset households, and one residual profit-income household. The factor-share matrix α_t stacks the 833 occupations and nine capital assets; firm profits are assigned to the residual profit-income household, so all profits are received by some household type and household budgets exhaust GDP. To assign consumption bundles to occupations, I use the Consumer Expenditure Survey (CEX). For each CEX household-year, I construct a smoothed 66-industry expenditure-share vector using the household's 100 nearest-income neighbors. Let $\bar{J}_{o,t}$ denote mean OEWS earnings for occupation o . I assign occupation o the CEX expenditure vector associated with the household income closest to $\bar{J}_{o,t}$, denoted $\beta_{o,t}^{CEX}$. The residual non-labor bundle is chosen to match aggregate Production Accounts final demand: $\beta_t^{NL} = (1 - \sum_o \Lambda_{o,t})^{-1} (\beta_t^{IO} - \sum_o \Lambda_{o,t} \beta_{o,t}^{CEX})$. I assign β_t^{NL} to the nine capital-asset households and to the residual profit-income household. Thus, the occupational model preserves aggregate final demand while allowing labor-income groups to differ in both industry exposure and consumption baskets. Relative to the representative-household benchmark, aggregate fit falls modestly but remains strong: the index of agreement falls from 0.923 to 0.878, and the correlation with BLS aggregate TFP growth falls from 0.875 to 0.824.

5.3 Empirical Sufficient Statistics and TFP Decomposition

I next evaluate the sufficient statistics from [Theorem 2](#). [Table 4](#) reports covariance shares, $100 \times \text{Cov}(X_t, d \log TFP_t) / \text{Var}(d \log TFP_t)$, for the components of aggregate TFP growth across three specifications: representative household, geographic state-level heterogeneity, and occupational heterogeneity. The geographic column corresponds to the state-level robustness specification in [Online Appendix F.3](#); it shows that the aggregate distributional-reallocation component is stable under a household partition based on geography rather than occupations. The table establishes the time-series importance of distributional reallocation before turning to the cross-sectional centrality rankings used in the transfer counterfactuals in [Section 6](#).

Three facts stand out. First, technology is the dominant source of TFP-growth fluctuations, but distributional reallocation is quantitatively relevant: it accounts for 17–20% of TFP-growth comovement across specifications. Second, the distributional component is driven mainly by firm-side recomposition. Direct household redistribution and final-demand recomposition are small in the aggregate, while intermediate-input and factor-demand terms are large and offsetting: intermediate-input recomposition offsets TFP fluctuations, whereas factor-demand recomposition amplifies them. Third, direct markup effects are largely offset by the competitive terms-of-trade channel. The net markup contribution is close to zero across all specifications. Thus, focusing only on the direct competitiveness term would miss the indirect reallocation channel, while focusing only on household redistribution would miss the dominant firm-side

Table 4: TFP Growth Decomposition: Covariance Shares

	Rep. Household	Geographic Heterogeneity	Occupational Heterogeneity
<i>Technology</i>	107.44%	108.79%	116.86%
<i>Competitiveness</i>	-15.94%	-18.19%	-17.52%
<i>Distributional Reallocation</i>	17.93%	20.04%	17.48%
<i>Residual</i>	-9.43%	-10.65%	-16.82%
<i>-Distributive TT</i>	0.00%	-0.06%	-0.56%
<i>-Final-Demand TT</i>	0.25%	-0.05%	1.17%
<i>-Intermediate-Demand TT</i>	-27.09%	-29.53%	-23.24%
<i>-Factor-Demand TT</i>	29.41%	32.10%	23.42%
<i>-Competitive TT</i>	15.37%	17.59%	16.69%
<i>Net Markup Contribution</i>	-0.58%	-0.59 %	-0.83%

Notes: Entries report $100 \times \text{Cov}(X_t, d \log TFP_t) / \text{Var}(d \log TFP_t)$, where $d \log TFP_t$ is BLS aggregate TFP growth. Columns correspond to the representative-household specification, the geographic-heterogeneity robustness specification, and the occupational-heterogeneity specification. Positive entries comove positively with TFP growth; negative entries offset TFP fluctuations. The residual is $d \log TFP_t - \text{Technology}_t - \text{Competitiveness}_t - \text{Distributional Reallocation}_t$. By construction, $\text{Technology} + \text{Competitiveness} + \text{Distributional Reallocation} + \text{Residual} = 100\%$. The lower panel reports the signed contributions of the terms-of-trade components, so $\text{Distributional Reallocation} = -\text{Distributive TT} - \text{Final-Demand TT} - \text{Intermediate-Input TT} - \text{Factor-Demand TT} - \text{Competitive TT}$. The net inverse-markup contribution is $\text{Competitiveness} - \text{Competitive TT}$.

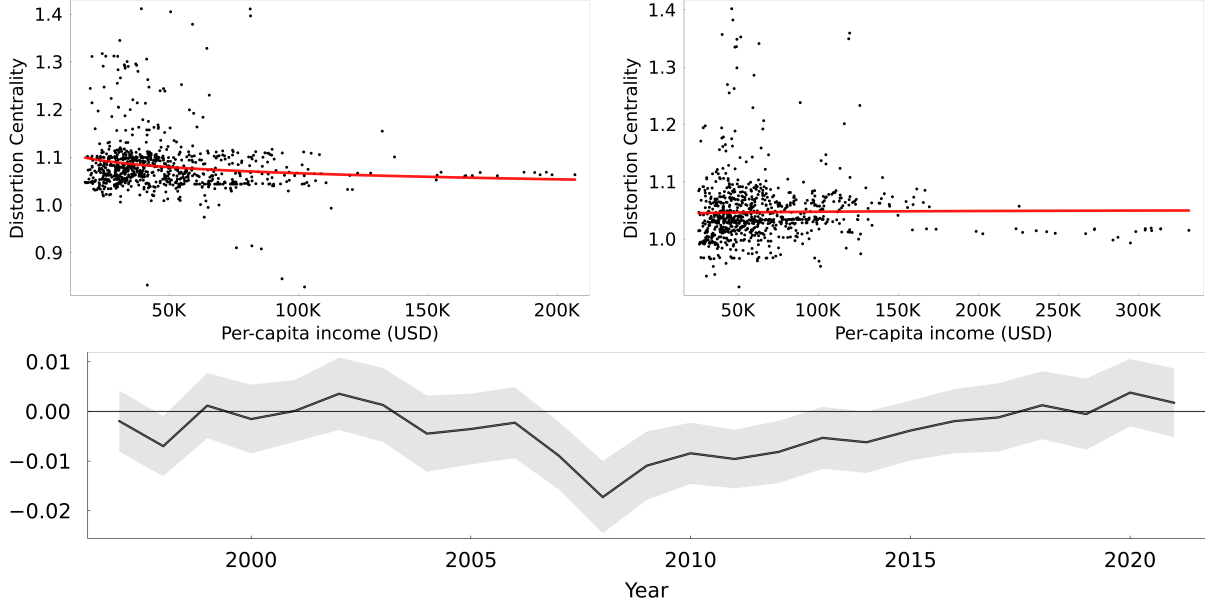
recomposition margins.

Because $F_{i,t}$ is the sufficient statistic for final- and intermediate-demand recomposition in [Theorem 2](#), I next ask what $F_{i,t}$ captures in the data. Using the baseline specification, the Online Appendix reports the full sector-year ranking of revenue centralities. The cross-sector mean of $F_{i,t}$ is 0.99. Sectoral time averages range from 0.77 for arts, sports, and museums to 1.04 for support activities for mining. The full sector-year range is wider: the sample minimum is 0.70 for arts, sports, and museums in 2004, while the maximum is 1.29 for securities, commodity contracts, and investments in 2008. Securities, commodity contracts, and investments is also the most volatile sector, with $F_{i,t}$ ranging from 0.83 to 1.29 over the sample. Empirically, $F_{i,t}$ is not a size ranking: its yearly correlation with revenue Domar weights $\lambda_{i,t}$ averages only 0.01. Instead, $F_{i,t}$ is tightly linked to inverse markups. The yearly correlation between $F_{i,t}$ and $\mu_{i,t}$ averages 0.94, and within-sector changes in own markups account for 79% of year-to-year movements in $F_{i,t}$. Thus, revenue centrality mainly captures pass-through and wedge exposure rather than sector size: high- F sectors tend to have lower markups, while F is nearly orthogonal to sales shares.

The small aggregate contribution of distributive terms of trade does not imply that household sufficient statistics are homogeneous. The Online Appendix reports yearly moments for labor and capital distortion centralities, $\delta_{o,t}$ and $\delta_{k,t}$, and occupational expenditure centralities, $M_{o,t}$. Average distortion centrality is 1.09 for labor and 1.07 for capital, and both decline over time: labor falls from 1.11 to 1.04, and capital from 1.09 to 1.03. Expenditure centrality averages 0.99 and declines slightly, from 1.00 to 0.98. Cross-sectional dispersion is much larger with

occupational than geographic heterogeneity; for example, in 2004 the max-min spread in δ is 53 percentage points across occupations, compared with 6.7 percentage points across states. Thus, even when direct distributive terms are modest in the aggregate time series, the cross-sectional centrality dispersion leaves scope for TFP gains from targeted expenditure reallocation.

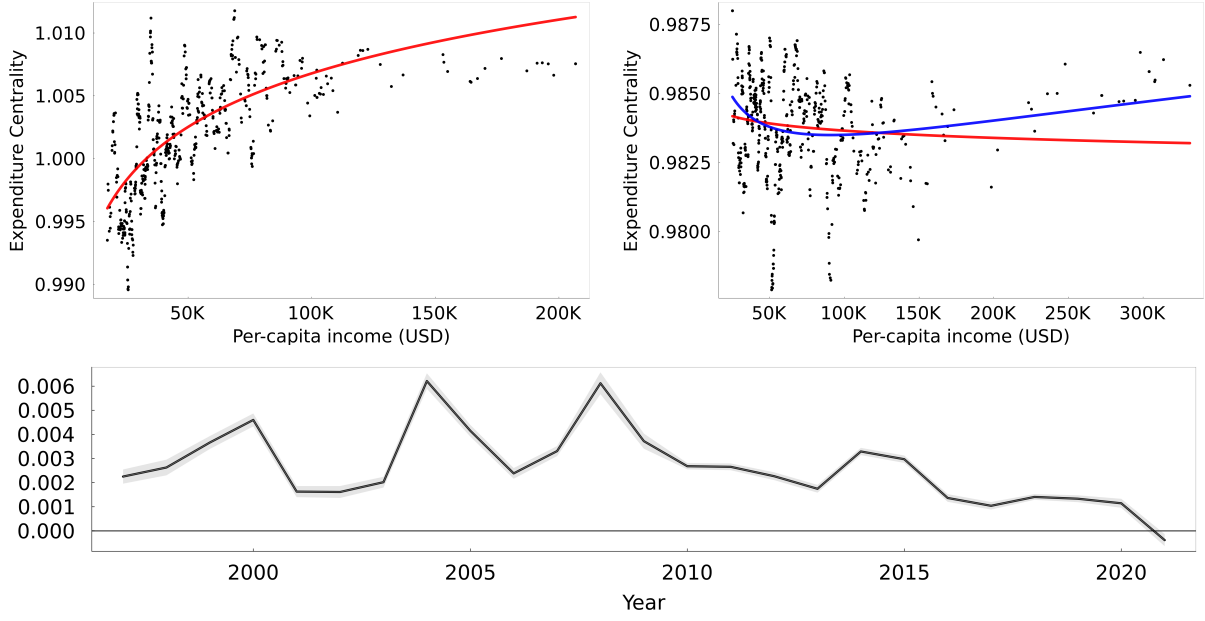
Figure 4: Occupation-Level Distortion Centrality and Earnings



Note: The top panels plot occupation-level distortion centrality against mean OEWS earnings in 2008 (left) and 2021 (right); each dot is a SOC occupation ($N = 833$). The fitted line is the year-specific log-log regression $\log \delta_{o,t} = \beta_{0,t}^\delta + \beta_{1,t}^\delta \log \bar{J}_{o,t} + \epsilon_{o,t}^\delta$, where $\bar{J}_{o,t}$ denotes mean OEWS earnings for occupation o . The bottom panel reports the estimated income elasticity $\beta_{1,t}^\delta$ for 1997–2021; shaded bands are 95% pointwise confidence intervals. Negative values imply that higher-earning occupations have lower distortion centrality, while positive values imply the opposite.

Figures 4 and 5 relate occupational earnings to distortion and expenditure centralities. For each year t , I estimate $\log \delta_{o,t} = \beta_{0,t}^\delta + \beta_{1,t}^\delta \log \bar{J}_{o,t} + \epsilon_{o,t}^\delta$ and $\log M_{o,t} = \beta_{0,t}^M + \beta_{1,t}^M \log \bar{J}_{o,t} + \epsilon_{o,t}^M$, where $\bar{J}_{o,t}$ denotes mean OEWS earnings for occupation o . The top panels show the 2008 and 2021 cross-sections with fitted lines; the bottom panels plot the yearly income elasticities, $\beta_{1,t}^\delta$ and $\beta_{1,t}^M$, with 95% confidence intervals. For distortion centrality, $\beta_{1,t}^\delta < 0$ in most years and is statistically different from zero in 1998 and 2007–2012, implying that higher-earning occupations tend to have lower $\delta_{o,t}$. In those years, factor-income shifts toward high earners raise TFP through the distributional-reallocation channel. For expenditure centrality, $\beta_{1,t}^M > 0$ and is statistically significant in all years before 2021, implying a pre-2021 *lower-income multiplier*: shifting expenditure from higher-earning-high- M occupations toward lower-earning-low- M occupations raises TFP. In 2021, the elasticity turns negative, and the quadratic fit becomes U-shaped, with a minimum around \$50–\$100k. This implies a *middle-income multiplier*: expenditure shifts toward middle-income occupations are productivity-enhancing relative to shifts toward either tail of the earnings distribution. The decline in $\beta_{1,t}^M$ through the 2010s foreshadows this transition from a monotone lower-income multiplier to a middle-income multiplier.

Figure 5: Occupation-Level Expenditure Centrality and Earnings



Note: The top panels plot occupation-level expenditure centrality against mean OEWS earnings in 2008 (left) and 2021 (right); each dot is a SOC occupation ($N = 833$). The fitted line is the year-specific log-log regression $\log M_{o,t} = \beta_{0,t}^M + \beta_{1,t}^M \log \bar{J}_{o,t} + \epsilon_{o,t}^M$, where $\bar{J}_{o,t}$ denotes mean OEWS earnings for occupation o . The fitted curve is the year-specific quadratic log-log regression $\log M_{o,t} = \gamma_{0,t}^M + \gamma_{1,t}^M \log \bar{J}_{o,t} + \gamma_{2,t}^M (\log \bar{J}_{o,t})^2 + u_{o,t}^M$. The bottom panel reports the estimated income elasticity $\beta_{1,t}^M$ for 1997–2021; shaded bands are 95% pointwise confidence intervals. Negative values imply that higher-earning occupations have lower expenditure centrality.

These results provide the empirical bridge to the counterfactuals. The covariance decomposition shows that distributional reallocation is quantitatively relevant for TFP fluctuations, while the cross-sectional M -hierarchies identify the local sign of expenditure redistribution: by [Theorem 2](#), shifting spending from high- M to low- M households raises TFP. [Section 6](#) asks whether this first-order ranking remains informative in general equilibrium.

6 From M to GE: Distributional Engines of Productivity

This section uses the occupational-heterogeneity CES economy to quantify the aggregate and distributional effects of M -guided transfers. I first estimate the common substitution elasticities by matching BLS aggregate TFP growth over 1997–2021. I then run two counterfactual exercises. The first isolates pure redistribution by holding productivity and markups fixed. The second superimposes the same transfers on the historical sequence of U.S. productivity and markup shocks. The main result is that the Okun tradeoff is conditional rather than mechanical. In the calibrated U.S. economy, M -guided transfers generate modest TFP gains while improving real-consumption incidence for lower-earning occupations.

6.1 Calibration and Elasticity Robustness

Table 5: Multi-Start GMM Estimates and Model Fit

Parameter	Mean	Std. Dev.	Best	95% Range	
θ	0.079	0.072	0.001	[0.001, 0.246]	
θ_ℓ	1.466	0.812	0.946	[0.191, 2.894]	
θ_x	1.959	0.465	2.319	[1.095, 2.849]	
ϱ	0.482	0.370	0.065	[0.014, 1.320]	
	MAE	RMSE	Bias	Corr	IA
Mean	5.32e-03	6.68e-03	3.00e-05	0.892	0.929
Best	5.21e-03	6.54e-03	2.52e-07	0.891	0.930

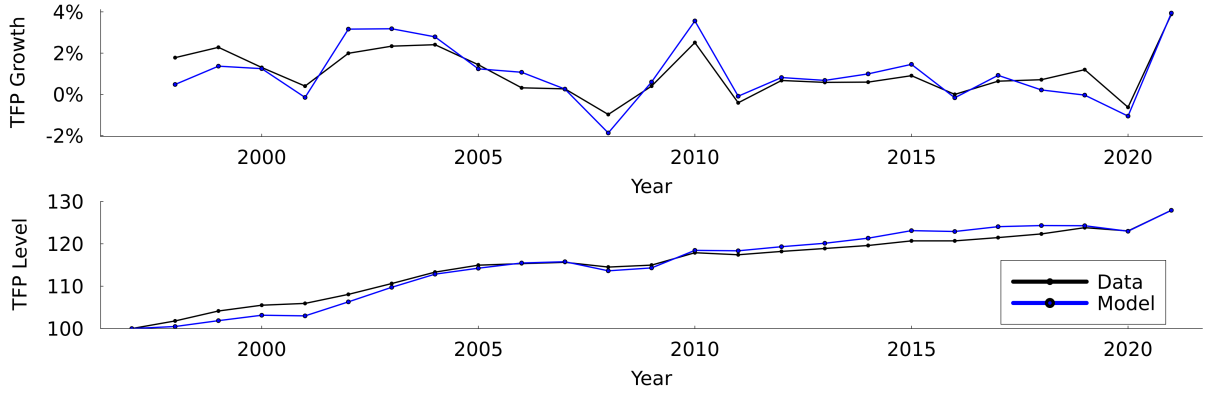
Note: First panel reports estimates from 500 independent two-stage GMM runs. Because the moments are growth rates, the GMM sample is 1998–2021. Each run uses adaptive differential evolution followed by L-BFGS with automatic differentiation. “Mean” is the average across runs; “Best” is the parameter vector with the lowest GMM objective. “Std. Dev.” and the central 95% range describe cross-run dispersion. The second panel reports fit statistics evaluated using Mean and Best θ .

Table 5 reports estimates from 500 independent two-stage GMM runs on the unchanged sample. “Mean” is the average across runs, and “Best” is the parameter vector with the lowest objective value. The best fit implies strong complementarity between primary factors and intermediates, $\theta = 0.001$, an approximately unit elasticity within the primary-factor nest, $\theta_\ell = 0.946$, high substitutability across intermediates, $\theta_x = 2.319$, and strong complementarity across household consumption goods, $\varrho = 0.065$. These values are best interpreted as economy-wide elasticities chosen to match aggregate TFP dynamics, not as direct micro estimates. For comparison, [Boehm et al. \(2014\)](#) estimate near-Leontief substitution around the 2011 Tōhoku shock, while [Atalay \(2017\)](#) estimates lower intermediate-input substitutability in industry-level IV regressions. The estimates here imply stronger factor-intermediate complementarity and higher intermediate-input substitutability than those benchmarks; neither study directly estimates θ_ℓ or ϱ . Online Appendix F.2 repeats the transfer counterfactuals under mean-GMM, Cobb–Douglas, low- and high-substitution, and literature-inspired elasticity calibrations.

Model fit is strong for both the mean and best parameters. Under the best fit, the correlation with BLS aggregate TFP growth is 0.89 and the index of agreement is 0.93, slightly above the non-parametric representative-household model. **Figure 6** plots annual TFP growth and the implied TFP level. The model matches the pre-2008 expansion, the 2008–09 contraction, the 2010 rebound, and the 2021 surge, and the level index tracks the BLS series closely through the sample. I therefore use the best-fit parameter vector for the counterfactual transfer exercises.

I calibrate the nested-CES economy by estimating four elasticities: between factors and intermediates, θ ; within factors, θ_ℓ ; within intermediates, θ_x ; and across consumption goods, ϱ . Let $\theta = (\theta, \theta_\ell, \theta_x, \varrho)$ and restrict $\theta \in [0, 3]^4$. I estimate θ by GMM, minimizing $Q(\theta) = g(\theta)'g(\theta)$, where $g_t(\theta) = d \log TFP_t^{model}(\theta) - d \log TFP_t^{BLS}$ stacks annual observations over 1998–2021. For each candidate θ , I compute model TFP growth using the same sectoral productivity and inverse-markup shock sequences, $d \log A_t$ and $d \log \mu_t$. To handle nonconvexity, I use adaptive differential evolution followed by L-BFGS with automatic differentiation.

Figure 6: U.S. TFP: Data versus Model Fit, 1997–2021

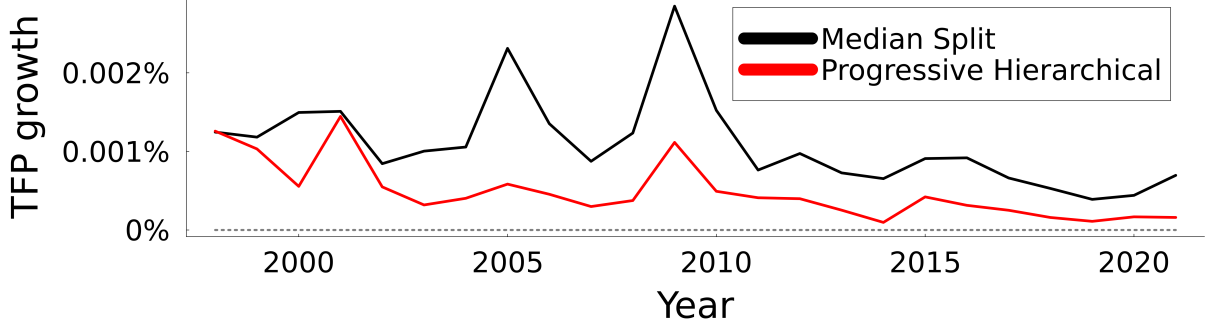


Note: The model series uses the best-fit parameter vector reported in Table 5. Model growth uses the estimated sectoral productivity and inverse-markup shock sequences, $\{d \log A_t, d \log \mu_t\}$. The top panel plots annual TFP growth; the bottom panel plots the TFP level index, normalized to 100 in 1997. Data are BLS aggregate TFP growth.

6.2 Pure Redistribution

To evaluate the general equilibrium effects of pure expenditure redistribution, I hold productivity and markup shocks fixed and apply budget-neutral transfers across occupations in each year. Transfers are defined at the worker level: a \$1,000 transfer to occupation o in year t equals $1000 \times N_{o,t}$ dollars in the aggregate, where $N_{o,t}$ is occupation-level employment. The transfer is expressed as a share of that year's GDP. I consider two rules based on expenditure centrality M . First, the *median-split rule* taxes workers in occupations with M above the employment-weighted median and rebates to workers in occupations with M below it. Second, the *progressive hierarchical-matching rule* allows transfers only from higher-income-higher- M occupations to lower-income-lower- M occupations. Matches are formed sequentially along the M -ranking, exhausting high- M donors and low- M recipients first; unmatched workers are left untreated. In both cases, transfers are budget neutral within year.⁸

Figure 7: TFP Growth Response to a \$1,000 Per-Worker Transfer



Note: The figure reports the change in annual TFP growth from budget-neutral occupation-level transfers with no productivity or inverse-markup shocks. The policy unit is \$1,000 per worker. For occupation o with employment $N_{o,t}$, the aggregate dollar transfer is $1000 N_{o,t}$, expressed in the model as a share of GDP_t .

Figure 7 plots the TFP-growth response to the two transfer rules. Under both rules, the TFP

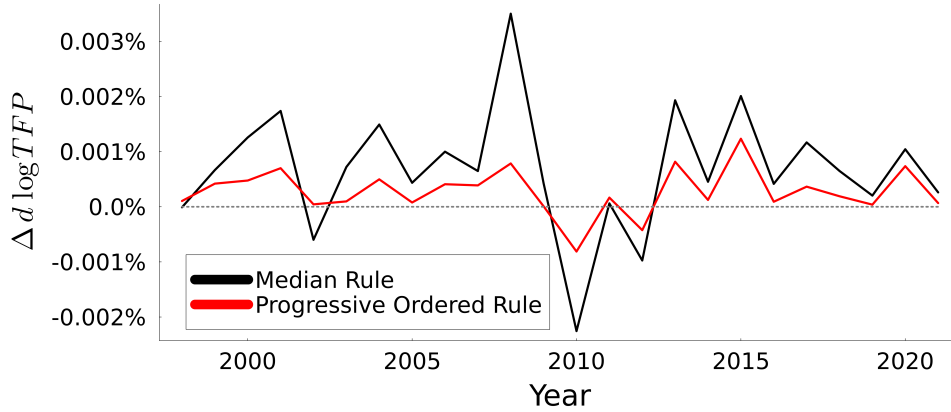
⁸Online Appendix F.4 compares these M -guided transfers with same-volume reverse- M , income-only progressive, and random transfer rules, and reports effects per dollar transferred.

effect is positive in every year but modest in magnitude, reflecting the small size of the transfer relative to GDP. The median-split rule delivers larger aggregate gains because it reallocates expenditure across a larger matched employment mass, while the progressive hierarchical rule imposes the additional income-ordering constraint and leaves some workers unmatched. The response peaks in the early-to-mid 2000s and declines thereafter, consistent with the weakening of the measured M -gradient documented in [Section 5](#). The Online Appendix shows that this qualitative pattern is robust to alternative elasticity calibrations and compares the M -guided rules with same-fiscal-volume reverse- M and income-only progressive rules. The placebo exercises show that the sign and magnitude of the TFP effects are driven by M -targeting rather than by fiscal volume or progressivity alone.

6.3 Transfers with Historical Shocks

I next ask whether the same M -guided transfers remain productivity-enhancing when they interact with the historical sequence of shocks. For each year t , I compare two general equilibrium paths. The baseline uses the observed productivity and markup shocks, $\{d \log A_t, d \log \mu_t\}$, with no transfers. The counterfactual uses the same shocks but adds the budget-neutral transfers defined above. I estimate $\Delta d \log TFP_t \equiv d \log TFP_t(\tau) - d \log TFP_t(0)$, so the comparison isolates the incremental effect of redistribution holding the historical shock sequence fixed.

Figure 8: TFP Growth Effects of M -Guided Transfers with Historical Shocks

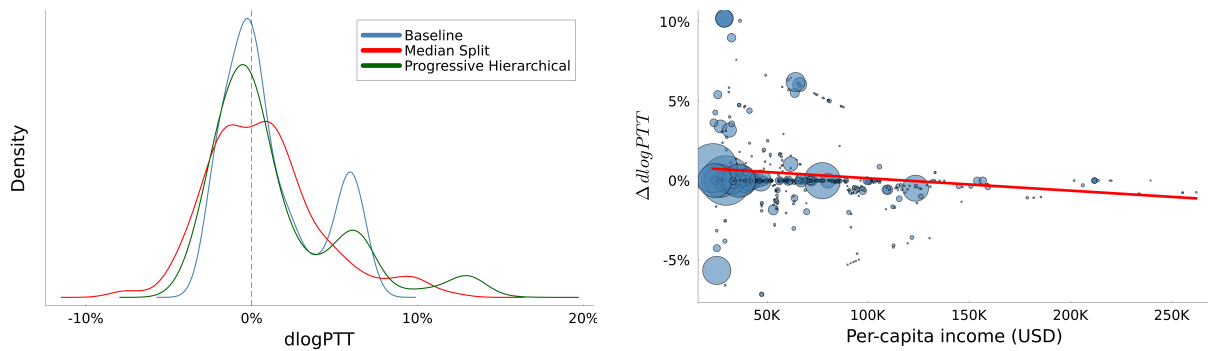


Note: $\Delta d \log TFP_t = d \log TFP_t(\tau) - d \log TFP_t(0)$ is the annual difference between the transfer counterfactual and the no-transfer baseline. The baseline and counterfactual use the same estimated sectoral productivity and inverse-markup shock sequences, $\{d \log A_t, d \log \mu_t\}$. Transfers are budget neutral and employment weighted. A \$1,000 transfer to occupation o in year t equals $1000 N_{o,t}$ dollars in the aggregate, expressed as a share of GDP_t .

[Figure 8](#) shows that the transfer effect is positive in 21 of 24 years under both rules. The magnitudes are modest, as expected given the small fiscal scale, but the sign is informative: the measured M -ranking typically remains productivity-enhancing even after prices, sales, factor payments, and consumption shares adjust to historical shocks. [Figure 9](#) shows that the

incidence effects are larger than the aggregate TFP effects. The left panel reports occupation-level $d \log PTT$ distributions in 2019. Both transfer rules move part of the distribution to the right relative to the no-transfer baseline, but the median-split rule also thickens the left tail, indicating larger losses for some occupations. The progressive rule produces a cleaner rightward shift and places more mass in the positive-PTT region. The right panel focuses on the progressive rule and plots the occupation-level PTT gain relative to the baseline against mean earnings. The employment-weighted fitted line is negative and statistically significant at the 1 percent level, implying that lower-earning occupations receive larger real-consumption gains from the progressive M -guided transfer.

Figure 9: Occupation-Level PTT Incidence under Historical Shocks, 2019



Note: The left panel plots the employment-weighted distribution of occupation-level $d \log PTT$ in 2019 under three scenarios: no transfers, median-split transfers, and progressive hierarchical transfers. Densities are trimmed at the 2.5 and 97.5 percent employment-weighted quantiles. The dashed line marks zero. The right panel plots $\Delta d \log PTT_0 = d \log PTT_0^{Prog} - d \log PTT_0^{Baseline}$ against occupation mean earnings in 2019. Each point is an occupation and marker size is proportional to employment. The fitted line is an employment-weighted OLS regression.

Taken together, the counterfactuals show that the Okun tradeoff is conditional, not mechanical. Budget-neutral expenditure reallocations guided by the measured M -hierarchies raise TFP in most years, even when combined with historical productivity and markup shocks. The aggregate effects are small, but the incidence effects are economically meaningful: progressive M -guided transfers improve occupation-level PTTs more for lower-earning occupations. The result is not that redistribution always raises efficiency. It is that, in a distorted production network, the efficiency cost or gain from redistribution depends on where the induced expenditure moves in the network.

7 Conclusion

This paper shows that the income-expenditure distribution is a first-order state variable for productivity in distorted production networks. When wedges break Hulten neutrality, reallocations of factor income, household expenditure, and firm demand affect TFP through allocative

efficiency. The theory derives sufficient statistics for these margins: distortion centrality δ ranks factor-income reallocations, expenditure centrality M ranks household-expenditure reallocations, and revenue centrality F ranks firm-demand reallocations. The household-level PTT decomposition extends the same logic to real-consumption incidence and links incidence back to aggregate efficiency through $d \log TFP = \sum_h \chi_h d \log PTT_h$.

The U.S. implementation shows that these margins are quantitatively relevant. Over 1997–2021, distributional reallocation accounts for roughly 17–20% of TFP-growth variation, and the measured M -rankings identify redistribution margins with nontrivial efficiency implications. Counterfactual budget-neutral transfers guided by M generate modest TFP gains and improve real-consumption incidence for lower-earning occupations. The central message is therefore not that redistribution always raises productivity, but that the equity-efficiency tradeoff is conditional: in a distorted network economy, the aggregate effect of redistribution depends on where expenditure moves.

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Online Appendix to Inequality and Misallocation under Production Networks

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A Proofs for the nonparametric model

A.1 Firms

Firm $i \in \mathcal{N}$ chooses $\{\ell_{ih}\}_{h \in \mathcal{H}}$ and $\{x_{ij}\}_{j \in \mathcal{N}}$ to minimize $\bar{c}_i = p_i^\ell L_i + p_i^x X_i = \sum_{h \in \mathcal{H}} w_h \ell_{ih} + \sum_{j \in \mathcal{N}} p_j x_{ij}$ subject to $1 = A_i Q_i(L_i, X_i)$, $L_i = A_i^\ell Q_i^\ell(\{A_{ih}^\ell \ell_{ih}\}_{h \in \mathcal{H}})$, $X_i = A_i^x Q_i^x(\{A_{ij}^x x_{ij}\}_{j \in \mathcal{N}})$, $mc_i = \mu_i p_i$, taking prices $\{w_h\}_{h \in \mathcal{H}}$ and $\{p_j\}_{j \in \mathcal{N}}$ as given. The optimality conditions for L_i and X_i are: $p_i^\ell = \kappa_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i}$ and $p_i^x = \kappa_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i}$, where κ_i is the lagrange multiplier. Multiplying both conditions by L_i and X_i and summing, we obtain: $\mu_i p_i = p_i^\ell L_i + p_i^x X_i = \kappa_i A_i Q_i(L_i, X_i) = \kappa_i$. The first-order conditions are given by

$$p_i^\ell = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i} \quad \text{if} \quad \frac{\partial Q_i(L_i, X_i)}{\partial L_i} > 0, \quad (1)$$

$$p_i^x = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i} \quad \text{if} \quad \frac{\partial Q_i(L_i, X_i)}{\partial X_i} > 0, \quad (2)$$

$$w_h = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial L_i} A_i^\ell \frac{\partial Q_i^\ell(\{A_{ib}^\ell \ell_{ib}\}_{b \in \mathcal{H}})}{\partial \ell_{ih}} \quad \text{if} \quad \frac{\partial y_i}{\partial \ell_{ih}} > 0, \quad (3)$$

$$p_j = \mu_i p_i A_i \frac{\partial Q_i(L_i, X_i)}{\partial X_i} A_i^x \frac{\partial Q_i^x(\{A_{im}^x x_{im}\}_{m \in \mathcal{N}})}{\partial x_{ij}} \quad \text{if} \quad \frac{\partial y_i}{\partial x_{ij}} > 0. \quad (4)$$

Let $e(a, b) \equiv (\partial a / \partial b)(b/a)$. Then: $\omega_i^\ell = e(y_i, L_i) = \frac{1}{\mu_i} \frac{p_i^\ell L_i}{p_i^x X_i}$, $\omega_i^x = e(y_i, X_i) = \frac{1}{\mu_i} \frac{p_i^x X_i}{p_i^\ell L_i}$, $\alpha_{ih} = e(L_i, \ell_{ih}) = \frac{w_h \ell_{ih}}{p_i^\ell L_i}$, $\mathcal{W}_{ij} = e(X_i, x_{ij}) = \frac{p_j x_{ij}}{p_i^x X_i}$, $\tilde{\Omega}_{ih}^\ell = e(y_i, \ell_{ih})$, and $\tilde{\Omega}_{ij}^x = e(y_i, x_{ij})$.

A.2 Households

Household $h \in \mathcal{H}$ chooses $\{C_{hi}\}_{i \in \mathcal{N}}$, L_h to maximize $U_h(C_h, L_h)$ subject to $C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}})$ and the budget constraint $E_h = p_h^c C_h = \sum_{i \in \mathcal{N}} p_i C_{hi} \leq w_h L_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \pi_i + T_h$. The household takes as given $\{w_h, T_h, \{p_i, \kappa_{ih}, \pi_i\}_{i \in \mathcal{N}}\}$.

The first-order conditions for household $h \in \mathcal{H}$ are given by

$$\frac{w_h}{p_h^c} = -\frac{U_{L_h}}{U_{C_h}}, \quad (5)$$

$$\frac{p_i}{p_h^c} = \frac{\partial C_h}{\partial C_{hi}} \quad \text{if} \quad \frac{\partial C_h}{\partial C_{hi}} > 0. \quad (6)$$

It follows that $\beta_{hi} = e(C_h, C_{hi}) = \frac{p_i C_{hi}}{p_h^c C_h}$. Substitutabilities are captured by $e(C_h, L_h) = \frac{w_h L_h}{p_h^c C_h}$, and cross-elasticities by $e(C_{hi}, C_{hm}) + \frac{p_m C_{hm}}{p_i C_{hi}} = 0$.

A.3 General Equilibrium Conditions

Proposition 2. The allocation and price system are an equilibrium iff markets clear, the fiscal budget is balanced, and the following conditions are jointly satisfied:

$$\frac{\partial C_h}{\partial C_{hj}} \bigg/ \frac{\partial C_h}{\partial C_{hi}} = \mu_i \frac{\partial y_i}{\partial x_{ij}} \quad \forall i, j \in \mathcal{N} : C_{hi} > 0, C_{hj} > 0, x_{ij} > 0, \quad (7)$$

$$-\frac{w_b}{w_h} \frac{U_{L_h}}{U_{C_{hi}}} = \mu_i \frac{\partial y_i}{\partial \ell_{ib}} \quad \forall i \in \mathcal{N}, \forall h, b \in \mathcal{H} : C_{hi} > 0, \ell_{ib} > 0, U_{L_h} < 0. \quad (8)$$

Necessity. First, using (4) and (6), derive (7) from [Proposition 2](#)

$$\frac{\partial C_h}{\partial C_{hj}} \bigg/ \frac{\partial C_h}{\partial C_{hi}} = \frac{p_j}{p_i} = \mu_i \frac{\partial y_i}{\partial x_{ij}} \quad \forall i, j \in \mathcal{N}, \forall h \in \mathcal{H} : C_{hi} > 0, C_{hj} > 0, x_{ij} > 0.$$

This condition states that, for any firm i and any household h , the household's MRS between goods i and j must match the cost ratio implied by firm i 's optimal sourcing of inputs. Notice that the first equality requires h to consume from both i and j , while the second requires only that i demands inputs from j .

Second, combining (3) and (5) yields (8)

$$-\frac{w_b}{w_h} \frac{U_{L_h}}{U_{C_{hi}}} = \frac{w_b}{p_i} = \mu_i \frac{\partial y_i}{\partial \ell_{ib}} \quad \forall i \in \mathcal{N}, \forall h, b \in \mathcal{H} : C_{hi} > 0, U_{L_h} \neq 0, \text{ and } \ell_{ib} > 0.$$

This second subset of equilibrium conditions links firm i 's demand for factor b to household h 's MRS between factor supply and consumption of goods from firm i . It is sufficient that firm i demands factor b and that household h consumes from firm i . When $b \neq h$, the wedge w_b/w_h reflects distributional differences in factor prices across household types.

Finally, $y_i = \sum_{h \in \mathcal{H}} C_{hi} + \sum_{j \in \mathcal{N}} x_{ji} \quad \forall i \in \mathcal{N}$, $L_h = \sum_{i \in \mathcal{N}} \ell_{ih} \quad \forall h \in \mathcal{H}$, and $\sum_{h \in \mathcal{H}} T_h = 0$.

Sufficiency. For any set $\{\mu_i, \{\kappa_{ih}\}_{h \in \mathcal{H}}\}_{i \in \mathcal{N}}$ and $\{T_h\}_{h \in \mathcal{H}}$, there exists a strictly positive system $\{p_i\}_{i \in \mathcal{N}}, \{w_h\}_{h \in \mathcal{H}}$ implementing $\{y_i, \{\ell_{ih}\}_{h \in \mathcal{H}}, \{x_{ij}\}_{j \in \mathcal{N}}\}_{i \in \mathcal{N}}$ and $\{C_h, \{C_{hi}\}_{i \in \mathcal{N}}, L_h\}_{h \in \mathcal{H}}$ as a competitive equilibrium.

Assume, without loss of generality, prices are normalized such that a CRS function defines the GDP deflator $\bar{p}_Y = Q^p(\{p_i\}_{i \in \mathcal{N}}) = 1$. Also assume there exists at least one “essential” factor of production $h_* \in \mathcal{H}$ – meaning all firms are directly or indirectly exposed to its value-added. Using equations (3) and (4), the price for firm i is given by

$$p_i = \frac{w_{h_*}}{\mu_i} \left(\frac{\partial y_i}{\partial \ell_{ih_*}} \right)^{-1} \quad \text{if } \ell_{ih_*} > 0 \quad \text{otherwise} \quad (9)$$

$$p_i = \frac{w_{h_*}}{\mu_i} \left(\frac{\partial y_i}{\partial x_{ij}} \right)^{-1} \left(\frac{\partial y_{\bar{j}}}{\partial \ell_{\bar{j}h_*}} \right)^{-1} \prod_{j \in \mathcal{N}_{z_i}} \frac{1}{\mu_j} \prod_{j \in \mathcal{N}_i \setminus \{\bar{j}\}} \left(\frac{\partial y_j}{\partial x_{jj+1}} \right)^{-1}$$

where $\mathcal{N}_{z_i} = \{\bar{j}, \bar{j}+1, \dots, \bar{j}-1, \bar{j}\}$ is an intermediate demand chain connecting i to h_* .

From the price normalization, the price for factor h_* is given by

$$w_{h_*}^{-1} = Q^p \left(\left\{ \frac{1}{\mu_i} \left(\mathbb{1} \{ \ell_{ih_*} > 0 \} \left(\frac{\partial y_i}{\partial \ell_{ih_*}} \right)^{-1} + \mathbb{1} \{ \ell_{ih_*} = 0 \} \left(\frac{\partial y_i}{\partial x_{ij}} \right)^{-1} \left(\frac{\partial y_{\bar{j}}}{\partial \ell_{\bar{j}h_*}} \right)^{-1} \prod_{j \in \mathcal{N}_{z_i}} \frac{1}{\mu_j} \prod_{j \in \mathcal{N}_i \setminus \{\bar{j}\}} \left(\frac{\partial y_j}{\partial x_{jj+1}} \right)^{-1} \right) \right\}_{i \in \mathcal{N}} \right). \quad (10)$$

Normalizing with $w_{h_*} = 1$ and equation (9) implies that all firm prices are strictly positive, since marginal productivities of both factors and intermediate inputs are strictly positive. It then follows from equation (3) that all factor wages are strictly positive as well.

It remains to show that, under the system of prices defined by (9) and (10), the optimality conditions hold given the equilibrium conditions in (7) and (8), the market clearing conditions, and the balanced budget constraint. In particular, I must verify that the allocation satisfies the first-order conditions for both producers and consumers, thereby establishing that the allocation and prices together constitute a competitive equilibrium.

To derive (5) and (6), assume that firm i directly or indirectly demands factor h , and firm j demands factor b . This assumption is made without loss of generality, as it holds for any combination of $i, j \in \mathcal{N}$ and $h, b \in \mathcal{H}$. Substituting (7) and (8) into (9) yields $p_i = -w_b \frac{U_{C_{bi}}}{U_{L_b}}$ and $p_j = -w_b \frac{U_{C_{bj}}}{U_{L_b}}$. From this, I obtain that $\frac{U_{C_{bj}}}{U_{C_{bi}}} = \frac{p_j}{p_i}$. Multiplying by C_{hi} and adding overall $j \in \mathcal{N}$ yields $\frac{\partial C_h}{\partial C_{hi}} \sum_{j \in \mathcal{N}} p_j C_{hj} = p_i \sum_{j \in \mathcal{N}} C_{hi} \frac{\partial C_h}{\partial C_{hj}} \rightarrow \frac{\partial C_h}{\partial C_{bi}} p_h^c C_h = p_i C_h \rightarrow \frac{p_i}{p_h^c} = \frac{\partial C_h}{\partial C_{hi}}$, which is (6). Moreover, from (8), I recover $\frac{w_h}{p_h^c} = -\frac{U_{L_h}}{U_{C_h}}$, which corresponds to (5).

For firms, begin by noting that $\frac{U_{C_{bj}}}{U_{C_{bi}}} = \frac{p_j}{p_i}$. Substituting into (7) yields $\frac{p_j}{p_i} = \mu_i \frac{\partial y_i}{\partial x_{ij}}$, which corresponds to (4). To derive (2), sum (4) across all intermediate suppliers $\sum_{j \in \mathcal{N}} p_j x_{ij} = \mu_i p_i \sum_{j \in \mathcal{N}} \frac{\partial y_i}{\partial x_{ij}} x_{ij} \rightarrow p_i^x X_i = \mu_i p_i \frac{\partial y_i}{\partial X_i} X_i \rightarrow p_i^x = \mu_i p_i \frac{\partial y_i}{\partial X_i}$. Now, take $p_i = -w_h \frac{U_{C_{hi}}}{U_{L_h}}$ and use equation (8) to derive $\frac{w_h}{p_i} = \mu_i \frac{\partial y_i}{\partial \ell_{ih}}$, which corresponds to equation (3). To derive equation (1), sum (3) across all households $\sum_{h \in \mathcal{H}} w_h \ell_{ih} = \mu_i p_i \sum_{h \in \mathcal{H}} \frac{\partial y_i}{\partial \ell_{ih}} x_{ij} \rightarrow p_i^\ell L_i = \mu_i p_i \frac{\partial y_i}{\partial L_i} i \rightarrow p_i^x = \mu_i p_i \frac{\partial y_i}{\partial L_i}$.

What remains to be verified is that the budget constraints hold. Summing across households $\sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} p_i C_{hi} = \sum_{h \in \mathcal{H}} (w_h L_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \pi_i + T_h) \rightarrow \sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} p_i C_{hi} = \sum_{h \in \mathcal{H}} w_h L_h + \sum_{i \in \mathcal{N}} \pi_i \sum_{h \in \mathcal{H}} \kappa_{ih} + \sum_{h \in \mathcal{H}} T_h \rightarrow 0 = \sum_{i \in \mathcal{N}} p_i (y_i - \sum_{h \in \mathcal{H}} C_{hi} - \sum_{j \in \mathcal{N}} x_{ji}) + \sum_{h \in \mathcal{H}} w_h (L_h - \sum_{i \in \mathcal{N}} \ell_{ih})$.

A.4 Equilibrium Centralities

Goods market. Substituting (2), (4), and (6) into market clearing for firm $i \in \mathcal{N}$: $S_i = \sum_{h \in \mathcal{H}} p_i C_{hi} + \sum_{j \in \mathcal{N}} p_i x_{ji} = \sum_{h \in \mathcal{H}} \beta_{hi} E_h + \sum_{j \in \mathcal{N}} \Omega_{ji}^x S_j$, where $\Omega_{ji}^x \equiv \mu_i \omega_i^x \mathcal{W}_{ij}$. Then $(I_N - \tilde{\Omega}'_x \text{diag}(\mu))S = \beta' E$ or $S = \mathcal{B}' E$, where $S \equiv [S_1, \dots, S_N]'$, $E \equiv [E_1, \dots, E_H]'$, and $\mu \equiv [\mu_1, \dots, \mu_N]'$. Dividing by nominal GDP yields $\lambda = \mathcal{B}' \chi$.

Define the matrix $\tilde{\mathcal{B}} \equiv \beta \tilde{\Psi}_x \equiv \beta \Psi_x (I_N - \Omega_x) \tilde{\Psi}_x \equiv \mathcal{B} (I_N - \Omega_x) \tilde{\Psi}_x$, where $\tilde{\Psi}_x \equiv (I_N - \tilde{\Omega}_x)^{-1}$. Then $\lambda = \Psi'_x (I_N - \tilde{\Omega}'_x) \tilde{\mathcal{B}}' \chi$, which motivates the definition of the cost-based Domar weights $\tilde{\lambda} \equiv \tilde{\Psi}'_x (I_N - \Omega'_x) \lambda \equiv \tilde{\mathcal{B}}' \chi$. To understand them, observe that $\tilde{S}_i \equiv \sum_{h \in \mathcal{H}} p_i C_{hi} + \sum_{j \in \mathcal{N}} \tilde{\Omega}_{ji}^x \tilde{S}_j = \sum_{h \in \mathcal{H}} \tilde{\mathcal{B}}_{hi} E_h$, where $\tilde{S}_i = \tilde{\lambda}_i \text{GDP}$. In equilibrium, $\tilde{\Omega}_{ji}^x$ captures the cost share in j allocated to intermediates from i . As a result, $\tilde{S}_i$ is the total value-added that passes through firm i . For a given expenditure distribution χ , $\tilde{\lambda}_i$ represents the share of aggregate value-added that flows through i . Moreover, the identity $\omega'_\ell \tilde{\lambda} = \mathbb{1}'_N (I_N - \tilde{\Omega}'_x) \tilde{\Psi}'_x \beta' \chi = 1$ implies that $\omega'_\ell \tilde{\lambda}_i$ corresponds to the share of aggregate value-added generated by factors directly employed in i .

Finally, I need to proof that $\tilde{\lambda}_i \geq \lambda_i$ holds $\forall i \in \mathcal{N}$. Begin with the identity $\tilde{\Psi}_x - \Psi_x = \tilde{\Psi}_x - \Psi_x = \sum_{q=1}^{\infty} (\tilde{\Omega}_x^q - \Omega_x^q)$. Note that $\tilde{\Omega}_x - \Omega_x = (I_N - \text{diag}(\mu))\tilde{\Omega}_x \succcurlyeq 0_N 0'_N$, since $\mu_i \in (0, 1]$ and $\tilde{\Omega}_x \succcurlyeq 0_N 0'_N$. Proceeding by induction, suppose that $\tilde{\Omega}_x^{q-1} - \Omega_x^{q-1} \succcurlyeq 0_N 0'_N$ for some $q > 1$. Then $\tilde{\Omega}_x^q - \Omega_x^q = (\tilde{\Omega}_x^{q-1} - \Omega_x^{q-1} \text{diag}(\mu))\tilde{\Omega}_x = (\tilde{\Omega}_x^{q-1} - \Omega_x^{q-1} + \Omega_x^{q-1}(I_N - \text{diag}(\mu)))\tilde{\Omega}_x \succcurlyeq 0_N 0'_N$. Thus, $\tilde{\Psi}_x \succcurlyeq \Psi_x$. It follows that $\tilde{\mathcal{B}} - \mathcal{B} = \beta(\tilde{\Psi}_x - \Psi_x) \succcurlyeq 0_H 0'_N$, and therefore $\tilde{\lambda} - \lambda = (\tilde{\mathcal{B}} - \mathcal{B})'\chi \succcurlyeq 0_N$.

Factor markets. Substituting (1) and (3) into the market clearing condition for factor $h \in \mathcal{H}$: $J_h = \sum_{i \in \mathcal{N}} w_h \ell_{ih} = \sum_{i \in \mathcal{N}} \mu_i \omega_i^\ell \alpha_{ih} S_i$, where $\tilde{\Omega}_{ih}^\ell \equiv \omega_i^\ell \alpha_{ih}$. In matrix form $J = \tilde{\Omega}_\ell' \text{diag}(\mu) S = \Omega_\ell' S$, where $J \equiv [J_1, \dots, J_H]'$. Dividing by nominal GDP yields $\Lambda = \Omega_\ell' \lambda$. Similarly, the cost-based factor weights is given by $\tilde{\Lambda} \equiv \tilde{\Omega}_\ell' \tilde{\lambda}$, where $\mathbf{1}_H' \tilde{\Lambda} = \mathbf{1}_H' \alpha' \text{diag}(\omega_\ell) \tilde{\lambda} = \omega_\ell' \tilde{\lambda} = 1$.

Notice that $\tilde{\Lambda} \succcurlyeq \Lambda$ because $\tilde{\Lambda} - \Lambda = \tilde{\Omega}_\ell' \tilde{\lambda} - \Omega_\ell' \lambda = \underbrace{\tilde{\Omega}_\ell'}_{\succcurlyeq 0_H 0'_N} \underbrace{(\tilde{\lambda} - \lambda)}_{\succcurlyeq 0_N} + \tilde{\Omega}_\ell' \underbrace{(I_N - \text{diag}(\mu)) \lambda}_{\succcurlyeq 0_N 0'_N}$.

The firm-to-factor and factor-to-firm centrality matrices are respectively given by $\Psi_\ell = \Psi_x \Omega_\ell$ and $\tilde{\Psi}_\ell = \tilde{\Psi}_x \tilde{\Omega}_\ell$, where $\tilde{\Psi}_\ell \mathbf{1}_H = \tilde{\Psi}_x \tilde{\Omega}_\ell \mathbf{1}_H = \tilde{\Psi}_x \omega_\ell = \tilde{\Psi}_x (I_N - \tilde{\Omega}_x) \mathbf{1}_N = \mathbf{1}_N$. Additionally $\tilde{\Psi}_\ell \succcurlyeq \Psi_\ell$ because $\tilde{\Psi}_\ell - \Psi_\ell = \underbrace{(\tilde{\Psi}_x - \Psi_x)}_{\succcurlyeq 0_N 0'_N} \underbrace{\tilde{\Omega}_\ell}_{\succcurlyeq 0_N 0'_H} + \underbrace{\Psi_x}_{\succcurlyeq 0_N 0'_N} \underbrace{(I_N - \text{diag}(\mu)) \tilde{\Omega}_\ell}_{\succcurlyeq 0_N 0'_N}$.

Similarly, the consumer-to-factor and factor-to-consumer centrality matrices are respectively given by $\mathcal{C} = \mathcal{B} \Omega_\ell$ and $\tilde{\mathcal{C}} = \tilde{\mathcal{B}} \tilde{\Omega}_\ell$, where $\tilde{\mathcal{C}} \mathbf{1}_H = \tilde{\mathcal{B}} \tilde{\Omega}_\ell \mathbf{1}_H = \beta \tilde{\Psi}_x \omega_\ell = \beta \tilde{\Psi}_x (I_N - \tilde{\Omega}_x) \mathbf{1}_N = \mathbf{1}_H$, $\tilde{\mathcal{C}}' \chi = \tilde{\Omega}_\ell' \tilde{\mathcal{B}}' \chi = \tilde{\Omega}_\ell' \tilde{\lambda} = \tilde{\Lambda}$, $\mathcal{C}' \chi = \Omega_\ell' \mathcal{B}' \chi = \Omega_\ell' \lambda = \Lambda$, and $\tilde{\mathcal{C}} \succcurlyeq \mathcal{C}$ because $\tilde{\mathcal{C}} - \mathcal{C} = \underbrace{(\tilde{\mathcal{B}} - \mathcal{B})}_{\succcurlyeq 0_H 0'_N} \underbrace{\tilde{\Omega}_\ell}_{\succcurlyeq 0_N 0'_H} + \underbrace{\mathcal{B}}_{\succcurlyeq 0_H 0'_N} \underbrace{(I_N - \text{diag}(\mu)) \tilde{\Omega}_\ell}_{\succcurlyeq 0_N 0'_N}$.

Household budget equilibrium conditions. Using $mc_i = \mu_i p_i$, firm costs are given $\bar{c}_i = \mu_i S_i$, and profits are $\pi_i = (1 - \mu_i) S_i$. Applying (3), $E_h = \sum_{i \in \mathcal{N}} (\mu_i \omega_i^\ell \alpha_{ih} + \kappa_{ih} (1 - \mu_i)) p_i y_i + T_h = \sum_{i \in \mathcal{N}} (\mu_i \tilde{\Omega}_{ih}^\ell + \kappa_{ih} (1 - \mu_i)) S_i + T_h$. In matrix form, $E = (\Omega_\ell + \Omega_\pi)' S + T$, where $T = [T_1, \dots, T_H]'$. Dividing by nominal GDP $\chi = (\Omega_\ell + \Omega_\pi)' \lambda + \tau$. It follows that $\mathbf{1}_H' (\Omega_\ell + \Omega_\pi)' \lambda = \mathbf{1}_H' \Lambda + \mathbf{1}_N' \text{diag}(\mathbf{1}_N - \mu) \lambda = \sum_{h \in \mathcal{H}} \Lambda_h + \sum_{i \in \mathcal{N}} (1 - \mu_i) \lambda_i = 1$.

Nominal GDP. Aggregating across all firms $\sum_{i \in \mathcal{N}} S_i = \sum_{i \in \mathcal{N}} (\sum_{h \in \mathcal{H}} p_i C_{hi} + \sum_{j \in \mathcal{N}} p_i x_{ij})$. Then $GDP \equiv \sum_{h \in \mathcal{H}} E_h = \sum_{i \in \mathcal{N}} (S_i - \sum_{j \in \mathcal{N}} p_j x_{ij})$. Using (2) and (4) $GDP = \sum_{i \in \mathcal{N}} (1 - \mu_i \sum_{j \in \mathcal{N}} \tilde{\Omega}_{ij}^x) S_i = \sum_{i \in \mathcal{N}} (1 - \omega_i^x \mu_i) S_i$.

A.5 Proof for Proposition 1

Prices are given by $p_i^\ell = \frac{\sum_{h \in \mathcal{H}} w_h \ell_{ih}}{A_i^\ell Q_i^\ell(\{A_{ih}^\ell \ell_{ih}\}_{h \in \mathcal{H}})}$, $p_i^x = \frac{\sum_{j \in \mathcal{N}} p_j x_{ij}}{A_i^x Q_i^x(\{A_{ij}^x x_{ij}\}_{j \in \mathcal{N}})}$, $p_i = \frac{(p_i^\ell L_i + p_i^x X_i)}{\mu_i A_i Q_i(L_i, X_i)}$, and $p_h^c = \frac{\sum_{i \in \mathcal{N}} p_i C_{hi}}{Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}})}$. The first-order approximation $\hat{p}_i^\ell = \frac{A_i^\ell}{p_i^\ell} \frac{\partial p_i^\ell}{\partial A_i^\ell} \hat{A}_i^\ell + \sum_{h \in \mathcal{H}} \left(\frac{w_h}{p_i^\ell} \frac{\partial p_i^\ell}{\partial w_h} \hat{w}_h + \frac{A_{ih}^\ell}{p_i^\ell} \frac{\partial p_i^\ell}{\partial A_{ih}^\ell} \hat{A}_{ih}^\ell + \frac{\ell_{ih}}{p_i^\ell} \frac{\partial p_i^\ell}{\partial \ell_{ih}} \hat{\ell}_{ih} \right)$, where the log deviation of any variable x around its steady-state value is $\hat{x} = \log(x/\bar{x})$. Using $\frac{A_i^\ell}{p_i^\ell} \frac{\partial p_i^\ell}{\partial A_i^\ell} = -1$, $\frac{w_h}{p_i^\ell} \frac{\partial p_i^\ell}{\partial w_h} = \alpha_{ih}$, $\frac{A_{ih}^\ell}{p_i^\ell} \frac{\partial p_i^\ell}{\partial A_{ih}^\ell} = -\alpha_{ih}$, and $\frac{\ell_{ih}}{p_i^\ell} \frac{\partial p_i^\ell}{\partial \ell_{ih}} = \alpha_{ih} - e(L_i, \ell_{ih}) = 0$,

we have that $\hat{p}_i^\ell = -\hat{A}_i^\ell + \sum_{h \in \mathcal{H}} \alpha_{ih}(\hat{w}_h - \hat{A}_{ih}^\ell)$, $\hat{p}_i^x = -\hat{A}_i^x + \sum_{j \in \mathcal{N}} \mathcal{W}_{ij}(\hat{p}_j - \hat{A}_{ij}^x)$, $\hat{p}_i = \omega_i^\ell \hat{p}_i^\ell + \omega_i^x \hat{p}_i^x - \hat{A}_i - \hat{\mu}_i$, and $\hat{p}_h^c = \sum_{i \in \mathcal{N}} \beta_{hi} \hat{p}_i$. In matrix form, these relationships are written as $\hat{p}_\ell = \alpha \hat{w} - \hat{A}_\ell - (\alpha \circ \hat{\underline{A}}_\ell) \mathbf{1}_H$, $\hat{p}_x = \mathcal{W} \hat{p} - \hat{A}_x - (\mathcal{W} \circ \hat{\underline{A}}_x) \mathbf{1}_N$, $\hat{p} = \text{diag}(\omega_\ell) \hat{p}_\ell + \text{diag}(\omega_x) \hat{p}_x - \hat{A} - \hat{\mu}$, $\hat{p} = \text{diag}(\omega_\ell) \hat{p}_\ell + \text{diag}(\omega_x) \hat{p}_x - \hat{A} - \hat{\mu}$, and $\hat{p}_c = \beta \hat{p}$. Then $\hat{p} = \tilde{\Psi}_x(\tilde{\Omega}_\ell \hat{w} - \hat{A} - \hat{\mu})$ and $\hat{p}_c = \tilde{\mathcal{B}}(\tilde{\Omega}_\ell \hat{w} - \hat{A} - \hat{\mu})$. The matrices previously used are: $\hat{\mathcal{A}} \equiv \hat{A} + \text{diag}(\omega_\ell) \hat{A}_\ell + (\tilde{\Omega}_\ell \circ \hat{\underline{A}}_\ell) \mathbf{1}_H + \text{diag}(\omega_x) \hat{A}_x + (\tilde{\Omega}_x \circ \hat{\underline{A}}_x) \mathbf{1}_N$, $\hat{A} \equiv [\hat{A}_1, \dots, \hat{A}_N]'$, $\hat{A}_\ell \equiv [\hat{A}_1^\ell, \dots, \hat{A}_N^\ell]'$, $\hat{A}_x \equiv [\hat{A}_1^x, \dots, \hat{A}_N^x]'$, $\hat{\underline{A}}_\ell = [\hat{\underline{A}}_1^\ell, \dots, \hat{\underline{A}}_N^\ell]'$, $\hat{\underline{A}}_x = [\hat{\underline{A}}_1^x, \dots, \hat{\underline{A}}_N^x]'$, $\hat{\underline{A}}_i = [\hat{A}_{i1}^\ell, \dots, \hat{A}_{iH}^\ell]'$, $\hat{\underline{A}}_i = [\hat{A}_{i1}^x, \dots, \hat{A}_{iN}^x]'$, $\hat{p} \equiv [\hat{p}_1, \dots, \hat{p}_N]'$, $\hat{p}_\ell \equiv [\hat{p}_1^\ell, \dots, \hat{p}_N^\ell]'$, $\hat{p}_x \equiv [\hat{p}_1^x, \dots, \hat{p}_N^x]'$, and $\hat{w} \equiv [\hat{w}_1, \dots, \hat{w}_H]'$.

A.6 Proof for Theorem 1

Revenue-based Λ . From $\Lambda = \Omega'_\ell \lambda$, $\Lambda_h = \mathbf{1}'_N \text{diag}(\tilde{\Omega}_\ell o_H(h)) (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)^{-1} \beta' \chi$.

$$\begin{aligned} d \Lambda_h &= \mathbf{1}'_N \text{diag}(\Omega_\ell o_H(h)) \left(\mathcal{B}' \begin{pmatrix} d \chi_1 \\ \vdots \\ d \chi_H \end{pmatrix} + \Psi'_x \begin{pmatrix} \sum_{b \in \mathcal{H}} \chi_b d \beta_{b1} \\ \vdots \\ \sum_{b \in \mathcal{H}} \chi_b d \beta_{bN} \end{pmatrix} \right) + \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \lambda_i (\hat{w}_i^\ell + \hat{\alpha}_{ih}) \\ &\quad + \mathbf{1}'_N \text{diag}(\tilde{\Omega}_\ell o_H(h)) \left(\frac{d (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)^{-1}}{d \log \tilde{\Omega}_x} + \frac{d (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)^{-1}}{d \log \mu} \right) \beta' \chi. \end{aligned}$$

Using $\lambda = \mathcal{B}' \chi$, $\Lambda = \Omega'_\ell \lambda$, and $\frac{d A^{-1}}{d x} = -A^{-1} \frac{d A}{d x} A^{-1}$ for any invertible A , then

$$\begin{aligned} d \Lambda_h &= \sum_{b \in \mathcal{H}} \mathcal{C}_{bh} d \chi_b + \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \lambda_i (\hat{w}_i^\ell + \hat{\alpha}_{ih}) + \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \sum_{j \in \mathcal{N}} \psi_{ji}^x \sum_{b \in \mathcal{H}} \chi_b d \beta_{bj} \\ &\quad - \mathbf{1}'_N \text{diag}(\Omega_\ell o_H(h)) \Psi'_x \left(\frac{d (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)}{d \log \tilde{\Omega}_x} + \frac{d (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)}{d \log \mu} \right) \text{diag}(\mu) \lambda \\ &= \sum_{b \in \mathcal{H}} \mathcal{C}_{bh} d \chi_b + \sum_{b \in \mathcal{H}} \mathcal{C}_{bh} d \chi_b = \sum_{b \in \mathcal{H}} \mathcal{C}_{bh} d \chi_b + \sum_{i \in \mathcal{N}} \psi_{ih}^\ell \lambda_i d \log \mu_i \\ &\quad + \sum_{j \in \mathcal{N}} \psi_{jh}^\ell \left(\sum_{b \in \mathcal{H}} \chi_b d \beta_{bj} + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d \tilde{\Omega}_{ij}^x \right) + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d \tilde{\Omega}_{ih}^\ell. \end{aligned}$$

Revenue-based λ . $\lambda_i = o_N(i)' \Psi'_x \beta' \chi = o_N(i)' \text{diag}(\mu)^{-1} (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)^{-1} \beta' \chi$. Following the same steps $d \lambda_i = \sum_{j \in \mathcal{N}} \psi_{ji}^x (\sum_{b \in \mathcal{H}} \chi_b d \beta_{bj} + \sum_{m \in \mathcal{N}} \mu_m \lambda_m d \tilde{\Omega}_{mj}^x) + \sum_{h \in \mathcal{H}} \mathcal{B}_{hi} d \chi_h + \sum_{j \in \mathcal{N}} \psi_{ji}^x \lambda_j d \log \mu_j - \lambda_i d \log \mu_i$.

Cost-based $\tilde{\Lambda}$. $\tilde{\Lambda}_h = \mathbf{1}'_N \text{diag}(\tilde{\Omega}_\ell o_H(h)) (I - \tilde{\Omega}'_x)^{-1} \beta' \chi$. Following the same steps $d \tilde{\Lambda}_h = \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{bh} d \chi_b + \sum_{b \in \mathcal{H}} \chi_b d \tilde{\mathcal{C}}_{bh} = \sum_{j \in \mathcal{N}} \tilde{\psi}_{jh}^\ell (\sum_{b \in \mathcal{H}} \chi_b d \beta_{bj} + \sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \tilde{\Omega}_{ij}^x) + \sum_{b \in \mathcal{H}} \tilde{\mathcal{C}}_{bh} d \chi_b + \sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \tilde{\Omega}_{ih}^\ell$.

Cost-based $\tilde{\lambda}$. $d \tilde{\lambda}_i = \sum_{j \in \mathcal{N}} \tilde{\psi}_{ji}^x (\sum_{b \in \mathcal{H}} \chi_b d \beta_{bj} + \sum_{m \in \mathcal{N}} \tilde{\lambda}_m d \tilde{\Omega}_{mj}^x) + \sum_{h \in \mathcal{H}} \tilde{\mathcal{B}}_{hi} d \chi_h$.

A.7 Proof for Theorem 4

The first order approximation to the household budget constraint is $E_h \hat{E}_h = \Lambda_h(\hat{w}_h + \hat{L}_h) + \sum_{i \in \mathcal{N}} \kappa_{ih}((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{S}_i) - \mu_i \hat{\mu}_i) S_i + T_h \hat{T}_h$. Using $\hat{p}_c = \tilde{\mathcal{B}}(\tilde{\Omega}_\ell \hat{w} - \hat{A} - \hat{\mu})$ yields $\hat{C}_h = \hat{E}_h - \hat{p}_h^c = \frac{\Lambda_h}{\chi_h}(\hat{w}_h + \hat{L}_h) - \tilde{\mathcal{C}}'_h \hat{w} + \tilde{\mathcal{B}}'_h(\hat{A} + \hat{\mu}) + \sum_{i \in \mathcal{N}} \kappa_{ih} \frac{\lambda_i}{\chi_h}((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{S}_i) - \mu_i \hat{\mu}_i) + \frac{T_h}{\chi_h} \hat{T}_h$, where

$\tilde{\mathcal{B}}_h = [\tilde{\mathcal{B}}_{h1}, \dots, \tilde{\mathcal{B}}_{hN}]'$ and $\tilde{\mathcal{C}}_h = [\tilde{\mathcal{C}}_{h1}, \dots, \tilde{\mathcal{C}}_{hH}]'$. Then

$$\hat{C}_h = \tilde{\mathcal{B}}'_h(\hat{A} + \hat{\mu}) + \frac{\Lambda_h}{\chi_h} \hat{J}_h - \tilde{\mathcal{C}}'_h \hat{J} + \sum_{i \in \mathcal{N}} \frac{\kappa_{ih} \lambda_i}{\chi_h} ((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{S}_i) - \mu_i \hat{\mu}_i) + \frac{\tau_h}{\chi_h} \hat{T}_h + \tilde{\mathcal{C}}'_h \hat{L}. \quad (11)$$

Hence $\frac{C_h}{\bar{C}_h} = \bar{\eta}_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(J) \mathcal{D}_h(\Pi) \mathcal{D}_h(T) f_h(\{L_b\}_{b \in \mathcal{H}})$, where $f_h(\{L_b\}_{b \in \mathcal{H}})$ is a CRS function such that $\frac{d \log f_h(\{L_b\}_{b \in \mathcal{H}})}{d \log L_b} = \tilde{\mathcal{C}}_{hb}$, and $\mathcal{D}_h(\mathcal{A}) = \exp\{\tilde{\mathcal{B}}'_h \hat{A}\}$, $\mathcal{D}_h(\mu) = \exp\{\tilde{\mathcal{B}}'_h \hat{\mu}\}$, $\mathcal{D}_h(\Pi) = \exp\{\sum_{i \in \mathcal{N}} \kappa_{ih} \frac{\lambda_i}{\chi_h} ((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{S}_i) - \mu_i \hat{\mu}_i)\}$, $\mathcal{D}_h(J) = \exp\{\frac{\Lambda_h}{\chi_h} \hat{J}_h - \tilde{\mathcal{C}}'_h \hat{J}\}$, $\mathcal{D}_h(T) = \exp\{\frac{\tau_h}{\chi_h} \hat{T}_h\}$, and $\bar{\eta}_h$ is a constant specific to household h . It follows that

$$C_h = \eta_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(J) \mathcal{D}_h(\Pi) \mathcal{D}_h(T) f_h(\{L_b\}_{b \in \mathcal{H}}) = PTT_h f_h(\{L_b\}_{b \in \mathcal{H}}),$$

where $PTT_h = \eta_h \mathcal{D}_h(\mathcal{A}) \mathcal{D}_h(\mu) \mathcal{D}_h(J) \mathcal{D}_h(\Pi) \mathcal{D}_h(T)$ and $\eta_h = \bar{\eta}_h \bar{C}_h$.

Add and subtract $\widehat{GDP}$ to express (11) as

$$\begin{aligned} \hat{C}_h &= \tilde{\mathcal{B}}'_h \hat{A} + (\tilde{\mathcal{B}}_h - \chi_h^{-1} \text{diag}(\lambda) \text{diag}(\mu) \kappa o_H(h))' \hat{\mu} + \Gamma_h \hat{\Lambda}_h - \tilde{\mathcal{C}}'_h \hat{\Lambda} \\ &\quad + \chi_h^{-1} o_H(h)' \kappa' \text{diag}(\lambda) \text{diag}(\mathbb{1}_N - \mu) (\hat{\kappa} o_H(h) + \hat{\lambda}) + \frac{\tau_h}{\chi_h} \hat{T}_h + \tilde{\mathcal{C}}'_h \hat{L}. \end{aligned}$$

The $N + 2$ vector $\mathcal{R}_h$ summarizes income sources for h : $\mathcal{R}'_h = \chi_h^{-1} [\Lambda_h \lambda' \text{diag}(\Omega_\pi o_H(h)) \tau_h] = \chi_h^{-1} [\Lambda_h \kappa_{1h}(1 - \mu_1) \lambda_1 \dots \kappa_{Nh}(1 - \mu_N) \lambda_N \tau_h]$. Relative to household h 's expenditure: the first entry corresponds to the share of factor income, the next N elements to the share of firms' profits, and the last element the share of lump-sum transfers. Since the elements of $\mathcal{R}_h$ sum to one by construction, its first-order approximation is $\chi_h \hat{\chi}_h = \Lambda_h \hat{\Lambda}_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \lambda_i ((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{\lambda}_i) - \mu_i \hat{\mu}_i) + \tau_h \hat{T}_h$. This implies the following first-order approximation for C_h : $\hat{C}_h = \tilde{\mathcal{B}}'_h \hat{A} + \tilde{\mathcal{B}}'_h \hat{\mu} + \hat{\chi}_h - \tilde{\mathcal{C}}'_h \hat{\Lambda} + \tilde{\mathcal{C}}'_h \hat{L}$. Now, using $\delta_{b|h} \equiv \tilde{\mathcal{C}}_{hb} / \tilde{\Lambda}_b$, $M_{q|h} \equiv \sum_{b \in \mathcal{H}} \mathcal{C}_{qb} \delta_{b|h}$, and $F_{i|h} \equiv \sum_{q \in \mathcal{H}} \psi_{iq}^\ell \delta_{q|h}$:

$$\begin{aligned} \widehat{PTT}_h &= \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} (\hat{A}_i + \hat{\mu}_i) + \hat{\chi}_h - \sum_{b \in \mathcal{H}} F_{b|h} d\chi_b - \sum_{q \in \mathcal{H}} \chi_q \sum_{b \in \mathcal{H}} \delta_{b|h} d\mathcal{C}_{qb} \\ &= \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} \hat{A}_i + \sum_{i \in \mathcal{N}} \tilde{\mathcal{B}}_{hi} \hat{\mu}_i + \hat{\chi}_h - \sum_{b \in \mathcal{H}} M_{b|h} d\chi_b - \sum_{i \in \mathcal{N}} \lambda_i F_{i|h} \hat{\mu}_i \\ &\quad - \sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{b \in \mathcal{H}} \delta_{b|h} d\tilde{\Omega}_{ib}^\ell - \sum_{b \in \mathcal{H}} \chi_b \sum_{i \in \mathcal{N}} F_{i|h} d\beta_{bi} - \sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{j \in \mathcal{N}} F_{j|h} d\tilde{\Omega}_{ij}^x. \end{aligned}$$

A.8 Proof for Theorem 2

The first-order approximation for GDP: $\widehat{GDP} = \sum_{h \in \mathcal{H}} \chi_h \hat{E}_h = \sum_{h \in \mathcal{H}} \chi_h (\hat{p}_h^c + \hat{C}_h)$. Taking the GDP deflator as the Divisia-weighted average in households' price indices $\hat{p}_Y = \sum_{h \in \mathcal{H}} \chi_h \hat{p}_h^c = \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h \hat{w}_h - \sum_{i \in \mathcal{N}} \tilde{\lambda}_i (\hat{A}_i + \hat{\mu}_i)$. Thus, the first-order approximation of real GDP is $\hat{Y} = \widehat{GDP} - \hat{p}_Y = \sum_{h \in \mathcal{H}} \chi_h \hat{C}_h$, and using (11), $\sum_{h \in \mathcal{H}} \kappa_{ih} = 1$, $\sum_{h \in \mathcal{H}} \kappa_{ih} \hat{\kappa}_{ih} = 0$, $\sum_{h \in \mathcal{H}} \tau_h \hat{T}_h = 0$:

$$\hat{Y} = \tilde{\lambda}' \hat{A} + (\tilde{\lambda} - \text{diag}(\mu) \lambda)' \hat{\mu} + (\Lambda - \tilde{\Lambda})' \hat{J} + \lambda' \text{diag}(\mathbb{1}_N - \mu) \hat{S} + \tilde{\Lambda}' \hat{L}. \quad (12)$$

To derive the previous equation use

$$\begin{aligned}\sum_{h \in \mathcal{H}} (\Lambda_h \hat{J}_h - \chi_h \tilde{\mathcal{C}}_h' \hat{J}) &= \sum_{h \in \mathcal{H}} \Lambda_h \hat{J}_h - \sum_{b \in \mathcal{H}} \hat{J}_b \sum_{i \in \mathcal{N}} \tilde{\Omega}_{ib}^\ell \sum_{h \in \mathcal{H}} \chi_h \tilde{\mathcal{B}}_{hi} \\ &= \sum_{h \in \mathcal{H}} \Lambda_h \hat{J}_h - \sum_{b \in \mathcal{H}} \hat{J}_b \sum_{i \in \mathcal{N}} \tilde{\Omega}_{ib}^\ell \tilde{\lambda}_i = (\Lambda - \tilde{\Lambda})' \hat{J},\end{aligned}$$

where the second equality follows from $\tilde{\lambda} \equiv \tilde{\Psi}_x'(I_N - \Omega'_x)\lambda \equiv \tilde{\mathcal{B}}' \chi$ and $\tilde{\Lambda} \equiv \tilde{\Omega}'_\ell \tilde{\lambda}$.

Hence $Y = Q_Y(\{C_h\}_{h \in \mathcal{H}}) = \eta_Y \mathcal{D}(\mathcal{A}) \mathcal{D}(\mu) \mathcal{D}(J) \mathcal{D}(\Pi) F(\{L_h\}_{h \in \mathcal{H}})$, where η_Y is a constant, $Q_Y(\{C_h\}_{h \in \mathcal{H}})$ is a CRS aggregator with $\frac{d \log Q_Y(\{C_h\}_{h \in \mathcal{H}})}{d \log C_h} = \chi_h$. Similarly, $F(\{L_h\}_{h \in \mathcal{H}})$ is a CRS function with $\frac{d \log F(\{L_h\}_{h \in \mathcal{H}})}{d \log L_h} = \tilde{\Lambda}_h$. Its components are defined as $\mathcal{D}(\mathcal{A}) = \exp\{\tilde{\lambda}' \hat{\mathcal{A}}\}$, $\mathcal{D}(\mu) = \exp\{\tilde{\lambda}' \hat{\mu}\}$, $\mathcal{D}(J) = \exp\{(\Lambda - \tilde{\Lambda})' \hat{J}\}$, and $\mathcal{D}(\Pi) = \exp\{\lambda' \text{diag}(\mathbf{1}_N - \mu) \hat{S} - \lambda' \text{diag}(\mu) \hat{\mu}\}$. Then $Y = \eta_Y \mathcal{D}(\mathcal{A}) \mathcal{D}(\mu) \mathcal{D}(\Pi) \mathcal{D}(J) F = TFP F$ with $TFP = \eta_Y \mathcal{D}(\mathcal{A}) \mathcal{D}(\mu) \mathcal{D}(\Pi) \mathcal{D}(J)$.

Add and subtract $\widehat{GDP}$ to express (12) as

$$\hat{Y} = \tilde{\lambda}' \hat{\mathcal{A}} + (\tilde{\lambda} - \text{diag}(\mu) \lambda)' \hat{\mu} + (\Lambda - \tilde{\Lambda})' \hat{\Lambda} + \lambda' \text{diag}(\mathbf{1}_N - \mu) \hat{\lambda} + \tau' \hat{\tau} + \tilde{\Lambda}' \hat{L}.$$

The 3H-dimensional vector $\mathcal{R}$ decomposition sources of income across households: $\mathcal{R}' = [\lambda' \Omega_\ell \lambda' \Omega_\pi \tau'] = [\Lambda_1 \cdots \Lambda_H \sum_{i \in \mathcal{N}} \kappa_{i1}(1 - \mu_i) \lambda_i \cdots \sum_{i \in \mathcal{N}} \kappa_{iH}(1 - \mu_i) \lambda_i \tau_1 \cdots \tau_H]$. The first H entries represent the factor income shares, the next H capture the household's share of profits in total expenditure, and the final H entries correspond to the share of lump-sum transfers. Since the entries of this vector sum to one

$$\begin{aligned}0 &= \sum_{h \in \mathcal{H}} \Lambda_h \hat{\Lambda}_h + \sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} \kappa_{ih} \lambda_i ((1 - \mu_i)(\hat{\kappa}_{ih} + \hat{\lambda}_i) - \mu_i \hat{\mu}_i) + \underbrace{\sum_{h \in \mathcal{H}} \tau_h \hat{\tau}_h}_{=0} \\ &= \sum_{h \in \mathcal{H}} \Lambda_h \hat{\Lambda}_h + \sum_{i \in \mathcal{N}} \lambda_i ((1 - \mu_i) (\underbrace{\sum_{h \in \mathcal{H}} \kappa_{ih} \hat{\kappa}_{ih}}_{=0} + \underbrace{\hat{\lambda}_i \sum_{h \in \mathcal{H}} \kappa_{ih}}_{=1}) - \mu_i \hat{\mu}_i \underbrace{\sum_{h \in \mathcal{H}} \kappa_{ih}}_{=1}) \\ &= \sum_{h \in \mathcal{H}} \Lambda_h \hat{\Lambda}_h + \sum_{i \in \mathcal{N}} \lambda_i ((1 - \mu_i) \hat{\lambda}_i - \mu_i \hat{\mu}_i).\end{aligned}$$

This implies that $\hat{Y} = \tilde{\lambda}' \hat{\mathcal{A}} + \tilde{\lambda}' \hat{\mu} - \tilde{\Lambda}' \hat{\Lambda} + \tilde{\Lambda}' \hat{L}$, which, under the additional assumption of inelastic factor supply, coincides with Theorem 1 in [Baqee & Farhi \(2020\)](#). Hence, using $d \Lambda_h$ from Theorem 1

$$\begin{aligned}\widehat{TFP} &= \sum_{i \in \mathcal{N}} \tilde{\lambda}_i (\hat{\mathcal{A}}_i + \hat{\mu}_i) - \sum_{h \in \mathcal{H}} M_h d \chi_h - \sum_{h \in \mathcal{H}} \delta_h \sum_{b \in \mathcal{H}} \chi_b d \mathcal{C}_{bh} \\ &= \sum_{i \in \mathcal{N}} \tilde{\lambda}_i \hat{\mathcal{A}}_i + \sum_{i \in \mathcal{N}} (\tilde{\lambda}_i - \lambda_i F_i) \hat{\mu}_i - \sum_{h \in \mathcal{H}} M_h d \chi_h - \sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{h \in \mathcal{H}} \delta_h d \tilde{\Omega}_{ih}^\ell \\ &\quad - \sum_{h \in \mathcal{H}} \chi_h \sum_{i \in \mathcal{N}} F_i d \beta_{hi} - \sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{j \in \mathcal{N}} F_j d \tilde{\Omega}_{ij}^x.\end{aligned}$$

Proof for Corollary 1. If $M_h = M$ for all $h \in \mathcal{H}$, then the distributive terms of trade satisfy $M \sum_{h \in \mathcal{H}} d \chi_h = 0$. **(i) Undistorted economy.** $\delta_h = 1$ and $M_h = 1 \forall h \in \mathcal{H}$. **(ii) Symmetric consumption bundles.** Let $\beta = \mathbf{1}_H \bar{\beta}'$ where $\bar{\beta}$ denotes the common consumption vector. Then, $\mathcal{C} = \beta \Psi_x \Omega_\ell = \mathbf{1}_H \bar{\beta}' \Psi_x \Omega_\ell = \mathbf{1}_H \bar{\mathcal{C}}'$. This implies that $M_h = \bar{\mathcal{C}}' \delta$ for all $h \in \mathcal{H}$, where

δ denotes the vector of distortion centralities. **(iii) No intermediate inputs and symmetric distortions.** In this case, $\mathcal{C} = \beta \Omega_\ell = \mu \beta \alpha$. Then, $\mathcal{C} \mathbf{1}_H = \mu \beta \alpha \mathbf{1}_H = \mu \beta \mathbf{1}_H = \mu \mathbf{1}$. Hence $M_h = \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} \delta_b = \mu^{-1} \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} = 1$ for all $h \in \mathcal{H}$, since $\delta_b = \mu^{-1}$ for all $b \in \mathcal{H}$. **(iv) No intermediate inputs and firm specific factor supply.** Let $\tilde{\Omega}_\ell = I_N$. Then, $\mathcal{C} = \beta \text{diag}(\mu)$, and $\delta_i = \mu_i^{-1}$. Therefore, $\mathcal{C} \delta = \beta \mathbf{1}_N = \mathbf{1}_N$.

Proof for Corollary 2. In the absence of distortions, we have $\delta_h = 1$ for all $h \in \mathcal{H}$. Consequently, $M_h = \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} = 1 \forall h \in \mathcal{H}$, and $F_i = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell = 1$ for all $i \in \mathcal{N}$. As a result, the final demand terms of trade simplifies to $\sum_{h \in \mathcal{H}} \chi_h \sum_{i \in \mathcal{N}} d \beta_{hi} = 0$. Similarly, the sum of the factor $TT + \text{intermediate } TT$ reduces to $\sum_{i \in \mathcal{N}} \lambda_i \sum_{i \in \mathcal{N}} d \beta_{hi} = 0$. Cases 2 and 3 follow directly from the preceding arguments and are omitted for brevity.

A.9 Proof for Theorem 3

The constrained planner chooses $\{Y, L, \{C_h, L_h, T_h, \{C_{hi}\}_{i \in \mathcal{N}}\}_{h \in \mathcal{H}}\}$ to maximize $W(Y, L)$ subject to $Y = Q_Y(\{C_h\}_{h \in \mathcal{H}})$, $C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}})$, $L = F(\{L_h\}_{h \in \mathcal{H}})$, $\sum_{h \in \mathcal{H}} T_h = 0$, and each household budget constraint. The FOCs for T_h equalize the multipliers on the budget constraints across h . Hence, the problem is equivalent to choosing $\{Y, L, \{C_h, L_h, \{C_{hi}\}_{i \in \mathcal{N}}\}_{h \in \mathcal{H}}\}$ subject to the consolidated budget constraint $GDP = p_Y Y = \sum_{h \in \mathcal{H}} p_h^c C_h = \sum_{i \in \mathcal{N}} p_i \sum_{h \in \mathcal{H}} C_{hi} \leq w L + \sum_{i \in \mathcal{N}} \pi_i$ where $w L = \sum_{h \in \mathcal{H}} w_h L_h$. The planner takes $\{w_h\}_{h \in \mathcal{H}}$ and $\{p_i, \pi_i\}_{i \in \mathcal{N}}$ as given.

The planner FOCs are $\frac{w_h}{p_Y} = -\frac{W_L}{W_Y} \frac{\partial L}{\partial L_h}$ and $\frac{p_i}{p_Y} = \frac{\partial Y}{\partial C_h} \frac{\partial C_h}{\partial C_{hi}} \forall h \in \mathcal{H} \forall i \in \mathcal{N}$ such that $C_{hi} > 0$. I first link aggregate output Y to aggregate factor supply L . The optimality condition is $-\frac{W_L}{W_Y} = \frac{w_L}{p_Y Y} \frac{Y}{L} = \Gamma \frac{Y}{L}$. This equates the MRS between Y and L to the MRT scaled by Γ , which equals the aggregate factor share in equilibrium. With $Y = TFP L$, the MRT is $TFP = Y/L = -(W_L/W_Y) \Gamma^{-1}$. Under the planner optimum, the MRS relates aggregate output to factor- h supply as $-\frac{W_L}{W_Y} = \frac{\Lambda_h}{\tilde{\Lambda}_h} \frac{Y}{L}$. This follows from $\tilde{\Lambda}_h = \frac{L_h}{L} \frac{\partial L}{\partial L_h}$. Combining $-\frac{W_L}{W_Y} = \Gamma \frac{Y}{L}$ and $-\frac{W_L}{W_Y} = \frac{\Lambda_h}{\tilde{\Lambda}_h} \frac{Y}{L}$ implies the planner requires $\Gamma = \frac{\Lambda_h}{\tilde{\Lambda}_h} = \delta_h^{-1} \forall h \in \mathcal{H}$. Any deviation indicates an inefficient composition of aggregate factor supply across h . Summing over all factors yields $\Gamma \sum_{h \in \mathcal{H}} \tilde{\Lambda}_h = \sum_{h \in \mathcal{H}} \Lambda_h$ and $\Gamma = \sum_{h \in \mathcal{H}} \Lambda_h$.

Define $\beta_i \equiv \frac{d \log GDP}{d \log p_i}$ as the real GDP elasticity with respect to good- i 's price. By Shephard's lemma, $\beta_i = \frac{\sum_{h \in \mathcal{H}} p_i C_{hi}}{GDP} = \sum_{h \in \mathcal{H}} \frac{C_{hi}}{Y} \frac{\partial Y}{\partial C_h} \frac{\partial C_h}{\partial C_{hi}} = \sum_{h \in \mathcal{H}} \chi_h \beta_{hi}$. Hence, $\lambda = \mathcal{B}' \chi = \Psi'_x \bar{\beta}$, where $\bar{\beta} = [\beta_1, \dots, \beta_N]'$, and $\Lambda = \Omega'_\ell \Psi'_x \bar{\beta}$. Therefore, $\Lambda_h = \mathbf{1}'_N \text{diag}(\tilde{\Omega}_\ell \circ_H(h)) (\text{diag}(\mu)^{-1} - \tilde{\Omega}'_x)^{-1} \bar{\beta}$

and

$$\begin{aligned}\Lambda_h \widehat{\Lambda}_h &= \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \lambda_i (\widehat{\omega}_i^\ell + \widehat{\alpha}_{ih}) + \mathbf{1}'_N \text{diag}(\Omega_{\ell o_H}(h)) \Psi'_x \begin{pmatrix} \beta_1 \widehat{\beta}_1 \\ \vdots \\ \beta_N \widehat{\beta}_N \end{pmatrix} \\ &\quad + \mathbf{1}'_N \text{diag}(\widetilde{\Omega}_{\ell o_H}(h)) \left(\frac{d(\text{diag}(\mu)^{-1} - \widetilde{\Omega}'_x)^{-1}}{d \log \widetilde{\Omega}_x} + \frac{d(\text{diag}(\mu)^{-1} - \widetilde{\Omega}'_x)^{-1}}{d \log \mu} \right) \bar{\beta}.\end{aligned}$$

Using $\frac{dA^{-1}}{dx} = -A^{-1} \frac{dA}{dx} A^{-1}$: $d\Lambda_h = \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\widetilde{\Omega}_{ih}^\ell + \sum_{i \in \mathcal{N}} \psi_{ih}^\ell \lambda_i d \log \mu_i + \sum_{j \in \mathcal{N}} \psi_{jh}^\ell (d\beta_j + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\widetilde{\Omega}_{ij}^x)$. Then $d\Lambda = \sum_{h \in \mathcal{H}} d\Lambda_h = \sum_{i \in \mathcal{H}} \psi_i^\ell \lambda_i d \log \mu_i + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\omega_i^\ell + \sum_{j \in \mathcal{N}} \psi_j^\ell (d\beta_j + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d\widetilde{\Omega}_{ij}^x)$ with $\psi_i^\ell = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell$.

B Proofs for the normalized nested-CES model

B.1 Environment

Firms. $i \in \mathcal{N}$ chooses $\{y_i, \{\ell_{ih}\}_{h \in \mathcal{H}}, \{x_{ij}\}_{j \in \mathcal{N}}\}$ to minimize $\bar{c}_i$ subject to $\frac{y_i}{\bar{y}_i} = A_i(\omega_i^\ell (\frac{L_i}{\bar{L}_i})^{\frac{\theta_i-1}{\theta_i}} + \omega_i^x (\frac{X_i}{\bar{X}_i})^{\frac{\theta_i-1}{\theta_i}})^{\frac{\theta_i}{\theta_i-1}}$, $\frac{L_i}{\bar{L}_i} = (\sum_{h \in \mathcal{H}} \alpha_{ih} (\frac{\ell_{ih}}{\bar{\ell}_{ih}})^{\frac{\theta_i^\ell-1}{\theta_i^\ell}})^{\frac{\theta_i^\ell}{\theta_i^\ell-1}}$ and $\frac{X_i}{\bar{X}_i} = (\sum_{j \in \mathcal{N}} \omega_{ij} (\frac{x_{ij}}{\bar{x}_{ij}})^{\frac{\theta_i^x-1}{\theta_i^x}})^{\frac{\theta_i^x}{\theta_i^x-1}}$. The FOCs are:

$$p_i^\ell L_i = (\mu_i \omega_i^\ell)^{\theta_i} (A_i \frac{p_i \bar{y}_i}{p_i^\ell \bar{L}_i})^{\theta_i-1} p_i y_i,$$

$$p_i^x X_i = (\mu_i \omega_i^x)^{\theta_i} (A_i \frac{p_i \bar{y}_i}{p_i^x \bar{X}_i})^{\theta_i-1} p_i y_i,$$

$$w_h \ell_{ih} = (\mu_i \omega_i^\ell)^{\theta_i} \alpha_{ih}^{\theta_i} (A_i \frac{p_i \bar{y}_i}{p_i^\ell \bar{L}_i})^{\theta_i-1} (\frac{p_i^\ell \bar{L}_i}{w_h \bar{\ell}_{ih}})^{\theta_i^\ell-1} p_i y_i,$$

,

$$p_j x_{ij} = (\mu_i \omega_i^x)^{\theta_i} \mathcal{W}_{ij}^{\theta_i^x} (A_i \frac{p_i \bar{y}_i}{p_i^x \bar{X}_i})^{\theta_i-1} (\frac{p_i^x \bar{X}_i}{p_j \bar{x}_{ij}})^{\theta_i^x-1} p_i y_i.$$

At the point of normalization, assume $A_i = 1$ for all $i \in \mathcal{N}$. Then, the first-order conditions satisfy in equilibrium: (i) $\bar{p}_i^\ell \bar{L}_i = \mu_i \omega_i^\ell \bar{p}_i \bar{y}_i$, (ii) $\bar{p}_i^x \bar{X}_i = \mu_i \omega_i^x \bar{p}_i \bar{y}_i$, (iii) $\bar{w}_h \bar{\ell}_{ih} = \alpha_{ih} \bar{p}_i^\ell \bar{L}_i$, and (iv) $\bar{p}_j \bar{x}_{ij} = \mathcal{W}_{ij} \bar{p}_i^x \bar{X}_i$. Prices are given by

$$p_i^\ell = \frac{1}{\bar{L}_i} \left(\sum_{h \in \mathcal{H}} \alpha_{ih}^{\theta_i} (w_h \bar{\ell}_{ih})^{1-\theta_i^\ell} \right)^{\frac{1}{1-\theta_i^\ell}},$$

$$p_i^x = \frac{1}{\bar{X}_i} \left(\sum_{j \in \mathcal{N}} \mathcal{W}_{ij}^{\theta_i^x} (p_j \bar{x}_{ij})^{1-\theta_i^x} \right)^{\frac{1}{1-\theta_i^x}},$$

$$p_i = \frac{1}{A_i \mu_i \bar{y}_i} (\omega_i^\ell \theta_i (p_i^\ell \bar{L}_i)^{1-\theta_i} + \omega_i^x \theta_i (p_i^x \bar{X}_i)^{1-\theta_i})^{\frac{1}{1-\theta_i}}.$$

Households. $h \in \mathcal{H}$ chooses $\{\{C_{hi}\}_{i \in \mathcal{N}}, L_h\}$ to maximize $U(C_h) = \frac{C_h}{\bar{C}_h} = (\sum_{i \in \mathcal{N}} \beta_{hi} (\frac{C_{hi}}{\bar{C}_{hi}})^{\frac{q_h-1}{q_h}})^{\frac{q_h}{q_h-1}}$ subject to $E_h = p_h^c C_h = \sum_{i \in \mathcal{N}} p_i C_{hi} \leq w_h L_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \pi_i + T_h$. The FOC is given by $\frac{p_i}{p_h^c} = \frac{\partial C_h}{\partial C_{hi}} = \beta_{hi} (\frac{\bar{C}_h}{\bar{C}_{hi}})^{\frac{q_h-1}{q_h}} (\frac{C_h}{\bar{C}_{hi}})^{\frac{1}{q_h}}$. Then $p_i C_{hi} = \beta_{hi}^{q_h} (\frac{p_h^c \bar{C}_h}{p_i \bar{C}_{hi}})^{q_h-1} E_h$ or $\bar{p}_i \bar{C}_{hi} = \beta_{hi} \bar{C}_h$ at the point of normalization. Finally, the price index for household h 's is given by $p_h^c = \frac{1}{\bar{C}_h} (\sum_{i \in \mathcal{N}} \beta_{hi}^{q_h} (p_i \bar{C}_{hi})^{1-q_h})^{\frac{1}{1-q_h}}$.

B.2 Equilibrium conditions

Goods markets: $S_i = \sum_{j \in \mathcal{N}} (\mu_j \omega_j^x)^{\theta_j} \mathcal{W}_{ji}^{\theta_j^x} (A_j \frac{p_j \bar{y}_j}{p_j^x \bar{X}_j})^{\theta_j-1} (\frac{p_j^x \bar{X}_j}{p_i \bar{x}_{ji}})^{\theta_j^x-1} S_j + \sum_{h \in \mathcal{H}} \beta_{hi}^{q_h} (\frac{p_h^c \bar{C}_h}{p_i \bar{C}_{hi}})^{q_h-1} E_h$. Then the log-linearized goods-market condition is $\lambda_i \hat{S}_i = \sum_{j \in \mathcal{N}} \Omega_{ji}^x \lambda_j [\theta_j (\hat{\mu}_j + \hat{\omega}_j^x) + \theta_j^x \hat{\omega}_{ji} + (\theta_i - 1)(\hat{A}_j + \hat{p}_j) + (\theta_j^x - \theta_j) \hat{p}_j^x - (\theta_j^x - 1) \hat{p}_i + \hat{S}_j] + \sum_{h \in \mathcal{H}} \beta_{hi} \chi_h (q_h \hat{\beta}_{hi} + (q_h - 1)(\hat{p}_h^c - \hat{p}_i) + \hat{E}_h)$.

In matrix form

$$\begin{aligned} \text{diag}(\lambda)\widehat{S} &= \Psi'_x \{ (\beta \circ \widehat{\beta})' \text{diag}(\varrho)\chi + \beta' \text{diag}(\chi)(\text{diag}(\varrho - \mathbf{1}_H)\widehat{p}_c + \widehat{E}) \\ &\quad - \text{diag}(\beta' \text{diag}(\varrho - \mathbf{1}_H)\chi)\widehat{p} + (\Omega_x \circ \widehat{\mathcal{W}})' \text{diag}(\theta_x)\lambda - \text{diag}(\Omega'_x \text{diag}(\theta_x - \mathbf{1}_N)\lambda)\widehat{p} \\ &\quad + \Omega'_x \text{diag}(\lambda)(\text{diag}(\theta)(\widehat{\omega}_x + \widehat{\mu}) + \text{diag}(\theta - \mathbf{1}_N)(\widehat{A} + \widehat{p}) + \text{diag}(\theta_x - \theta)\widehat{p}_x) \}. \end{aligned} \quad (13)$$

Budget constraints: The log-linearized version is $\chi_h \widehat{E}_h = \Lambda_h \widehat{J}_h + \sum_{i \in \mathcal{N}} \kappa_{ih} \lambda_i ((1 - \mu_i)(\widehat{\kappa}_{ih} + \widehat{S}_i) - \mu_i \widehat{\mu}_i) + \tau_h \widehat{T}_h$ In matrix form

$$\begin{aligned} \text{diag}(\chi)\widehat{E} &= \text{diag}(\Lambda)\widehat{J} + \kappa' \text{diag}(\lambda)(\text{diag}(\mathbf{1}_N - \mu)\widehat{S} - \text{diag}(\mu)\widehat{\mu}) \\ &\quad + (\kappa \circ \widehat{\kappa})' \text{diag}(\mathbf{1}_N - \mu)\lambda + \text{diag}(\tau)\widehat{T}. \end{aligned} \quad (14)$$

Factor Markets: FOC imply that $w_h L_h = \sum_{i \in \mathcal{N}} (\mu_i \omega_i^\ell)^{\theta_i} \alpha_{ih}^\ell (A_i \frac{p_i \bar{y}_i}{p_i^\ell L_i})^{\theta_i - 1} (\frac{p_i^\ell \bar{L}_i}{w_h \bar{\ell}_{ih}})^{\theta_i^\ell - 1} S_i$. The log-linearized version is $\Lambda_h \widehat{J}_h = \sum_{i \in \mathcal{N}} \Omega_{ih}^\ell \lambda_i [\theta_i (\widehat{\mu}_i + \widehat{\omega}_i^\ell) + \theta_i^\ell \widehat{\alpha}_{ih} + (\theta_i - 1)(\widehat{A}_i + \widehat{p}_i) + (\theta_i^\ell - \theta_i) \widehat{p}_i^\ell - (\theta_i^\ell - 1) \widehat{w}_h + \widehat{S}_i]$. In matrix form

$$\begin{aligned} \text{diag}(\Lambda)\widehat{J} &= (\Omega_\ell \circ \widehat{\alpha})' \text{diag}(\theta_\ell)\lambda - \text{diag}(\Omega'_\ell \text{diag}(\theta_\ell - \mathbf{1}_N)\lambda)\widehat{w} \\ &\quad + \Omega'_\ell \text{diag}(\lambda)(\text{diag}(\theta)(\widehat{\omega}_\ell + \widehat{\mu}) + \text{diag}(\theta - \mathbf{1}_N)(\widehat{A} + \widehat{p}) + \text{diag}(\theta_\ell - \theta)\widehat{p}_\ell + \widehat{S}). \end{aligned} \quad (15)$$

Prices. The first-order approximation for p_i^ℓ , p_i^x , p_i , and p_h^c are given by $\widehat{p}_i^\ell = \sum_{h \in \mathcal{H}} \alpha_{ih} \widehat{w}_h$, $\widehat{p}_i^x = \sum_{j \in \mathcal{N}} \mathcal{W}_{ij} \widehat{p}_j$, $\widehat{p}_i = \omega_i^\ell \widehat{p}_i^\ell + \omega_i^x \widehat{p}_i^x - \widehat{A}_i - \widehat{\mu}_i$, and $\widehat{p}_h^c = \sum_{i \in \mathcal{N}} \beta_{hi} \widehat{p}_i$. In matrix form $\widehat{p}_\ell = \alpha \widehat{w}$, $\widehat{p} = \widetilde{\Psi}_\ell \widehat{w} - \widetilde{\Psi}_x (\widehat{A} + \widehat{\mu})$, $\widehat{p}_x = \mathcal{W} \widetilde{\Psi}_x (\widetilde{\Omega}_\ell \widehat{w} - \widehat{A} - \widehat{\mu})$.

Sufficient Equations. The following equations characterize a $(2H + N)$ -dimensional linear system for the endogenous first-order responses of household expenditure $d \log E_h$, factor prices $d \log w_h$, and firm sales $d \log S_i$. In a CES economy, the first-order response of consumption expenditure, factor prices, and firm sales, in response to shocks in productivity, markups, preferences, lump-sum taxes, and equity composition are, to a first order:

$$d \log E_h = \underbrace{\frac{\Lambda_h}{\chi_h} d \log w_h}_{\text{Factor Price Effect on Expenditure (GE)}} + \underbrace{\frac{\tau_h}{\chi_h} d \log T_h}_{\text{Lump-sum Effect on Expenditure (PE)}} + \underbrace{\sum_{i \in \mathcal{N}} \frac{\lambda_i}{\chi_h} ((1 - \mu_i)(d \kappa_{ih} + \kappa_{ih} d \log S_i) - \kappa_{ih} \mu_i d \log \mu_i)}_{\text{Corporate Income Effect on Expenditure (PE + GE)}}; \quad (16)$$

$$\begin{aligned} d \log w_h &= \underbrace{\sum_{i \in \mathcal{N}} \frac{\mu_i \lambda_i}{\Lambda_h} (\theta_i \alpha_{ih} d \omega_i^\ell + \theta_i^\ell \omega_i^\ell d \alpha_{ih})}_{\text{Firm Technology Effect on Factor Prices (PE)}} + \underbrace{\sum_{i \in \mathcal{N}} \frac{\Omega_{ih}^\ell \lambda_i}{\Lambda_h} ((\theta_i - 1) d \log A_i + \theta_i d \log \mu_i)}_{\text{Direct Effect on Factor Prices (PE)}} \\ &\quad - \underbrace{\sum_{j \in \mathcal{N}} \left(\sum_{i \in \mathcal{N}} \frac{\Omega_{ih}^\ell \lambda_i}{\Lambda_h} (\theta_i - 1) \widetilde{\psi}_{ij}^x \right) (d \log A_j + d \log \mu_j)}_{\text{Supplier Effect on Factor Prices (PE)}} + \underbrace{\sum_{i \in \mathcal{N}} \frac{\Omega_{ih}^\ell \lambda_i}{\Lambda_h} d \log S_i}_{\text{Sales Effect on Factor Prices (GE)}} \\ &\quad - \underbrace{\left(\sum_{i \in \mathcal{N}} \frac{\Omega_{ih}^\ell \lambda_i}{\Lambda_h} (\theta_i^\ell - 1) \right) d \log w_h}_{\text{Direct Substitution Effect on Factor Prices (GE)}} + \underbrace{\sum_{b \in \mathcal{H}} \left(\sum_{i \in \mathcal{N}} \frac{\Omega_{ih}^\ell \lambda_i}{\Lambda_h} (\theta_i - 1) (\widetilde{\psi}_{ib}^\ell - \alpha_{ib}) + (\theta_i^\ell - 1) \alpha_{ib} \right) d \log w_b}_{\text{Supplier Substitution Effect on Factor Prices (GE)}}. \end{aligned} \quad (17)$$

$$\begin{aligned}
d \log S_i = & \underbrace{\sum_{h \in \mathcal{H}} \frac{\beta_{hi} \chi_h}{\lambda_i} d \log E_h}_{\text{Expenditure Effect on Sales (GE)}} + \underbrace{\sum_{j \in \mathcal{N}} \frac{\Omega_{ji}^x \lambda_j}{\lambda_i} d \log S_j}_{\text{Sales Effect on Sales (GE)}} + \underbrace{\sum_{j \in \mathcal{N}} \frac{\Omega_{ji}^x \lambda_j}{\lambda_i} ((\theta_j - 1) d \log A_j + \theta_j d \log \mu_j)}_{\text{Direct Effect on Sales (PE)}} \\
& + \underbrace{\sum_{j \in \mathcal{N}} \left(\sum_{h \in \mathcal{H}} \frac{\beta_{hi} \chi_h}{\lambda_i} (e_h - 1) (\tilde{\psi}_{ij}^x - \tilde{\mathcal{B}}_{hj}) + \sum_{q \in \mathcal{N}} \frac{\Omega_{qi}^x \lambda_q}{\lambda_i} (\theta_q - 1) (\tilde{\psi}_{ij}^x - \tilde{\psi}_{qj}^x) \right) (d \log A_j + d \log \mu_j)}_{\text{Supplier Effect on Sales (PE)}} \\
& + \underbrace{\sum_{j \in \mathcal{N}} \left(\sum_{q \in \mathcal{N}} \frac{\Omega_{qi}^x \lambda_q}{\lambda_i} (\theta_q^x - \theta_q) \left(\tilde{\psi}_{ij}^x - \sum_{m \in \mathcal{N}} \mathcal{W}_{qm} \tilde{\psi}_{mj}^x \right) \right) (d \log A_j + d \log \mu_j)}_{\text{Indirect Supplier Effect on Sales (PE)}} + \underbrace{\sum_{h \in \mathcal{H}} \frac{\chi_h}{\lambda_i} e_h d \beta_{hi}}_{\text{Household Preference Effect on Sales (PE)}} \\
& + \underbrace{\sum_{j \in \mathcal{N}} \frac{\mu_j \lambda_j}{\lambda_i} (\theta_j \mathcal{W}_{ji} d \omega_j^x + \theta_j^x \omega_j^x d \mathcal{W}_{ji})}_{\text{Firm Technology Effect on Sales (PE)}} + \underbrace{\sum_{h \in \mathcal{H}} \left(\sum_{j \in \mathcal{N}} \frac{\Omega_{ji}^x \lambda_j}{\lambda_i} (\theta_j^x - \theta_j) \left(\sum_{q \in \mathcal{N}} \mathcal{W}_{iq} \tilde{\psi}_{qh}^\ell - \tilde{\psi}_{ih}^\ell \right) \right) d \log w_h}_{\text{Indirect Supplier Substitution Effect on Sales (GE)}} \\
& + \underbrace{\sum_{h \in \mathcal{H}} \left(\sum_{b \in \mathcal{H}} \frac{\beta_{bi} \chi_b}{\lambda_i} (e_b - 1) (\tilde{\mathcal{C}}_{bh} - \tilde{\psi}_{ih}^\ell) + \sum_{j \in \mathcal{N}} \frac{\Omega_{ji}^x \lambda_j}{\lambda_i} (\theta_j - 1) (\tilde{\psi}_{jh}^\ell - \tilde{\psi}_{ih}^\ell) \right) d \log w_h}_{\text{Supplier Substitution Effect on Sales (GE)}}
\end{aligned} \tag{18}$$

B.3 Example 1

Assume inelastic labor supply, $\beta_h \neq \beta_l$, $\mu_m \neq \mu_a$, $\alpha_{mh} = \alpha_{ah} = \frac{1}{2}$, and $\kappa_{mh} = \kappa_{ah} = \frac{1}{2}$. Then $\tilde{\lambda}_m = \lambda_m = \beta_l + (\beta_h - \beta_l) \chi_h$, $\tilde{\lambda}_a = \lambda_a = 1 - \lambda_m$, $\tilde{\Lambda}_h = \tilde{\Lambda}_l = \frac{1}{2}$, $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + (\mu_m - \mu_a) \frac{\lambda_m}{2}$, $\chi_h = \Lambda_h + \frac{1}{2} (1 - \mu_a + (\mu_a - \mu_m) \lambda_m) + \tau_h$, and $\chi_l = 1 - \chi_h$. In equilibrium: $\chi_h = \frac{1}{2} + \tau_h$, $\chi_l = \frac{1}{2} - \tau_h$, $\lambda_m = \frac{1}{2} (\beta_h + \beta_l) + (\beta_h - \beta_l) \tau_h$, $\lambda_a = 1 - \frac{1}{2} (\beta_h + \beta_l) - (\beta_h - \beta_l) \tau_h$, $\Lambda_h = \Lambda_l = \frac{\mu_a}{2} + \frac{1}{4} (\beta_h + \beta_l) (\mu_m - \mu_a) + \frac{1}{2} (\beta_h - \beta_l) (\mu_m - \mu_a) \tau_h$.

From (16), (18), and (17):

- (i) $\Lambda_h d \log w_h = \Lambda_l d \log w_l = \frac{1}{2} (\mu_m \lambda_m d \log S_m + \mu_a \lambda_a d \log S_a)$,
- (ii) $\chi_h d \log E_h = \Lambda_h d \log w_h + \frac{1}{2} ((1 - \mu_m) \lambda_m d \log S_m + (1 - \mu_a) \lambda_a d \log S_a) + \tau_h \hat{T}_h$,
- (iii) $\chi_l d \log E_l = \Lambda_l d \log w_l + \frac{1}{2} ((1 - \mu_m) \lambda_m d \log S_m + (1 - \mu_a) \lambda_a d \log S_a) + \tau_l \hat{T}_l$,
- (iv) $\lambda_m d \log S_m = \beta_h \chi_h d \log E_h + \beta_l \chi_l d \log E_l$,
- (v) $\lambda_a d \log S_a = (1 - \beta_h) \chi_h d \log E_h + (1 - \beta_l) \chi_l d \log E_l$.

Add and subtract $d \log \text{GDP}$:

- (i) $d \Lambda_h = d \Lambda_l = \frac{1}{2} (\mu_m d \lambda_m + \mu_a d \lambda_a)$,
- (ii) $d \chi_h = d \Lambda_h + \frac{1}{2} ((1 - \mu_m) d \lambda_m + (1 - \mu_a) d \lambda_a) + d \tau_h$,
- (iii) $d \chi_l = d \Lambda_l + \frac{1}{2} ((1 - \mu_m) d \lambda_m + (1 - \mu_a) d \lambda_a) - d \tau_h$,
- (iv) $d \lambda_m = \beta_h d \chi_h + \beta_l d \chi_l$,
- (v) $d \lambda_a = (1 - \beta_h) d \chi_h + (1 - \beta_l) d \chi_l$.

Using $d \lambda_m + d \lambda_a = 0$ and $d \chi_h = d \chi_l$, the previous system of equations can be simplified

into: (i) $d\Lambda_h = d\Lambda_l = \frac{1}{2}(\mu_m - \mu_a)d\lambda_m$, (ii) $d\chi_h = d\Lambda_h + \frac{1}{2}(\mu_a - \mu_m)d\lambda_m + d\tau_h$, and (iii) $d\lambda_m = (\beta_h - \beta_l)d\chi_h$. Thus (i) $d\Lambda_h = d\Lambda_l = \frac{1}{2}(\beta_h - \beta_l)(\mu_m - \mu_a)d\tau_h$, (ii) $d\lambda_m = (\beta_h - \beta_l)d\tau_h$, and (iii) $d\chi_h = d\tau_h$.

Using the *entropic terms of trade* from Theorem 2:

$$d \log Y = -\frac{d\Lambda_h + d\Lambda_l}{2\Lambda_h} = \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + (\mu_m - \mu_a)((\beta_h - \beta_l)\tau_h + \frac{1}{2}(\beta_h + \beta_l))} d\tau_h.$$

Additionally, we have $\delta_h = \delta_l = \frac{1}{2\Lambda_h}$, $\mathcal{C}_{hh} = \mathcal{C}_{hl} = \frac{\mu_a}{2} + \frac{1}{2}(\mu_m - \mu_a)\beta_h$, $\mathcal{C}_{lh} = \mathcal{C}_{ll} = \frac{\mu_a}{2} + \frac{1}{2}(\mu_m - \mu_a)\beta_l$, $M_h = \frac{\mu_a + (\mu_m - \mu_a)\beta_h}{\mu_a + (\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}$, $M_l = \frac{\mu_a + (\mu_m - \mu_a)\beta_l}{\mu_a + (\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}$.

Using the *distributive terms of trade* from Theorem 2

$$d \log Y = -(M_h - M_l)d\chi_h = \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + (\mu_m - \mu_a)((\beta_h - \beta_l)\tau_h + \frac{1}{2}(\beta_h + \beta_l))} d\tau_h,$$

which is the same expression as before.

Household-specific δ s are given by $\delta_{h|b} = \frac{1}{2\Lambda_h}$ and $\delta_{l|h} = \frac{1}{2\Lambda_l}$ for $b \in \{h, l\}$, and their *expenditure centralities* are $M_{h|h} = M_{h|l} = M_h = \frac{\mu_a + (\mu_m - \mu_a)\beta_h}{\mu_a + (\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}$ and $M_{l|h} = M_{l|l} = M_l = \frac{\mu_a + (\mu_m - \mu_a)\beta_l}{\mu_a + (\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)}$.

From Theorem 4

$$d \log C_h = d \log \chi_h - \sum_{b \in \{h, l\}} M_{b|h} d\chi_b = \left(\frac{2}{1 + 2\tau_h} + \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + (\mu_m - \mu_a)((\beta_h - \beta_l)\tau_h + \frac{1}{2}(\beta_h + \beta_l))} \right) d\tau_h,$$

$$d \log C_l = d \log \chi_l - \sum_{b \in \{h, l\}} M_{b|l} d\chi_b = \left(-\frac{2}{1 - 2\tau_h} + \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + (\mu_m - \mu_a)((\beta_h - \beta_l)\tau_h + \frac{1}{2}(\beta_h + \beta_l))} \right) d\tau_h.$$

With $d\tau_h > 0$ we have that $d \log C_l > 0$ as long as $(\beta_h - \beta_l)(\mu_a - \mu_m)(1 - 2\tau_h) > 2\mu_a + (\beta_h + \beta_l)(\mu_m - \mu_a) + 2(\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h \rightarrow \beta_h(\mu_a - \mu_m) > \mu_a$. **Case 1:** $\mu_a > \mu_m$. In this case, for $d \log C_l > 0$ we require $\beta_h > \frac{\mu_a}{\mu_a - \mu_m}$, which is never satisfied because $\frac{\mu_a}{\mu_a - \mu_m} > 1$ and $\beta_h \leq 1$. **Case 2:** $\mu_m > \mu_a$. In this case, for $d \log C_l > 0$ we require $-\frac{\mu_a}{\mu_m - \mu_a} > \beta_h$, which is never satisfied because $\beta_h \geq 0$.

B.4 Example 2

Consider the same economy as in Example 1, but now centralize household decisions under the constrained planner in Theorem 3. The planner chooses $\{C_{hm}, C_{ha}, C_{lm}, C_{la}\}$ to equalize households' MRS with the aggregate MRS between each good $i \in \{m, a\}$ and aggregate output: $\frac{p_m}{p_Y Y} = \chi_h \frac{\beta_h}{C_{hm}}$ and $\frac{p_a}{p_Y Y} = \chi_h \frac{1 - \beta_h}{C_{ha}} = (1 - \chi_h) \frac{1 - \beta_l}{C_{la}}$. Here we use the property of the aggregator Q_Y that $\frac{d \log Y}{d \log C_h} = \chi_h$, so $\partial Y / \partial C_h = \chi_h Y / C_h$; with Cobb-Douglas C_h , $\partial C_h / \partial C_{hi} = \beta_{hi} C_h / C_{hi}$.

Labor demand for firms m and a is given by $\ell_{mh} = \frac{1}{2} \frac{\mu_m p_m y_m}{w_h}$, $\ell_{ml} = \frac{1}{2} \frac{\mu_m p_m y_m}{w_l}$, $\ell_{ah} = \frac{1}{2} \frac{\mu_a p_a y_a}{w_h}$, and $\ell_{al} = \frac{1}{2} \frac{\mu_a p_a y_a}{w_l}$. Hence, prices and wages for firms are related by $y_i = \frac{1}{2} \mu_i p_i y_i (w_h w_l)^{-\frac{1}{2}}$. Equivalently $p_i = \frac{2}{\mu_i} \sqrt{w_h w_l}$ for $i \in \{m, a\}$. By definition, $GDP = p_m y_m + p_a y_a$, with $\lambda_i = \frac{p_i y_i}{GDP}$,

without intermediates $\tilde{\lambda}_i = \lambda_i$, and by symmetry $\tilde{\Lambda}_h = \tilde{\Lambda}_l = \frac{1}{2}(\tilde{\lambda}_m + \tilde{\lambda}_a) = \frac{1}{2}$.

Using the labor market clearing conditions and normalizing each type's endowment to one, $L_h = L_l = 1$: $1 = \ell_{mh} + \ell_{ah} = \frac{1}{2w_h}(\mu_m p_m y_m + \mu_a p_a y_a)$ and $1 = \ell_{ml} + \ell_{al} = \frac{1}{2w_l}(\mu_m p_m y_m + \mu_a p_a y_a)$. Since the bracketed term is common across both equations, it follows that $w_h = w_l$. Define the common wage as $w = w_h = w_l = \frac{1}{2}(\mu_m p_m y_m + \mu_a p_a y_a)$.

Consequently, $\Lambda_h = \Lambda_l$, and total labor income equals $2w = \mu_m p_m y_m + \mu_a p_a y_a$. Wage symmetry implies homogeneous labor demand: $\ell_m = \ell_{mh} = \ell_{ml}$ and $\ell_a = \ell_{ah} = \ell_{al}$. Hence, output and prices satisfy: $y_m = \ell_m$, $y_a = \ell_a = 1 - \ell_m$, $p_m = \frac{2w}{\mu_m}$ and $p_a = \frac{2w}{\mu_a}$.

Definitions $P_Y Y = p_m y_m + p_a y_a = 2w \left(\frac{\ell_m}{\mu_m} + \frac{1-\ell_m}{\mu_a} \right)$ and $\Gamma = \frac{2w}{P_Y Y}$ imply $1 = \Gamma \left(\frac{\ell_m}{\mu_m} + \frac{1-\ell_m}{\mu_a} \right)$. Note $\lambda_m = \chi_h \beta_h + (1 - \chi_h) \beta_l$ and $\Gamma = \mu_m \lambda_m + \mu_a (1 - \lambda_m) = \mu_a + \lambda_m (\mu_m - \mu_a)$. Therefore $y_m = \ell_m = \frac{\mu_m}{\Gamma} \lambda_m$. Moving forward, normalize by setting $w = 1$.

Case 1: No distortions ($\mu_m = \mu_a = 1$) and no transfers ($\tau_h = 0$)

In the frictionless benchmark, prices equal marginal costs, and the decentralized allocation coincides with the planner's allocation. The expenditure distribution is given by $\chi_h = \chi_l = \frac{1}{2}$ by symmetry and the absence of transfers. Then, $\lambda_m = \frac{\beta_h + \beta_l}{2}$, $\Gamma = 1$, $y_m = \frac{\beta_h + \beta_l}{2}$, $w = 1$, and $p_m = p_a = 2$. Because $y_m + y_a = 1$, it follows that $P_Y Y = 2$, and hence $\Gamma = 1$. Hence, the consumption allocations are $C_{hm} = \frac{\beta_h}{2}$, $C_{ha} = \frac{1-\beta_h}{2}$, $C_{lm} = \frac{\beta_l}{2}$, and $C_{la} = \frac{1-\beta_l}{2}$. This implies that $C_h = \frac{1}{2} \beta_h^{\beta_h} (1 - \beta_h)^{1-\beta_h}$ and $C_l = \frac{1}{2} \beta_l^{\beta_l} (1 - \beta_l)^{1-\beta_l}$. Define the aggregator Q_Y as $Y = (C_h C_l)^{1/2}$, which is consistent with the assumption $\frac{d \log Y}{d \log C_h} = \chi_h$. Then, real GDP is given by $Y_{eff} = \frac{1}{2} \beta_h^{\frac{\beta_h}{2}} (1 - \beta_h)^{\frac{1-\beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1 - \beta_l)^{\frac{1-\beta_l}{2}}$.

Case 2: Distortions and no transfers ($\tau_h = 0$)

The expenditure distribution is given by $\chi_h = \chi_l = \frac{1}{2}$ by symmetry and zero transfers. Let $\bar{\beta} = \frac{\beta_h + \beta_l}{2} \in [0, 1]$. Then $\lambda_m = \bar{\beta}$, $p_m = \frac{2}{\mu_m}$, $p_a = \frac{2}{\mu_a}$, $w = 1$, $\Gamma = \bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a$, $y_m = \ell_m = \frac{\bar{\beta} \mu_m}{\Gamma}$, $y_a = 1 - \ell_m = \frac{(1 - \bar{\beta}) \mu_a}{\Gamma}$, and $P_Y Y = \frac{2}{\Gamma}$. These imply the following real consumption allocations $C_{hm} = \frac{\mu_m \beta_h}{2\Gamma}$, $C_{lm} = \frac{\mu_m \beta_l}{2\Gamma}$, $C_{ha} = \frac{\mu_a (1 - \beta_h)}{2\Gamma}$, $C_{la} = \frac{\mu_a (1 - \beta_l)}{2\Gamma}$. Then $C_h = \frac{(\mu_m \beta_h)^{\beta_h} (\mu_a (1 - \beta_h))^{1-\beta_h}}{2\Gamma}$ and $C_l = \frac{(\mu_m \beta_l)^{\beta_l} (\mu_a (1 - \beta_l))^{1-\beta_l}}{2\Gamma}$. Using the same aggregator provides $Y = \frac{\mu_m^{\bar{\beta}} \mu_a^{1-\bar{\beta}} \beta_h^{\frac{\beta_h}{2}} (1 - \beta_h)^{\frac{1-\beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1 - \beta_l)^{\frac{1-\beta_l}{2}}}{2\Gamma}$. The TFP wedge relative to the efficient economy is given by $\frac{Y}{Y_{eff}} = \frac{\mu_m^{\bar{\beta}} \mu_a^{1-\bar{\beta}}}{\bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a}$.

It remains to show that $Y_{\mu, \tau=0} \leq Y_{\mu=1, \tau=0}$. When $\mu_m = \mu_a$, then $Y_{\mu, \tau=0} = Y_{\mu=1, \tau=0}$. For $\mu_m \neq \mu_a$, we need $\mu_m^{\bar{\beta}} \mu_a^{1-\bar{\beta}} \leq \bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a$ which is satisfied by definition.

Case 3: Efficient labor allocation via lump-sum transfers

The expenditure distribution is $\chi_h = \frac{1}{2} + \tau_h$ and $\chi_l = \frac{1}{2} - \tau_h$. Hence, $\lambda_m = \bar{\beta} + (\beta_h - \beta_l) \tau_h$, $p_m = \frac{2}{\mu_m}$, $p_a = \frac{2}{\mu_a}$, $w = 1$, and $\Gamma = \bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a + (\beta_h - \beta_l) (\mu_m - \mu_a) \tau_h$. Firm-level output $y_m = \ell_m = \frac{\mu_m (\bar{\beta} + (\beta_h - \beta_l) \tau_h)}{\bar{\beta} \mu_m + (1 - \bar{\beta}) \mu_a + (\beta_h - \beta_l) (\mu_m - \mu_a) \tau_h}$. The transfer τ_h that restores the undistorted labor

allocation ($y_m = \bar{\beta}$) is given by $\frac{\mu_m(\bar{\beta} + (\beta_h - \beta_l)\tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a + (\beta_h - \beta_l)(\mu_m - \mu_a)\tau_h} = \bar{\beta} \rightarrow \tau_h = \tilde{\tau} = \frac{(\mu_a - \mu_m)\bar{\beta}(1 - \bar{\beta})}{(\beta_h - \beta_l)(\bar{\beta}\mu_a + (1 - \bar{\beta})\mu_m)}$. Therefore, $\tilde{\tau} > 0$ when $\frac{\mu_a - \mu_m}{\beta_h - \beta_l} > 0$. Feasibility requires $|\tilde{\tau}| < \frac{1}{2}$. Consequently $\Gamma = \frac{\mu_m\mu_a}{\aleph}$, $\lambda_m = \frac{\bar{\beta}\mu_a}{\aleph}$, $GDP = \frac{2\aleph}{\mu_m\mu_a}$, $C_{hm} = \aleph \frac{\beta_h}{\mu_a} (\frac{1}{2} + \tilde{\tau})$, $C_{lm} = \aleph \frac{\beta_l}{\mu_a} (\frac{1}{2} - \tilde{\tau})$, $C_{am} = \aleph \frac{1 - \beta_h}{\mu_m} (\frac{1}{2} + \tilde{\tau})$, $C_{al} = \aleph \frac{1 - \beta_l}{\mu_m} (\frac{1}{2} - \tilde{\tau})$, where $\aleph = \bar{\beta}\mu_a + (1 - \bar{\beta})\mu_m$. Hence $C_h = \aleph \left(\frac{\beta_h}{\mu_a}\right)^{\beta_h} \left(\frac{1 - \beta_h}{\mu_m}\right)^{1 - \beta_h} (\frac{1}{2} + \tilde{\tau})$, $C_l = \aleph \left(\frac{\beta_l}{\mu_a}\right)^{\beta_l} \left(\frac{1 - \beta_l}{\mu_m}\right)^{1 - \beta_l} (\frac{1}{2} - \tilde{\tau})$, and $Y = (C_h C_l)^{1/2} = \aleph \frac{\beta_h^{\frac{\beta_h}{2}} (1 - \beta_h)^{\frac{1 - \beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1 - \beta_l)^{\frac{1 - \beta_l}{2}}}{\mu_a^{\bar{\beta}} \mu_m^{1 - \bar{\beta}}} \sqrt{\frac{1}{4} - \tilde{\tau}^2}$.

Case 4: Efficient lump-sum transfers

We have that $y_m = \frac{\mu_m(\bar{\beta} + (\beta_h - \beta_l)\tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}$, $y_a = \frac{\mu_a(1 - \bar{\beta} - (\beta_h - \beta_l)\tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}$, and $GDP = \frac{2}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}$. Consumption allocations are given by

$$\begin{aligned} C_{hm} &= \frac{\beta_h \mu_m (\frac{1}{2} + \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}, \\ C_{lm} &= \frac{\beta_l \mu_m (\frac{1}{2} - \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}, \\ C_{ha} &= \frac{(1 - \beta_h) \mu_m (\frac{1}{2} + \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}, \\ C_{la} &= \frac{(1 - \beta_l) \mu_m (\frac{1}{2} - \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}, \\ C_h &= \frac{(\beta_h \mu_m)^{\beta_h} ((1 - \beta_h) \mu_a)^{1 - \beta_h} (\frac{1}{2} + \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}, \\ C_l &= \frac{(\beta_l \mu_m)^{\beta_l} ((1 - \beta_l) \mu_a)^{1 - \beta_l} (\frac{1}{2} - \tau_h)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}. \end{aligned}$$

Hence $Y = \frac{\beta_h^{\frac{\beta_h}{2}} (1 - \beta_h)^{\frac{1 - \beta_h}{2}} \beta_l^{\frac{\beta_l}{2}} (1 - \beta_l)^{\frac{1 - \beta_l}{2}} \mu_a^{\bar{\beta}} \mu_m^{1 - \bar{\beta}} \sqrt{\frac{1}{4} - \tau_h^2}}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a - (\beta_h - \beta_l)(\mu_a - \mu_m)\tau_h}$. Then $\tau_h^* = \frac{1}{4} \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\bar{\beta}\mu_m + (1 - \bar{\beta})\mu_a}$.

C Nonlinear Effects using Differentiated Hat Algebra

Nonlinear effects are handled with differentiated hat algebra. Rather than solving the nonlinear fixed point for the finite shock (exact hat algebra), I trace the adjustment path by applying the shock in small increments. After each increment, I (i) update Jacobians at the new equilibrium, (ii) solve a linear system for first-order changes in endogenous variables, and (iii) update and renormalize shares.

Notation:

- The algorithm allows distinct index sets for firms/goods $i \in \{1, \dots, N\}$; households $h \in \{1, \dots, H\}$; and factors $f \in \{1, \dots, F\}$.
- Structural objects (updated across iterations): final demand shares $\beta \in [0, 1]^{H \times N}$; intermediate and factor cost shares $\tilde{\Omega}_\ell \in [0, 1]^{N \times F}$, $\tilde{\Omega}_x \in [0, 1]^{N \times N}$, $\alpha \in [0, 1]^{N \times F}$, $W \in [0, 1]^{N \times N}$, $\omega_\ell \in [0, 1]^N$, and $\omega_x = \mathbb{1}_N - \omega_\ell$; inverse markups $\mu \in \mathbb{R}_+^N$, equity ownership $\kappa \in [0, 1]^{N \times H}$, and lump-sum transfers $\tau \in [-1, 1]^H$.
- Elasticities and parameters (held fixed when computing Jacobians): $\theta, \theta_\ell, \theta_x, \varrho$.
- Endogenous objects: $\lambda \in \mathbb{R}^N$, $\tilde{\lambda} \in \mathbb{R}^N$, $\Lambda \in \mathbb{R}^F$, $\tilde{\Lambda} \in \mathbb{R}^F$, and $\chi \in \mathbb{R}^H$.
- Shocks: $d \log A$ (productivity), $d \log \mu$ (markup variation), $d T$ (transfers), $d \beta$ (preferences), $d \kappa$ (equity ownership), $d \omega_\ell$, $d \alpha$, $d W$ (technology).

Numerical Steps:

- **Shock:** To apply a finite shock, step along a path with maximum step size u (e.g., 1%). Let the remaining shock vector at iteration k be s_k . The algorithm computes the max shock $m_k \equiv \|s_k\|_\infty$. Set the step size $\eta_k \equiv \min\{1, u/m_k\}$, take the increment $\Delta s_k \equiv \eta_k s_k$, and update $s_{k+1} \equiv s_k - \Delta s_k$.
- **Equilibrium:** Solve for $\lambda \in \mathbb{R}^N$, $\tilde{\lambda} \in \mathbb{R}^N$, $\Lambda \in \mathbb{R}^F$, $\tilde{\Lambda} \in \mathbb{R}^F$, and $\chi \in \mathbb{R}^H$ using $\lambda = \mathcal{B}' \chi$, $\Lambda = \mathcal{C}' \chi$, $\tilde{\lambda} = \tilde{\mathcal{B}}' \chi$, $\tilde{\Lambda} = \tilde{\mathcal{C}}' \chi$, and $\chi = (\Omega_\ell + \Omega_\pi)' \lambda + \tau$. Replace one equation by imposing $\mathbb{1}_H' \chi = 1$. If needed, use ridge-regularized linear solves (Tikhonov) rather than explicitly inverting $I_N - \Omega_x$ or $I_N - \tilde{\Omega}_x$.
- **First-order approximation:** For the corresponding structure and s_k , assemble the Jacobian

linear system $\begin{pmatrix} S_S & S_E & S_w \\ E_S & E_E & E_w \\ w_S & w_E & w_w \end{pmatrix} \begin{pmatrix} d \log S_k \\ d \log E_k \\ d \log w_k \end{pmatrix} = \begin{pmatrix} S_{0,k} \\ E_{0,k} \\ w_{0,k} \end{pmatrix}$ with

$$\begin{aligned}
S_S &= \text{diag}(\lambda), & S_E &= -\mathcal{B}' \text{diag}(\chi), & E_S &= -\Omega'_\pi \text{diag}(\lambda), & E_E &= \text{diag}(\chi), \\
E_w &= -\text{diag} \left(\begin{smallmatrix} \Lambda \\ 0_{H-F} \end{smallmatrix} \right), & E_{0,k} &= d \kappa'_k \text{diag}(\mathbb{1}_N - \mu) \lambda - \kappa' \text{diag}(\mu) \text{diag}(\lambda) d \log \mu_k + d \tau_k, \\
w_S &= -\Omega'_\ell \text{diag}(\lambda), & w_E &= -0_H, & w_w &= \text{diag}(\Lambda) + \text{diag}(\Omega'_\ell \text{diag}(\theta_\ell - \mathbb{1}_N) \lambda) \\
&\quad - \Omega'_\ell \text{diag}(\Lambda) \text{diag}(\theta - \mathbb{1}_N) \tilde{\Psi}_\ell - \Omega'_\ell \text{diag}(\Lambda) \text{diag}(\theta_\ell - \theta) \alpha, \\
w_{0,k} &= \Omega'_\ell \text{diag}(\lambda) \text{diag}(\theta - \mathbb{1}_N) (I_N - \tilde{\Psi}_x) d \log A_k + (\text{diag}(\mu) \alpha)' \text{diag}(\lambda) \text{diag}(\theta) d \omega_{k,\ell} \\
&\quad + \Omega'_\ell \text{diag}(\lambda) (\text{diag}(\theta) - \text{diag}(\theta - \mathbb{1}_N) \tilde{\Psi}_\ell) d \log \mu_k + (\text{diag}(\mu) \text{diag}(\omega_\ell) d \alpha_k)' \text{diag}(\Omega_\ell) \lambda \\
S_w &= -\Psi'_x (\beta' \text{diag}(\chi) \text{diag}(\rho - \mathbb{1}_H) \mathcal{C} - \text{diag}(\beta' \text{diag}(\rho - \mathbb{1}_H) \chi) \tilde{\Psi}_\ell) \\
&\quad - \Psi'_x (\Omega'_x \text{diag}(\lambda) \text{diag}(\theta - \mathbb{1}_N) - \text{diag}(\Omega'_x \text{diag}(\theta - \mathbb{1}_N) \lambda)) \tilde{\Psi}_\ell \\
&\quad - \Psi'_x (\Omega'_x \text{diag}(\lambda) \text{diag}(\theta_x - \theta) W - \text{diag}(\Omega'_x \text{diag}(\theta_x - \theta) \lambda)) \tilde{\Psi}_\ell, \\
S_{0,k} &= \Psi'_x \Omega'_x \text{diag}(\lambda) (\text{diag}(\theta - \mathbb{1}_N) d \log A_k + \text{diag}(\theta) d \log \mu_k) \\
&\quad - \Psi'_x (\beta' \text{diag}(\chi) \text{diag}(\rho - \mathbb{1}_H) \mathcal{B} - \text{diag}(\beta' \text{diag}(\rho - \mathbb{1}_H) \chi) \tilde{\Psi}_x) (d \log A_k + d \log \mu_k) \\
&\quad - \Psi'_x (\Omega'_x \text{diag}(\lambda) \text{diag}(\theta - \mathbb{1}_N) - \text{diag}(\Omega'_x \text{diag}(\theta - \mathbb{1}_N) \lambda)) \tilde{\Psi}_x (d \log A_k + d \log \mu_k) \\
&\quad - \Psi'_x (\Omega'_x \text{diag}(\lambda) \text{diag}(\theta_x - \theta) W - \text{diag}(\Omega'_x \text{diag}(\theta_x - \theta) \lambda)) \tilde{\Psi}_x (d \log A_k + d \log \mu_k) \\
&\quad + \Psi'_x d \beta'_k \text{diag}(\rho) \chi - \Psi'_x W' \text{diag}(\mu) \text{diag}(\lambda) \text{diag}(\theta) d \omega_{k,\ell} \\
&\quad + \Psi'_x d W'_k \text{diag}(\omega_x) \text{diag}(\mu) \text{diag}(\theta_x) \lambda.
\end{aligned}$$

Replace the last row with $\tilde{\Lambda}' d \log w_k = \tilde{\lambda}' (d \log A + d \log \mu)$ to impose the normalization $d \log p_Y = 0$. At step k , solve for $d \log S_k$, $d \log E_k$, and $d \log w_k$.

- **Real GDP in step k :** From Theorem 2 $d \log Y_k = \tilde{\lambda}' d \log A_k + (\tilde{\lambda} - \text{diag}(\mu) \lambda)' d \log \mu_k - ((\delta - \mathbb{1}_F) \circ \Lambda)' d \log w_k + (\lambda \circ (\mathbb{1}_N - \mu))' d \log S_k$.
- **Distributional updates:** $d \log \lambda_k = d \log S_k - d \log Y_k$, $d \log \Lambda_k = d \log J_k - d \log Y_k$, $d \log \chi_k = d \log E_k - d \log Y_k$.
- **Price variation in k :** $d \log p_k = \tilde{\Psi}_\ell d \log w_k - \tilde{\Psi}_x (d \log A_k + d \log \mu_k)$, $d \log p_k^\ell = \alpha d \log w_k$, $d \log p_k^x = W d \log p_k$, $d \log p_k^c = \tilde{\mathcal{C}} d \log w_k - \tilde{\mathcal{B}} (d \log A_k + d \log \mu_k)$.
- **Updating:**

$$\begin{aligned}
d \log \tilde{\Omega}_{k,\ell} &= \text{diag}(\theta - \mathbb{1}_N) (d \log A_k + d \log \mu_k + d \log p_k - d \log p_k^\ell) \mathbb{1}'_F \\
&\quad + \text{diag}(\theta_\ell - \mathbb{1}_N) (d \log p_k^\ell \mathbb{1}'_F - \mathbb{1}_N d \log w'_k) + \tilde{\Omega}_\ell^{\circ-1} (\text{diag}(\omega_\ell) d \alpha_k + \text{diag}(d \omega_{k,\ell}) \alpha),
\end{aligned}$$

$$\begin{aligned}
d \log \tilde{\Omega}_{k,x} &= \text{diag}(\theta - \mathbb{1}_N) (d \log A_k + d \log \mu_k + d \log p_k - d \log p_k^x) \mathbb{1}'_N \\
&\quad + \text{diag}(\theta_x - \mathbb{1}_N) (d \log p_k^x \mathbb{1}'_N - \mathbb{1}_N d \log p'_k) + \tilde{\Omega}_x^{\circ-1} (\text{diag}(\mathbb{1}_N - \omega_\ell) d W_k - \text{diag}(d \omega_{k,\ell}) W),
\end{aligned}$$

$$d \log \beta_k = \text{diag}(\rho - \mathbb{1}_H) (d \log p_k^c \mathbb{1}'_N - \mathbb{1}_H d \log p'_k) + \beta^{\circ-1} d \beta_k.$$

Using these variations, update shares by the multiplicative rule and renormalize with $\text{Renorm}_1(\cdot)$ to keep shares in $[0, 1]$ and summing to one; updates apply only to active links: $\chi^* = \text{Renorm}_1(\chi + \chi \circ d \log \chi_k)$, $d \chi_k = \chi^* - \chi$, $\beta^* = \text{Renorm}_1(\beta + \beta \circ d \log \beta_k)$, $d \beta_k = \beta^* - \beta$, $\tilde{\Omega}_\ell^* = \tilde{\Omega}_\ell + \tilde{\Omega}_\ell \circ d \log \tilde{\Omega}_{k,\ell}$, and $\tilde{\Omega}_x^* = \tilde{\Omega}_x + \tilde{\Omega}_x \circ d \log \tilde{\Omega}_{k,x}$.

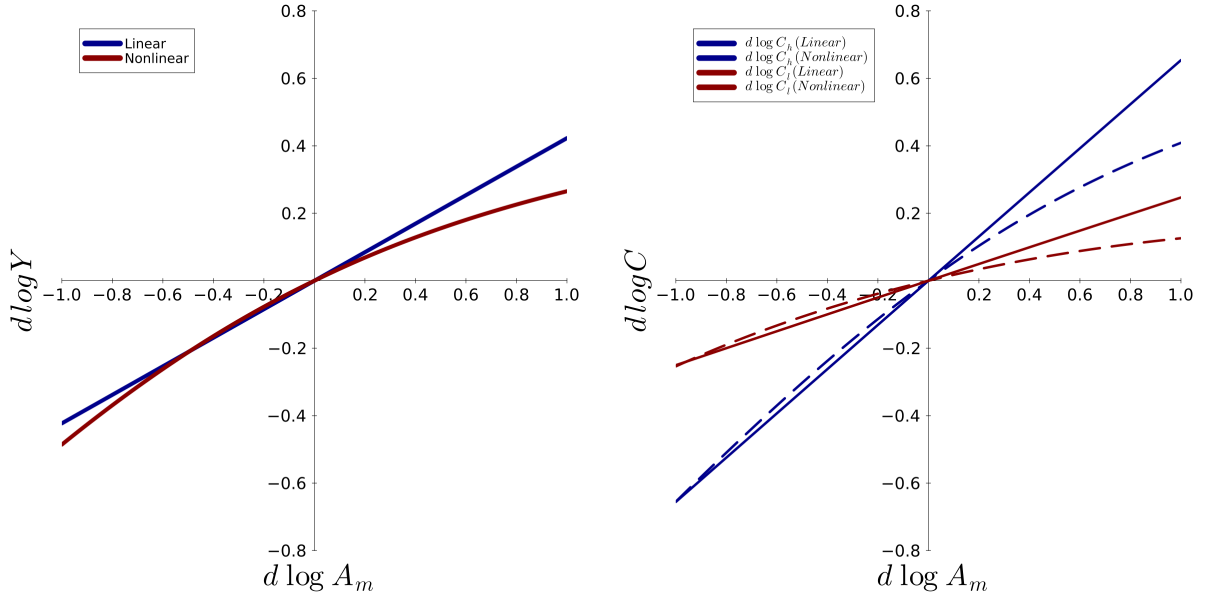
Normalize $\tilde{\Omega}_\ell$ and $\tilde{\Omega}_x$ by $\aleph = \tilde{\Omega}_\ell^* \mathbb{1}_F + \tilde{\Omega}_x^* \mathbb{1}_N$. Set $\tilde{\Omega}_\ell^* \leftarrow \text{diag}(\aleph^{\circ-1}) \tilde{\Omega}_\ell^*$ and $\tilde{\Omega}_x^* \leftarrow \text{diag}(\aleph^{\circ-1}) \tilde{\Omega}_x^*$, which guarantess CRS in cost shares. Then $\omega_\ell^* = \text{Clamp}_{[0,1]}(\tilde{\Omega}_\ell^* \mathbb{1}_F)$,

$$\omega_x^* = \mathbb{1}_N - \omega_\ell^*, \alpha_i^* = \begin{cases} \tilde{\Omega}_{\ell,i}^* / \omega_{\ell,i}^* & \omega_{\ell,i}^* > 0 \\ 0 & \text{otherwise} \end{cases}, \text{ and } W_i^* = \begin{cases} \tilde{\Omega}_{x,i}^* / \omega_{x,i}^* & \omega_{x,i}^* > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Finally, row-renormalize α and W to sum to one. Use the updated objects to compute $d \tilde{\Omega}_{k,\ell} = \tilde{\Omega}_\ell^* - \tilde{\Omega}_\ell$ and $d \tilde{\Omega}_{k,x} = \tilde{\Omega}_x^* - \tilde{\Omega}_x$.

- **Accumulate variations from step k :** For any variable estimates in stage k , total cumulated variation is given by $d \log X = \sum_{s=1}^k d \log X_s$.
- **Update:** $\mu \leftarrow \text{clamp}_{[0,1]}(\mu + \mu \circ d \log \mu_k)$, $\kappa_i \leftarrow \text{Renorm}_1(\kappa_i + d \kappa_i)$, and $T \leftarrow T + d T$.
- **Convergence:** stop when $|d \log \lambda_k| \leq \text{tol}$, $|d \log \Lambda_k| \leq \text{tol}$, $|d \log \chi_k| \leq \text{tol}$, and $m_k = 0$. Equivalently, stop when updates are $\leq \text{tol}$ and shocks are exhausted.

Figure 1: Linear vs. Nonlinear responses to a productivity shock



Note: Left panel plots aggregate output changes $d \log Y$ for the horizontal economy in Figure ?? in response to a manufacturing productivity shock $d \log A_m \in [-1, 1]$ under $(\beta_h, \beta_l) = (0.8, 0.3)$, $(\mu_m, \mu_a) = (0.2, 0.7)$, $\alpha_{mh} = \alpha_{al} = 0.8$, $\alpha_{ml} = \alpha_{ah} = 0.2$, $\omega_m^\ell = \omega_a^\ell = 1$, $\tau_h = \tau_l = 0$, $\kappa_{ih} = 0.5 \forall i \in \mathcal{N}, h \in \mathcal{H}$, $\theta_m = \theta_a = 0.5$, and $\rho_h = \rho_l = 0.5$. Right panel plots idiosyncratic consumption variation for households h and l in the same economy. *Linear* curve applies a single first-order approximation at the baseline; *nonlinear* curves aggregate incremental responses via differentiated hat algebra, updating cost and expenditure shares each step.

Figure 1 compares baseline first-order ("linear") predictions with the path-integrated via differentiated hat algebra for the horizontal economy in Example 1 in response to productiv-

ity shocks to manufacturing. The curves coincide for small shocks — reflecting first-order equivalence — but diverge as shock magnitudes increase. Along the nonlinear path, prices, Domar weight, and both cost and expenditure shares are re-updated each step, which introduces general equilibrium feedback that systematically modify the impact on aggregate output $d \log Y$ (left panel) relative to a single baseline linearization. The same effect is observed in household consumption, as shown by $d \log C_h$ and $d \log C_l$ (right panel), with differences across households arising from preference and income heterogeneity. Overall, the figure shows that nonlinear estimations bend responses away from linear predictions, so large shocks require the differentiated hat-algebra treatment to capture nonlinear general equilibrium effects.

D Data

D.1 Input Output Tables

These steps are implemented in the Julia script `DataCleanIO.jl`.

Capital Compensation. I use BEA’s Industry-Level Production Accounts (ILPA) and Extended Capital Detail from the BEA–BLS integrated accounts. **1. Capital asset typology.** Harmonize nine capital asset classes (i) communications equipment, (ii) computer hardware, (iii) R&D, (iv) software, (v) entertainment originals, (vi) instruments and other office equipment, (vii) structures, land, and inventories, (viii) transportation equipment, and (ix) other equipment. **2. Capital series harmonization.** For each asset class, import quantities and compensation, and construct a balanced asset-industry-year panel (1997-2021). Observations are mapped to $(year, NAICS)$.

Use and Make Input Output Tables. Use the BEA USE and MAKE tables (1997-2021). **1. Table cleaning.** Drop government and scrap/used categories – federal (defense, nondefense, enterprises), state/local (general government, enterprises), and scrap/used goods – leaving 66 private industries. Use is $industry \times commodity$ and Make is $commodity \times industry$. **2. Normalization.** Row-normalize Use and Make to form WUSE and WMAKE (rows sum to 1), and set the industry IO matrix to $W = WUSE, WMAKE$. **3. Industry accounts and capital.** For each industry-year, construct Sales (T018), Value Added (VABAS), Labor Cost (V001), and Intermediate Cost (column sums) from Use, and merge Capital Cost and Capital Quantity by asset from ILPA. For three ILPA aggregates split in Use/Make, allocate capital to subindustries as described below. (1) Retail trade (44RT) is split into 441, 445, 452, and 4A0. (2) Real estate (531) is split into HR and ORE. (3) 622H0 is split into 622 and 623. For these splits, allocate capital proportionally to Value Added–Labor Cost. For example, for HR, ORE allocate any capital series $K \in \{Cost, Quantity\}$ as $K(HR) = \frac{VA(HR) - LC(HR)}{\sum_{g \in \{HR, ORE\}} (VA(g) - LC(g))} K(531)$, and analogously for ORE. $Cost = Labor + Intermediates + \sum_{a=1}^9 Capital_a$. Compute Intermediate Sales as $IS = \min\{WMAKE'(USE'1), Sales\}$. Define Final Sales as $FS = Sales - IS$.

Inverse markups are computed as $\mu = Cost./Sales$. This procedure does not impose $\mu \leq 1$; values $\mu > 1$ indicate operating losses for the sector. The following is a list of the combinations of sectors, years, and inverse markups for which operational losses are estimated. **1999:** *Data processing, internet publishing, and other information services* (514) $\mu = 1.041$. **2000:** (i) *Data processing, internet publishing, and other information services* (514) $\mu = 1.256$; (ii) *securities, commodity contracts, and investment* (523) $\mu = 1.009$; and (iii) *computer systems design and related activities* (5415) $\mu = 1.032$. **2001:** (i) *Data processing, internet publishing, and other information*

services (514) $\mu = 1.035$; and (ii) computer systems design and related activities (5415) $\mu = 1.003$.

2008: Securities, commodity contracts, and investment (523) $\mu = 1.222$. **2009:** Motor vehicles, bodies and trailers, and parts (3361MV) $\mu = 1.027$. **2011:** Securities, commodity contracts, and investment (523) $\mu = 1.025$. **2013:** (i) Support activities for mining (213) $\mu = 1.029$; and (ii) rail transportation (482) $\mu = 1.005$. **2014:** (i) Support activities for mining (213) $\mu = 1.069$; (ii) motor vehicle and part dealers (441) $\mu = 1.001$, (iii) rail transportation (482) $\mu = 1.012$; and (iv) publishing industries, except internet (includes software) (511) $\mu = 1.004$. **2015:** (i) Support activities for mining (213) $\mu = 1.057$; (ii) motor vehicle and part dealers (441) $\mu = 1.01$; (iii) food and beverage stores (445) $\mu = 1.007$; (iv) general merchandise stores (452) $\mu = 1.001$; (v) other retail (4A0) $\mu = 1.008$; (vi) rail transportation (482) $\mu = 1.019$; (vii) publishing industries, except internet (includes software) (511) $\mu = 1.005$; (viii) hospitals (622) $\mu = 1.001$; and (ix) accommodation (721) $\mu = 1.006$. **2016:** (i) Support activities for mining (213) $\mu = 1.028$; (ii) food and beverage and tobacco products (311FT) $\mu = 1.001$; (iii) motor vehicle and part dealers (441) $\mu = 1.022$; (iv) food and beverage stores (445) $\mu = 1.019$; (v) general merchandise stores (452) $\mu = 1.007$; (vi) other retail (4A0) $\mu = 1.022$; (vii) rail transportation (482) $\mu = 1.026$; (viii) publishing industries, except internet (includes software) (511) $\mu = 1.017$; (ix) broadcasting and telecommunications (513) $\mu = 1.003$; (x) hospitals (622) $\mu = 1.003$; and (xi) accommodation (721) $\mu = 1.012$. **2017:** (i) Support activities for mining (213) $\mu = 1.083$; (ii) food and beverage and tobacco products (311FT) $\mu = 1.002$; (iii) motor vehicle and part dealers (441) $\mu = 1.034$; (iv) food and beverage stores (445) $\mu = 1.028$; (v) general merchandise stores (452) $\mu = 1.012$; (vi) other retail (4A0) $\mu = 1.034$; (vii) rail transportation (482) $\mu = 1.034$; (viii) publishing industries, except internet (includes software) (511) $\mu = 1.026$; (ix) broadcasting and telecommunications (513) $\mu = 1.006$; (x) insurance carriers and related activities (524) $\mu = 1.002$; (xi) hospitals (622) $\mu = 1.005$; (xii) nursing and residential care facilities (623) $\mu = 1.001$; and (xiii) accommodation (721) $\mu = 1.019$. **2018:** (i) Support activities for mining (213) $\mu = 1.138$; (ii) motor vehicles, bodies and trailers, and parts (3361MV) $\mu = 1.001$; (iii) chemical products (325) $\mu = 1.018$; (iv) motor vehicle and part dealers (441) $\mu = 1.034$; (v) food and beverage stores (445) $\mu = 1.028$; (vi) general merchandise stores (452) $\mu = 1.012$; (vii) other retail (4A0) $\mu = 1.035$; (viii) rail transportation (482) $\mu = 1.031$; (ix) publishing industries, except internet (includes software) (511) $\mu = 1.007$; (x) broadcasting and telecommunications (513) $\mu = 1.004$; (xi) housing (HS) $\mu = 1.01$; (xii) hospitals (622) $\mu = 1.004$; (xiii) nursing and residential care facilities (623) $\mu = 1.001$; and (xiv) accommodation (721) $\mu = 1.026$. **2019:** (i) Support activities for mining (213) $\mu = 1.117$; (ii) motor vehicles, bodies and trailers, and parts (3361MV) $\mu = 1.002$; (iii) plastics and rubber products (326) $\mu = 1.001$; (iv) motor vehicle and part dealers (441) $\mu = 1.037$; (v) food and beverage stores (445) $\mu = 1.028$; (vi) general merchandise stores (452) $\mu = 1.014$; (vii) other retail (4A0) $\mu = 1.037$; (viii) rail transportation (482) $\mu = 1.03$; (ix) publishing industries, except internet (includes software) (511) $\mu = 1.036$; (x) broadcasting and

telecommunications (513) $\mu = 1.017$; (xi) housing (HS) $\mu = 1.006$; (xii) hospitals (622) $\mu = 1.005$; (xii) nursing and residential care facilities (623) $\mu = 1.001$; and (xiii) accommodation (721) $\mu = 1.026$. **2020:** (i) Mining, except oil and gas (212) $\mu = 1.011$; (ii) support activities for mining (213) $\mu = 1.117$; (iii) apparel and leather and allied products (315AL) $\mu = 1.094$; (iv) printing and related support activities (323) $\mu = 1.006$; (v) plastics and rubber products (326) $\mu = 1.002$; (vi) wholesale trade (42) $\mu = 1.009$; (vii) motor vehicle and part dealers (441) $\mu = 1.064$; (viii) food and beverage stores (445) $\mu = 1.05$; (ix) general merchandise stores (452) $\mu = 1.011$; (x) other retail (4A0) $\mu = 1.062$; (xi) air transportation (481) $\mu = 1.042$; (xii) rail transportation (482) $\mu = 1.023$; (xiii) transit and ground and passenger transportation (484) $\mu = 1.005$; (xiv) other transportation and support activities (4870S) $\mu = 1.029$; (xv) publishing industries, except internet (includes software) (511) $\mu = 1.006$; (xvi) broadcasting and telecommunications (513) $\mu = 1.01$; (xvii) insurance carriers and related activities (524) $\mu = 1.002$; (xviii) housing (HS) $\mu = 1.023$; (xix) other real estate (ORE) $\mu = 1.005$; (xx) rental and leasing services and lessors of intangible assets (532RL) $\mu = 1.028$; (xxi) hospitals (622) $\mu = 1.015$; (xxii) nursing and residential care facilities (623) $\mu = 1.001$; (xxiii) accommodation (721) $\mu = 1.084$; and (xxiv) food services and drinking places (722) $\mu = 1.003$. **2021:** (i) Mining, except oil and gas (212) $\mu = 1.054$; (ii) support activities for mining (213) $\mu = 1.108$; (iii) apparel and leather and allied products (315AL) $\mu = 1.090$; (iv) printing and related support activities (323) $\mu = 1.015$; (v) petroleum and coal products (324) $\mu = 1.018$; (vi) chemical products (325) $\mu = 1.026$; (vii) motor vehicle and part dealers (441) $\mu = 1.067$; (viii) food and beverage stores (445) $\mu = 1.047$; (ix) general merchandise stores (452) $\mu = 1.021$; (x) other retail (4A0) $\mu = 1.059$; (xi) air transportation (481) $\mu = 1.068$; (xii) rail transportation (482) $\mu = 1.035$; (xiii) water transportation (483) $\mu = 1.073$; (xiv) transit and ground and passenger transportation (484) $\mu = 1.012$; (xv) other transportation and support activities (4870S) $\mu = 1.031$; (xvi) warehousing and storage (493) $\mu = 1.003$; (xvii) publishing industries, except internet (includes software) (511) $\mu = 1.024$; (xviii) broadcasting and telecommunications (513) $\mu = 1.0006$; (xix) housing (HS) $\mu = 1.049$; (xx) other real estate (ORE) $\mu = 1.014$; (xxi) rental and leasing services and lessors of intangible assets (532RL) $\mu = 1.059$; (xxii) waste management and remediation services (562) $\mu = 1.004$; (xxiii) hospitals (622) $\mu = 1.012$; (xxiv) nursing and residential care facilities (623) $\mu = 1.003$; (xxv) amusements, gambling, and recreation industries (713) $\mu = 1.006$; (xxvi) accommodation (721) $\mu = 1.053$; and (xxvii) food services and drinking places (722) $\mu = 1.016$.

D.2 Sectoral Productivities

The steps implemented are in the Julia script Data_Clean_TFP.jl. **1. Crosswalk.** ILPA/KLEMS NAICS sectors to the 66 Production Accounts industries.⁹ **2. Productivity growth.** For sector

⁹Agriculture/resources: 111-112 $\rightarrow$ 111CA, 113-115 $\rightarrow$ 113FF. Manufacturing: 311-312 $\rightarrow$ 311FT, 313-314 $\rightarrow$ 313TT, 315-316 $\rightarrow$ 315AL, 3361-3363 $\rightarrow$ 3361MV, 3364-3369 $\rightarrow$ 33640T. Retail: 44, 45 $\rightarrow$ { 441, 445, 452, 4A0}. Other

i in year $t \in \{1998, \dots, 2022\}$, use $d \log A_{i,t} = \log(TFP_{i,t} / TFP_{i,t-1})$ using ILPA sectoral TFP levels.

D.3 State Level Personal Consumer Expenditure Bundles

The Julia script `Data Clean PCE.jl` combines BEA state-level PCE tables (1997-2021) with the PCE Bridge to map NIPA categories into Production Accounts commodities. State-level expenditure-share vectors are constructed as follows. **1. Crosswalk weights.** For each year t and NIPA PCE line n , let PV_{cnt} be the PCE Bridge purchasers' value assigned to commodity c . Define $w_{c|n,t} \equiv PV_{cnt} / \sum_{c'} PV_{c'nt}$, so $\sum_c w_{c|n,t} = 1$. For example, in 1999, for *jewelry and watches* (NIPA code 20), the crosswalk vector is $w_{334|20,1999}, w_{339|20,1999}, w_{USED|20,1999} = (0.1119, 0.8637, 0.0244)$, i.e., 11.19% to *computer and electronic products* (334), 86.37% to *miscellaneous manufacturing* (339), and 2.44% to *used*. **2. Define expenditure grid.** Use 52 geographies g (50 states + DC + national), 42 non redundant NIPA lines n , and 70 commodity codes c (mapped to the 66 private Production Accounts industries, used, other, state and local enterprises, and federal government enterprises). **3. Commodity spending.** Let E_{gnt} be PCE spending in geography g on NIPA line n . Implied commodity spending is $X_{gct} = \sum_n w_{c|n,t} E_{gnt}$. **4. Shares.** Restrict to the 66 private industries and set $s_{gct} \equiv X_{gct} / \sum_{c' \in \mathcal{N}} X_{gc't}$, so $\sum_{c \in \mathcal{N}} s_{gct} = 1$.

D.4 Quarterly Census of Employment and Wages

The Julia script `Data Clean QCEW.jl` uses BLS QCEW data (1997–2021) to construct state-industry labor compensation. **1. NAICS selection and mapping.** Select the QCEW NAICS detail that can be mapped to the Production Accounts industries (105 NAICS codes), and crosswalk these codes to the Production Accounts classification. **2. Filtering and aggregation.** QCEW files are split by NAICS. For each year, I filter the selected NAICS files and aggregate to annual state-industry totals. **3. Outputs.** For each year t , state s , and industry i , compute annual average employment and total labor compensation, and store $(\text{emp}_{sit}, \text{comp}_{sit})$.

D.5 Consumer Expenditure Survey

The Julia script `Data Clean CEX.jl` maps CEX household expenditures to the 66 Production Accounts commodities for 1997-2021. The script uses the sample and filters in [Markevych \(2024\)](#).

1. NIPA-to-commodity bridge. Using the BEA PCE Bridge, construct year-specific weights mapping NIPA lines to the 66 Production Accounts commodities. **2. Income measure.** For

transport: 487,488,492 $\rightarrow$ 4870S. Broadcast/telecom: 515,517 $\rightarrow$ 513. Data processing/info: 518-519 $\rightarrow$ 514. Credit intermediation: 521-522 $\rightarrow$ 521CI. Real estate: 531 $\rightarrow$ { HS, ORE}. Rental and leasing maps 532-533 $\rightarrow$ 532RL. Other prof. services: 5412-5414,5416-5419 $\rightarrow$ 54120P. Hospitals/nursing: 622-623 $\rightarrow$ { 622, 623}. Arts/sports: 711-712 $\rightarrow$ 711AS.

each household-year, compute annual income by annualizing the mean monthly salary, and assign it to all records in that household-year. **3. Income neighborhoods.** Within each year, form 101-household income neighborhoods by selecting the 100 nearest-income households (plus the target household). **4. Group bundle construction.** Aggregate expenditures within each income neighborhood, map NIPA spending to commodities via the bridge weights, and sum to obtain commodity totals. **5. Normalization and storage.** Normalize commodity totals to a 66-share vector that sums to one, and store it with income, year, and occupation identifiers.

D.6 Occupational Employment Wage Statistics

The Julia script `Data_Clean_OEWS.jl` processes BLS OEWS data for 1997-2024, building employment and wage aggregates. **1. Data loading.** Load annual OEWS files for industry, state, national, and MSA tabulations (1997-2024), with SIC (1997-2001) and NAICS (2002-2024) industry codes. **2. Industry crosswalks.** Harmonize OEWS industries to Production Accounts by chaining SIC→NAICS and NAICS vintages (2002→2007→2012→2017→2022) using Census crosswalks. Then map Production Accounts industries to 4-digit NAICS (e.g., 3361MV → 3361,3362,3363). **3. Occupational crosswalks.** Harmonize occupational across OES (1997–1999) and SOC 2000/2010/2018 (2000–2024). allocate employment and wage bills using employment weights (plus residual weighting for many-to-many) to preserve occupation totals.¹⁰ **4. Aggregation.** For each year t , map industries and geographies to Production Accounts and occupation o to SOC 2018, then aggregate employment and wage bills.

D.7 Representative Household Specification

The Julia script `Representative_Household.jl` computes aggregate TFP growth (1997–2021) under eight specifications and compares them to BLS TFP growth using forecast metrics. Inputs are BEA USE/MAKE, ILPA sectoral TFP, and PCE. **1. GDP and shares.** Compute income-side GDP by summing labor, nine capital payments, and profits across industries; use GDP to form sector shares λ , factor shares Λ , and profit shares Π . **2. Markup variation.** Compute year-over-year $d \log \mu$ to measure changes in sectoral competitiveness. **3. Consumption bundles.**

¹⁰**SOC 2018 ↔ SOC 2010.** Three mapping types. **(a) One-to-one,** e.g., SOC 2018 *Chief Executive Officers* (11-1011) → SOC 2010 *Chief Executive Officers* (11-1011). **(b) Many-to-one:** multiple 2018 codes collapse to one 2010 code; e.g., SOC 2018 *Agents and Business Mgrs of Artists* (13-1011) and SOC 2018 *Agents and Business Mgrs of Athletes* (13-1012) → SOC 2010 *Agents and Business Managers of Artists, Performers, and Athletes* (13-1011). Allocate 2018 employment across the target 2010 code(s) using 2018 OEWS weights (e.g., 45,000 and 35,000 imply weights 0.5625 and 0.4375). **(c) Many-to-many:** a 2018 code maps to multiple 2010 codes (and vice versa); e.g., SOC 2018 *Administrative Services Managers* (11-3011) → SOC 2010 *Administrative Services Managers* (11-3011), *Facilities Managers* (11-3012), or *Transportation, Storage, Distribution Managers* (11-3013). For (c), apply a **residual weighting algorithm:** (i) classify mappings as “exclusive” (map to one target) or “ambiguous” (map to multiple targets); (ii) compute residual employment as total minus exclusive employment; (iii) allocate residual proportionally to 2018 employment in the target SOC 2010 codes (the last year SOC 2010 is observed). The SOC 2010 ↔ SOC 2000 and SOC 2000 ↔ OES crosswalks use the same procedure, yielding a SOC 2018 ↔ SOC 2010 ↔ SOC 2000 ↔ OES chain.

Construct β_{IO} from BEA final-demand shares and β_{PCE} from PCE commodity shares mapped to industries via WMAKE; vector is expressed in terms of industrial consumption shares. **4. Equilibrium.** Function `equilibrium()` solves for (λ, Λ, χ) from goods-, factor-, and budget-clearing, given $(\beta, \omega_\ell, \alpha, W, \mu, \kappa, T)$ (κ is a vector of ones and T a zero scalar), using the Leontief inverses Ψ_x and Ψ_ℓ . **5. Eight specifications.** Evaluate the $2 \times 2 \times 2$ designs: (labor+capital vs. labor-only) $\times$ (networks vs. no intermediates) $\times$ (β_{IO} vs. β_{PCE}). **6. TFP decomposition.** For each specification-year, decompose TFP growth into *technology*, *competitiveness*, and *distributional reallocation*; their sum is model TFP growth, which is then compared to observed BLS data. **7. Model validation and performance metrics.** The script evaluates each specification using eight standard forecast metrics (MAE, RMSE, bias, correlation, NSE, IA, Theil's U , KGE). **8. Variance decomposition.** For each component, compute its covariance share.

D.8 State-Level Specification

The Julia script `State_Level1.jl` extends the representative-household framework to state heterogeneity (50 states + D.C.). Unlike the representative-household model, the state specification fixes the best-performing case: multi-factor production with intermediates and state-specific consumption. This geographic disaggregation isolates how cross-state heterogeneity in labor income and consumption shapes aggregate outcomes. **1. QCEW labor heterogeneity.** Use QCEW employment and wage bills to parameterize state-specific labor types. For each industry-year i , compute state s labor-cost shares $\alpha_{is}^{QCEW} = \frac{\text{Labor Income}_{is}}{\sum_{s'} \text{Labor Income}_{is'}}$. This yields a 66×52 labor-share matrix. Labor types are 50 states + D.C. + a residual U.S. labor type that reconciles USE/MAKE with QCEW. Combine these labor shares with capital-cost shares to form α . **2. State consumption bundles.** Construct state PCE shares $\beta_{s,i}^{PCE}$. For the residual national worker, assign the national PCE bundle. Define the residual non-labor bundle to match aggregate final demand: $\beta^{NL} = (1 - \sum_s \Lambda_s)^{-1} (\beta^{IO} - \sum_s \Lambda_s \beta_s^{PCE})$. This captures non-labor consumption (capital/profits), enabling distributional decomposition beyond labor income. **3. Equilibrium.** Solve the same goods/budget/factor market conditions in the expanded state-industry-factor system. $\lambda \in \mathbb{R}^{66}$; $\Lambda \in \mathbb{R}^{61}$ (52 labor + 9 capital); $\chi \in \mathbb{R}^{62}$ (profits household included). Set $\kappa \in \mathbb{R}^{66 \times 62}$ with all profits assigned to the last household (last column ones). **4. TFP and PTT.** Compute $d \log TFP$, δ and M by state and household type.

D.9 Occupation-Level Specification

The Julia script `Occupation_Level1.jl` combines OEWS industry-by-occupation employment and wages with income-conditioned CEX bundles and solves the network equilibrium with an occupation-augmented factor block (833 occupations + 9 capital types) to produce λ , Λ ,

χ , and occupation-level TFP/PTT decompositions. **1. Labor shares (OEWS).** Using OEWS industry-by-occupation employment and wages, compute industry i labor shares by occupation o : $\alpha_{o,i}^{OEWS} = \frac{\text{Labor Income}_{o,i}}{\sum_o \text{Labor Income}_{o,i}}$. This yields $\alpha^{OEWS} \in [0, 1]^{66 \times 833}$, which is combined with capital shares to form the full factor-share matrix α . **2. Cleaning and crosswalks.** Drop 34 non-private occupations (public administration, military, and private services) and impute missing occupation-year cells via forward/backward fills (e.g., use 2004 values for 1997–2003 when earlier data is unavailable). **3. Occupation bundles (CEX).** For each (o, t) , map mean OEWS earnings to income-conditioned CEX expenditure-share vector via nearest-income bin to obtain $\beta_{o,t}^{CE}$. This yields occupation-specific consumption bundles aligned with empirically observed income-conditioned CEX shares. **4. Equilibrium.** The extended equilibrium() solves goods-, budget-, and factor-market clearing with $\lambda \in \mathbb{R}^{66}$, $\Lambda \in \mathbb{R}^{842}$ (833 occupations + 9 capital types), and $\chi \in \mathbb{R}^{843}$ (plus a profit-receiving household). Set $\kappa \in 0, 1^{66 \times 843}$ with only the last column equal to 1. **5. Outputs.** Compute $d \log TFP$, $d \log PTT$, δ and M series.

D.10 Generalized Method of Moments (GMM) Estimation: Global and Local Stages

Parameterization.jl estimates four substitution elasticities in a nested CES GE model via two-stage GMM. This section documents bootstrap and optimization steps for replication.

1. Parameters and moments. Elasticities: (i) factor-intermediate θ ; (ii) within-factor θ_ℓ ; (iii) within-intermediate θ_x ; (iv) consumption ρ . Stack in $\theta = (\theta, \theta_\ell, \theta_x, \rho)$. **2. Bounds.** Constrain $\theta \in [0, 3]^4$. **3. Moments.** Target BLS TFP growth with $g_t(\theta) = d \log TFP_t^{model}(\theta) - d \log TFP_t^{BLS}$, $t \in [1997, 2000]$. Stack $g(\theta) = [g_{1997}(\theta), g_{1998}(\theta), \dots, g_{2019}(\theta), g_{2020}(\theta)]' \in \mathbb{R}^{24}$. **4. Objective.** $Q(\theta) = g(\theta)' W g(\theta)$, set $W = I$. Stage 1 is global search and stage 2 local refinement.

5. Stage 1 - Global search (BlackBoxOptim). The GMM can be nonconvex due to: (i) Leontief inverses Ψ_x and Ψ_ℓ ; (ii) nonlinear parameter interaction (even under linearization); and (iii) network structure can generate multiple local minima and saddle points.

The code uses BlackBoxOptim's **adaptive differential evolution (DE)** to solve $\min_\theta Q(\theta)$ subject to $\theta \in [0, R_{\max}]^4$. **Algorithm: Input.** $Q(\theta)$, population size $N_p = 50$, and max steps $S_{\max} = 100$. **Initialize population.** Draw $\{\theta^i\}_{i=1}^{N_p}$ i.i.d. uniform on $[0, R_{\max}]^4$. **Fitness.** $f_i = Q(\theta^i)$. **Best.** $\theta^* = \arg \min_i f_i$. **Global loop.** For $s = 1, \dots, S_{\max}$ and each i draw $r_1, r_2, r_3 \in \{1, \dots, N_p\} \setminus \{i\}$; create a mutant $u^i = \theta^{r_1} + F(s)(\theta^{r_2} - \theta^{r_3})$ where $F(s)$ is the adaptive scaling factor; constraint to bounds $u^i \leftarrow \max\{0, \min\{u^i, R_{\max}\}\}$; crossover with probability $CR(s, i)$ to get trial $v^i = u^i$ with probability $CR(s, i)$ and $v^i = \theta^i$ with probability $1 - CR(s, i)$; if $f_v < f_i$, then set $\theta^i \leftarrow v^i$, $f_i \leftarrow f_v$, and update θ^* . **Output.** θ^* with corresponding f^* .

6. Stage 2 - Local Refinement with LBFGS. Solve $\min_{\theta \in [0, R_{\max}]^4} Q(\theta)$ by L-BFGS-B with automatic differentiation (AD) gradients. **Input.** $Q(\theta)$, starting with θ^* , bounds $[0, R_{\max}]^4$.

Gradient. $\nabla Q(\theta)$ with AD using `ForwardDiff.jl`, where $\partial Q / \partial \theta_j = 2 \sum_t g_t(\theta) \times \partial g_t(\theta) / \partial \theta_j$.

Settings: memory $M = 10$ (stores 10 most recent steps). For iteration $k = 0$ to $K_{\max} = 1000$.

Stop: when $|\nabla Q(\theta_k)|_{\infty} < \varepsilon$. LBFGS **search direction** d_k can be computed using *gradient descent* with $d_k^{GD} = -\nabla Q(\theta_k)$; however, convergence is slow when the objective function is elongated or curved. For this reason, *Newton's method* implements $d_k^{Newton} = -H_k^{-1} \nabla Q(\theta_k)$, where $H_k = \nabla^2 Q(\theta_k)$. H_k is approximated with $\hat{H}_k = \hat{H}_{k-1} + \frac{y_{k-1} y'_{k-1}}{y'_{k-1} s_{k-1}} - \frac{\hat{H}_{k-1} s_{k-1} s'_{k-1} \hat{H}_{k-1}}{s'_{k-1} \hat{H}_{k-1} s_{k-1}}$ with $s_i = \theta_i - \theta_{i-1}$ and $y_i = \nabla Q(\theta_i) - \nabla Q(\theta_{i-1})$. Use **line search** to find α_k used for $\theta_{k+1} = \theta_k + \alpha_k d_k$; if α_k is too small, convergence is slow; if α_k is too large, may overshoot/diverge. *Exact line search* solves $\alpha_k^* = \arg \min_{\alpha > 0} Q(\theta_k + \alpha d_k)$ but is computationally expensive. For this reason, LBFGS uses the *Hager-Zhang line search* algorithm, which combines: the *Armijo condition* that satisfies $Q(\theta_k + \alpha d_k) \leq Q(\theta_k) + 10^{-4} \alpha \nabla Q_k(\theta_k)' d_k$ guaranteeing a sufficient decrease in objective, while avoiding overshooting; and the *Wolfe curvature condition* $|\nabla Q(\theta_k + \alpha d_k)' d_k| \leq 0.1 |\nabla Q_k(\theta_k)' d_k|$. **Output.** θ^{**} with objective value f^{**} .

7. Bootstrap GMM. To assess uncertainty, the code runs 500 independent GMMs. For each run $b = 1, \dots, 500$, it stores θ^{**} , $Q(\theta^{**})$, and fit metrics (MAE, RMSE, etc.).

E Tables

Table 1: Sector-level Inverse Markups Statistics (1997-2021)

Sector	Mean	Std	Min	Max	Sector	Mean	Std	Min	Max
Arts, Sports & Museums	0.76	0.03	0.71	0.84	Insurance & Related	0.98	0.01	0.96	1.01
Farms	0.82	0.04	0.77	0.95	Data & Internet Services	0.98	0.07	0.87	1.26
Primary Metals	0.87	0.02	0.81	0.90	Wood Products	0.98	0.01	0.96	0.99
Legal Services	0.87	0.06	0.75	0.98	Motor vehicle parts dealers	0.98	0.04	0.93	1.07
Water Transport	0.90	0.04	0.85	1.07	Waste & Remediation	0.98	0.01	0.97	1.00
Transit & Ground Pass	0.91	0.03	0.80	0.94	Textiles & Textile Mills	0.98	0.01	0.97	0.99
Other Prof & Tech Svcs	0.92	0.02	0.88	0.95	Nonmetallic Minerals	0.98	0.01	0.96	0.99
Securities & Investments	0.92	0.08	0.83	1.22	Rental & Leasing	0.98	0.02	0.94	1.06
Construction	0.93	0.02	0.89	0.99	Gnrl merchandise stores	0.98	0.02	0.96	1.02
Other Services	0.93	0.02	0.91	0.98	Oil & Gas Extraction	0.98	0.02	0.95	1.00
Air Transport	0.93	0.04	0.90	1.07	Utilities	0.98	0.01	0.97	0.99
Social Assistance	0.93	0.03	0.89	0.98	Electrical Equip & Comp	0.99	0.01	0.95	0.99
Ambulatory Health Care	0.93	0.03	0.87	0.98	Credit & Banking	0.99	0.01	0.95	1.00
Funds & Trusts	0.93	0.05	0.83	1.00	Fabricated Metals	0.99	0.01	0.96	1.00
Forestry & Fishing	0.94	0.02	0.91	0.99	Broadcast & Telecom	0.99	0.01	0.97	1.02
Administrative Support	0.94	0.01	0.92	0.97	Machinery	0.99	0.01	0.96	1.00
Truck Transport	0.94	0.04	0.91	1.01	Petroleum & Coal Products	0.99	0.01	0.95	1.02
Film & Sound Recording	0.95	0.01	0.93	0.98	Publish Ind exc Internet	0.99	0.02	0.97	1.04
IT Systems Design	0.96	0.03	0.91	1.03	Warehousing & Storage	0.99	0.00	0.98	1.00
Computers & Electronics	0.96	0.04	0.88	1.00	Chemical Products	0.99	0.01	0.97	1.03
Amusements & Recreation	0.96	0.02	0.93	1.01	Housing	0.99	0.02	0.98	1.05
Education Services	0.96	0.01	0.95	0.98	Mining, except oil and gas	0.99	0.02	0.94	1.05
Misc Manufacturing	0.96	0.01	0.94	0.98	Accommodation	0.99	0.03	0.97	1.08
Other Transport & Support	0.97	0.02	0.94	1.03	Food, Beverage & Tobacco	0.99	0.01	0.98	1.00
Wholesale Trade	0.97	0.01	0.95	1.01	Company Management	0.99	0.00	0.99	1.00
Furniture & Related	0.97	0.01	0.96	0.98	Other real estate	0.99	0.01	0.99	1.01
Food Services & Drinking	0.97	0.01	0.96	1.02	Other Transport Equip	0.99	0.00	0.98	1.00
Pipeline Transport	0.97	0.03	0.91	1.00	Nursing & Residential Care	1.00	0.00	0.99	1.00
Other retail	0.97	0.04	0.93	1.06	Hospitals	1.00	0.01	0.99	1.02
Paper Products	0.98	0.01	0.96	0.98	Plastics & Rubber	1.00	0.00	0.99	1.00
Apparel & Leather	0.98	0.04	0.93	1.09	Motor Vehicles & Parts	1.00	0.01	0.98	1.03
Food and beverage stores	0.98	0.04	0.93	1.05	Mining Support	1.01	0.06	0.93	1.14
Printing & Support	0.98	0.01	0.97	1.01	Rail Transport	1.01	0.01	1.00	1.03

Notes: Sector-level inverse markups ($\mu_i = \text{Cost}_i / \text{Sales}_i$), 1997–2021 summary statistics Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) for the 66 private industries. Values are constructed from BEA USE/MAKE and ILPA Extended Capital Detail; $\mu_i > 1$ denotes operating losses.

Table 2: Inverse Markups Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	0.950	$5.559e^{-2}$	0.753	0.882	0.919	0.969	0.991	0.996	0.999
1998	0.953	$5.300e^{-2}$	0.757	0.892	0.932	0.969	0.991	0.996	0.999
1999	0.956	$5.222e^{-2}$	0.752	0.900	0.936	0.973	0.991	0.996	1.041
2000	0.962	$6.315e^{-2}$	0.748	0.901	0.939	0.976	0.993	0.997	1.256
2001	0.957	$5.132e^{-2}$	0.752	0.900	0.939	0.974	0.992	0.997	1.035
2002	0.956	$4.885e^{-2}$	0.742	0.904	0.943	0.973	0.991	0.995	0.999
2003	0.956	$4.889e^{-2}$	0.721	0.906	0.936	0.971	0.991	0.995	0.999
2004	0.957	$5.080e^{-2}$	0.707	0.909	0.943	0.972	0.991	0.995	0.999
2005	0.958	$4.935e^{-2}$	0.724	0.912	0.939	0.973	0.992	0.996	0.999
2006	0.959	$4.953e^{-2}$	0.735	0.907	0.942	0.978	0.991	0.995	0.999
2007	0.961	$4.643e^{-2}$	0.754	0.915	0.941	0.979	0.992	0.996	1.000
2008	0.964	$5.787e^{-2}$	0.744	0.914	0.946	0.980	0.993	0.996	1.222
2009	0.958	$4.905e^{-2}$	0.771	0.904	0.936	0.976	0.991	0.995	1.027
2010	0.961	$4.365e^{-2}$	0.780	0.912	0.938	0.977	0.992	0.995	1.000
2011	0.964	$4.180e^{-2}$	0.773	0.916	0.946	0.979	0.991	0.996	1.025
2012	0.962	$4.264e^{-2}$	0.776	0.915	0.946	0.978	0.991	0.996	1.000
2013	0.965	$4.359e^{-2}$	0.759	0.921	0.948	0.980	0.991	0.996	1.029
2014	0.966	$4.272e^{-2}$	0.779	0.926	0.950	0.977	0.993	0.997	1.069
2015	0.966	$4.267e^{-2}$	0.769	0.923	0.946	0.977	0.994	1.003	1.057
2016	0.964	$4.699e^{-2}$	0.768	0.917	0.946	0.973	0.995	1.009	1.028
2017	0.964	$5.117e^{-2}$	0.785	0.903	0.943	0.969	0.998	1.016	1.083
2018	0.964	$5.590e^{-2}$	0.778	0.872	0.946	0.974	0.999	1.015	1.138
2019	0.965	$5.577e^{-2}$	0.795	0.895	0.945	0.974	0.994	1.021	1.117
2020	0.983	$5.612e^{-2}$	0.814	0.913	0.974	0.986	1.006	1.035	1.117
2021	0.989	$5.634e^{-2}$	0.780	0.946	0.966	0.983	1.018	1.056	1.108

Notes: Inverse markups by year (1997–2021). Cross-sector distributional statistics for $\mu_{i,t} = \text{Cost}_{i,t} / \text{Sales}_{i,t}$ across the 66 private industries. Columns report mean, standard deviation, minimum, P10, Q1, median, Q3, P90, and maximum. Costs equal labor + capital (nine asset classes) + intermediate inputs, constructed from BEA USE/MAKE and ILPA. Values $\mu > 1$ indicate operating losses.

Table 3: Sector-level Productivities Statistics (1997—2023)

Sector	Mean	Std	Min	Max	Sector	Mean	Std	Min	Max
Education Services	-0.76%	3.08%	-7.18%	4.45%	Paper Products	0.46%	2.44%	-4.80%	4.72%
Rental & Leasing	-0.74%	3.57%	-9.09%	5.92%	Textiles & Textile Mills	0.47%	3.06%	-6.49%	8.87%
Mining, exct oil & gas	-0.64%	4.73%	-10.5%	9.42%	Warehousing & Storage	0.51%	4.65%	-8.25%	10.5%
Construction	-0.64%	1.40%	-4.44%	1.37%	Housing	0.56%	2.38%	-4.07%	5.40%
Funds & Trusts	-0.59%	5.11%	-11.7%	10.62%	Other real estate	0.56%	2.38%	-4.07%	5.40%
Credit & Banking	-0.59%	5.77%	-8.81%	15.2%	Ambul. Health Care	0.56%	1.32%	-2.41%	2.89%
Other Services	-0.53%	1.95%	-7.81%	1.87%	Company Management	0.68%	3.15%	-6.53%	5.76%
Amus. & Recreation	-0.47%	3.65%	-9.83%	8.44%	Transit & Ground Pass	0.69%	3.37%	-6.06%	6.04%
Chemical Products	-0.38%	3.27%	-6.33%	4.28%	Food and bever stores	0.72%	2.05%	-3.44%	5.40%
Other Transport & Supp	-0.36%	3.18%	-9.38%	3.95%	Gnrl merchandise stores	0.72%	2.05%	-3.44%	5.40%
Securities & Invest.	-0.30%	11.3%	-25.7%	32.9%	Motor vehi parts dealers	0.72%	2.05%	-3.44%	5.40%
Hospitals	-0.21%	1.06%	-2.56%	1.95%	Other retail	0.72%	2.05%	-3.44%	5.40%
Nursing & Resid. Care	-0.21%	1.06%	-2.56%	1.95%	Apparel & Leather	0.73%	4.70%	-7.10%	12.5%
Truck Transport	-0.10%	1.90%	-6.14%	3.44%	Insurance & Related	0.73%	5.90%	-11.37%	14.6%
Fabricated Metals	-0.09%	2.75%	-8.19%	4.84%	Arts Sports & Museums	0.76%	4.28%	-14.9%	5.6%
Furniture & Related	-0.08%	2.59%	-7.07%	4.30%	Misc Manufacturing	0.77%	2.68%	-4.68%	6.19%
Wood Products	-0.05%	2.14%	-5.33%	4.07%	Farms	0.77%	3.43%	-4.11%	7.73%
Legal Services	-0.01%	3.10%	-9.62%	5.40%	Pipeline Transport	0.91%	11.58%	-30.3%	33.1%
Food, Bev. & Tobacco	0.09%	1.54%	-3.15%	3.99%	Administrative Support	0.94%	1.71%	-1.68%	4.05%
Utilities	0.10%	5.81%	-14.3%	17.2%	Accommodation	0.97%	5.12%	-8.65%	19.3%
Primary Metals	0.13%	3.91%	-15.5%	4.10%	Motor Vehicles & Parts	1.01%	5.89%	-17.9%	16.3%
Other Transport Equip	0.14%	3.13%	-7.08%	6.24%	Rail Transport	1.06%	2.66%	-4.40%	5.55%
Nonmetallic Minerals	0.14%	2.18%	-4.95%	3.26%	Forestry & Fishing	1.09%	3.76%	-7.32%	10.5%
Plastics & Rubber	0.21%	1.84%	-3.63%	3.46%	Printing & Support	1.14%	2.01%	-4.09%	4.34%
Social Assistance	0.22%	1.50%	-2.33%	3.81%	Publish Ind exc Internet	1.20%	5.86%	-18.8%	9.40%
Water Transport	0.24%	5.62%	-8.79%	12.4%	Oil & Gas Extraction	1.26%	11.73%	-18.9%	24.7%
Petroleum & Coal Prod.	0.26%	6.07%	-8.23%	18.4%	Broadcast & Telecom	1.36%	3.29%	-4.68%	8.47%
Waste & Remediation	0.26%	3.70%	-9.69%	11.0%	Film & Sound Record	1.80%	5.59%	-9.78%	9.49%
Wholesale Trade	0.29%	2.14%	-4.34%	4.15%	Data & Internet Serv	2.04%	11.08%	-21.4%	29.7%
Food Serv & Drinking	0.32%	2.15%	-5.65%	7.33%	Air Transport	2.23%	16.29%	-64.9%	37.7%
Machinery	0.38%	3.18%	-5.75%	6.10%	Mining Support	3.24%	14.89%	-25.6%	31.5%
Elect Equip & Comp	0.38%	3.54%	-7.71%	6.02%	IT Systems Design	3.37%	2.50%	-1.49%	7.90%
Other Prof & Tech Svcs	0.42%	2.15%	-2.32%	5.51%	Computers & Electro	4.22%	4.97%	-4.78%	16.3%

Notes: Annual sectoral productivity growth is computed as $d \log A_{i,t} = \log (TFP_{i,t}/TFP_{i,t-1})$ using BEA-BLS Industry-Level Production Accounts (ILPA) TFP levels. Sectors are harmonized to the 66 private Production-Accounts industries (government and “scrap/used” categories excluded). One-to-many mappings follow the conventions described in the data section (e.g., retail split into 441, 445, 452, 4A0; housing split into HS, ORE; hospitals/nursing split into 622, 623; broadcasting/telecom merged into 513). For each sector, the table reports the time-series Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) of $d \log A_{i,t}$ over 1997–2021; values are reported in percent per year.

Table 4: Productivity Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1998	0.92%	4.62%	-7.95%	-4.64%	-1.75%	0.28%	3.17%	6.11%	17.06%
1999	0.68%	4.39%	-21.39%	-3.23%	-0.69%	0.78%	2.56%	5.35%	9.64%
2000	0.14%	5.16%	-18.95%	-4.52%	-1.00%	0.70%	1.84%	3.89%	16.27%
2001	0.87%	7.04%	-14.30%	-4.29%	-2.28%	-0.01%	2.02%	4.95%	33.07%
2002	0.99%	6.93%	-30.28%	-6.29%	-1.03%	1.17%	2.62%	4.89%	29.73%
2003	1.58%	4.14%	-13.79%	-2.04%	-0.04%	1.85%	3.25%	4.61%	18.36%
2004	1.83%	3.75%	-6.53%	-1.62%	-0.51%	1.42%	3.22%	6.87%	13.58%
2005	0.16%	4.20%	-21.37%	-3.80%	-1.43%	0.48%	2.18%	3.93%	8.61%
2006	0.84%	3.63%	-6.20%	-3.72%	-1.33%	0.10%	2.73%	5.68%	12.43%
2007	-0.10%	5.13%	-14.00%	-6.30%	-3.10%	-0.18%	2.42%	5.96%	18.75%
2008	-0.04%	5.21%	-25.74%	-4.04%	-1.81%	-0.31%	2.50%	5.37%	12.45%
2009	-0.67%	7.14%	-17.94%	-6.78%	-4.34%	-0.84%	1.25%	5.43%	32.90%
2010	2.31%	4.49%	-13.89%	-1.84%	-0.02%	2.48%	4.49%	5.80%	16.28%
2011	-0.13%	3.42%	-11.71%	-3.33%	-1.30%	-0.33%	1.44%	3.42%	12.62%
2012	0.41%	3.92%	-10.75%	-3.64%	-1.21%	-0.25%	1.96%	5.08%	11.53%
2013	0.75%	2.72%	-6.56%	-2.60%	-0.90%	0.44%	2.32%	3.60%	8.27%
2014	0.29%	3.14%	-11.69%	-2.17%	-0.72%	0.34%	0.89%	2.19%	14.58%
2015	0.18%	5.42%	-25.63%	-4.77%	-1.71%	0.14%	1.85%	4.05%	24.70%
2016	-0.30%	3.61%	-17.68%	-4.44%	-1.28%	-0.16%	1.60%	3.51%	6.02%
2017	0.95%	3.43%	-5.57%	-1.80%	-0.98%	0.60%	2.33%	3.53%	20.98%
2018	0.34%	3.11%	-6.83%	-2.72%	-0.96%	0.22%	1.68%	2.49%	15.82%
2019	-0.02%	3.45%	-17.80%	-3.18%	-1.57%	-0.10%	1.98%	2.53%	7.84%
2020	-1.50%	9.05%	-64.95%	-6.06%	-2.49%	-0.67%	2.22%	3.29%	15.92%
2021	3.37%	7.48%	-17.43%	-2.40%	0.11%	2.96%	5.74%	7.88%	37.73%
2022	-0.30%	6.10%	-15.82%	-6.61%	-3.44%	-0.07%	1.88%	3.45%	31.52%
2023	-0.10%	5.19%	-9.89%	-5.52%	-3.02%	-0.11%	2.68%	4.68%	27.10%

Notes: Annual sectoral productivity growth is $d \log A_{i,t} = \log(TFP_{i,t}/TFP_{i,t-1})$ from the BEA-BLS Industry-Level Production Accounts (ILPA), mapped to the 66 private Production Accounts industries. For each calendar year, the table reports unweighted cross-industry moments: Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth), P10/P90 (10th/90th percentiles), and Q1/Median/Q3 (25th/50th/75th percentiles).

Table 5: Sector-level Revenue Centrality Statistics (1997-2021)

Sector	Mean	Std	Min	Max	Sector	Mean	Std	Min	Max
Arts, Sports & Museums	0.77	0.03	0.70	0.84	Computers & Electronics	1.01	0.05	0.91	1.05
Farms	0.80	0.04	0.74	0.94	Other Transport & Support	1.01	0.02	0.99	1.05
Primary Metals	0.84	0.04	0.75	0.90	Education Services	1.01	0.01	1.00	1.02
Funds & Trusts	0.91	0.08	0.75	1.04	Other real estate	1.01	0.00	1.01	1.02
Legal Services	0.92	0.06	0.79	1.00	Wholesale Trade	1.01	0.01	0.98	1.03
Water Transport	0.92	0.03	0.87	1.05	Printing & Support	1.01	0.01	1.01	1.02
Food, Beverage & Tobacco	0.95	0.01	0.93	0.97	Broadcast & Telecom	1.02	0.01	1.00	1.04
Transit & Ground Pass	0.95	0.04	0.83	0.99	Waste & Remediation	1.02	0.01	1.00	1.03
Securities & Investments	0.95	0.10	0.83	1.29	Other retail	1.02	0.04	0.98	1.08
Construction	0.96	0.01	0.94	1.00	Oil & Gas Extraction	1.02	0.03	0.93	1.05
Other Prof & Tech Svcs	0.96	0.01	0.93	0.97	Petroleum & Coal Products	1.02	0.03	0.94	1.05
Air Transport	0.96	0.04	0.94	1.09	Nonmetallic Minerals	1.02	0.02	0.99	1.04
Other Services	0.96	0.01	0.95	0.99	Food and beverage stores	1.02	0.03	0.99	1.07
Forestry & Fishing	0.97	0.03	0.92	1.02	Insurance & Related	1.02	0.02	1.00	1.06
Social Assistance	0.97	0.03	0.92	1.01	Utilities	1.02	0.01	1.00	1.05
Ambulatory Health Care	0.97	0.02	0.92	1.01	Data & Internet Services	1.02	0.08	0.89	1.31
Film & Sound Recording	0.97	0.01	0.95	1.00	Other Transport Equip	1.03	0.01	0.99	1.05
Truck Transport	0.98	0.03	0.95	1.03	Motor vehicle parts dealers	1.03	0.04	0.98	1.09
Fabricated Metals	0.98	0.03	0.90	1.01	Plastics & Rubber	1.03	0.01	1.01	1.05
Administrative Support	0.99	0.01	0.97	1.00	Chemical Products	1.03	0.02	1.00	1.06
Furniture & Related	0.99	0.02	0.95	1.01	Rental & Leasing	1.03	0.02	0.99	1.07
Misc Manufacturing	0.99	0.01	0.97	1.01	Mining, except oil and gas	1.03	0.02	0.96	1.06
Wood Products	0.99	0.01	0.97	1.01	General merchandise stores	1.03	0.02	1.01	1.06
Amusements & Recreation	0.99	0.02	0.97	1.02	Credit & Banking	1.03	0.02	0.97	1.05
Electrical Equip & Comp	1.00	0.03	0.92	1.02	Accommodation	1.03	0.02	1.01	1.11
Paper Products	1.00	0.01	0.98	1.01	Housing	1.03	0.01	1.03	1.04
Textiles & Textile Mills	1.00	0.01	0.98	1.03	Publish Ind exc Internet	1.04	0.02	1.01	1.10
Apparel & Leather	1.00	0.04	0.95	1.11	Company Management	1.04	0.01	1.00	1.05
Food Services & Drinking	1.00	0.01	0.99	1.03	Hospitals	1.04	0.01	1.03	1.05
Pipeline Transport	1.00	0.03	0.93	1.05	Warehousing & Storage	1.04	0.02	1.02	1.08
Machinery	1.00	0.03	0.92	1.03	Nursing & Residential Care	1.04	0.01	1.03	1.05
IT Systems Design	1.00	0.03	0.96	1.08	Rail Transport	1.04	0.01	1.03	1.06
Motor Vehicles & Parts	1.01	0.03	0.92	1.05	Mining Support	1.04	0.06	0.99	1.18

Notes: Sector-level revenue centralities ($F_i = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell \delta_h$), 1997–2021 summary statistics Mean and Std (standard deviation), Min/Max (lowest/highest sectoral growth) for the 66 private industries.

Table 6: Revenue Centrality Statistics by Year

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	0.991	$6.315e^{-2}$	0.772	0.925	0.958	1.011	1.035	1.048	1.079
1998	0.993	$6.093e^{-2}$	0.767	0.939	0.966	1.009	1.036	1.045	1.077
1999	0.994	$5.980e^{-2}$	0.761	0.942	0.980	1.011	1.034	1.043	1.087
2000	0.999	$7.041e^{-2}$	0.755	0.942	0.982	1.010	1.033	1.043	1.309
2001	0.994	$5.860e^{-2}$	0.756	0.944	0.969	1.009	1.032	1.044	1.087
2002	0.993	$5.696e^{-2}$	0.745	0.943	0.968	1.007	1.030	1.041	1.052
2003	0.993	$5.674e^{-2}$	0.719	0.943	0.968	1.007	1.032	1.041	1.048
2004	0.992	$6.024e^{-2}$	0.703	0.940	0.975	1.006	1.033	1.042	1.051
2005	0.992	$5.823e^{-2}$	0.725	0.927	0.973	1.007	1.030	1.040	1.054
2006	0.994	$5.854e^{-2}$	0.740	0.941	0.971	1.012	1.031	1.042	1.051
2007	0.992	$5.482e^{-2}$	0.755	0.942	0.973	1.007	1.027	1.037	1.041
2008	0.991	$6.624e^{-2}$	0.738	0.943	0.974	1.002	1.020	1.030	1.294
2009	0.992	$6.018e^{-2}$	0.739	0.931	0.980	1.012	1.030	1.039	1.048
2010	0.992	$5.340e^{-2}$	0.793	0.944	0.975	1.006	1.027	1.034	1.041
2011	0.991	$4.898e^{-2}$	0.782	0.941	0.976	1.001	1.023	1.030	1.062
2012	0.995	$5.190e^{-2}$	0.796	0.936	0.985	1.009	1.029	1.037	1.045
2013	0.994	$5.306e^{-2}$	0.770	0.938	0.982	1.007	1.025	1.039	1.070
2014	0.993	$5.284e^{-2}$	0.795	0.948	0.979	1.001	1.018	1.039	1.109
2015	0.995	$5.174e^{-2}$	0.776	0.952	0.975	1.004	1.020	1.044	1.090
2016	0.992	$5.835e^{-2}$	0.779	0.936	0.972	0.997	1.027	1.052	1.073
2017	0.991	$6.478e^{-2}$	0.784	0.926	0.970	0.995	1.032	1.059	1.120
2018	0.988	$6.883e^{-2}$	0.755	0.905	0.966	0.995	1.026	1.056	1.175
2019	0.990	$6.971e^{-2}$	0.756	0.925	0.970	0.998	1.029	1.058	1.154
2020	0.992	$7.068e^{-2}$	0.748	0.913	0.972	1.002	1.026	1.064	1.132
2021	0.990	$6.882e^{-2}$	0.764	0.922	0.967	1.001	1.029	1.063	1.111

Notes: Revenue centralities by year (1997–2021). Cross-sector distributional statistics for $F_i = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell \delta_h$ across the 66 private industries. Columns report mean, standard deviation, minimum, P10, Q1, median, Q3, P90, and maximum.

Table 7: Distortion Centrality — Occupations

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max	Corr _{income}
1997	1.117	$4.617e^{-2}$	1.029	1.059	1.089	1.113	1.135	1.160	1.384	-0.016
1998	1.111	$4.493e^{-2}$	1.027	1.057	1.087	1.108	1.128	1.152	1.351	-0.070
1999	1.108	$5.145e^{-2}$	1.029	1.057	1.080	1.101	1.124	1.146	1.361	0.029
2000	1.101	$5.446e^{-2}$	1.014	1.051	1.072	1.093	1.115	1.134	1.370	-0.008
2001	1.104	$4.803e^{-2}$	1.025	1.056	1.077	1.099	1.115	1.141	1.328	-0.001
2002	1.105	$5.982e^{-2}$	1.016	1.056	1.070	1.093	1.116	1.153	1.448	0.024
2003	1.104	$6.134e^{-2}$	1.014	1.053	1.069	1.093	1.115	1.154	1.495	0.012
2004	1.104	$6.440e^{-2}$	1.013	1.054	1.070	1.091	1.114	1.153	1.545	-0.044
2005	1.101	$5.887e^{-2}$	1.012	1.054	1.069	1.091	1.112	1.150	1.494	-0.049
2006	1.101	$5.959e^{-2}$	1.011	1.050	1.068	1.090	1.112	1.150	1.468	-0.034
2007	1.094	$5.607e^{-2}$	1.010	1.046	1.062	1.082	1.107	1.143	1.414	-0.095
2008	1.083	$5.963e^{-2}$	0.828	1.041	1.051	1.071	1.099	1.121	1.411	-0.152
2009	1.096	$5.796e^{-2}$	1.013	1.048	1.060	1.087	1.109	1.139	1.399	-0.115
2010	1.091	$5.127e^{-2}$	1.010	1.046	1.061	1.082	1.110	1.131	1.359	-0.106
2011	1.083	$4.903e^{-2}$	1.001	1.042	1.053	1.074	1.104	1.124	1.361	-0.124
2012	1.091	$5.242e^{-2}$	1.009	1.044	1.055	1.085	1.108	1.137	1.366	-0.114
2013	1.086	$5.231e^{-2}$	1.000	1.038	1.052	1.078	1.101	1.129	1.401	-0.088
2014	1.082	$5.146e^{-2}$	0.993	1.036	1.050	1.075	1.098	1.123	1.374	-0.096
2015	1.080	$5.004e^{-2}$	0.985	1.032	1.052	1.073	1.095	1.123	1.375	-0.073
2016	1.082	$5.377e^{-2}$	0.978	1.031	1.054	1.073	1.098	1.129	1.377	-0.059
2017	1.082	$5.706e^{-2}$	0.970	1.029	1.052	1.072	1.101	1.136	1.401	-0.052
2018	1.081	$5.587e^{-2}$	0.929	1.030	1.052	1.069	1.096	1.138	1.357	-0.030
2019	1.080	$5.760e^{-2}$	0.945	1.029	1.050	1.066	1.094	1.150	1.388	-0.047
2020	1.046	$5.379e^{-2}$	0.912	0.999	1.017	1.034	1.062	1.102	1.417	-0.022
2021	1.048	$5.578e^{-2}$	0.916	0.998	1.019	1.037	1.066	1.091	1.402	-0.032

Notes: Each row reports cross-sectional summary statistics of the *occupation-level distortion centrality* index $\delta_{o,t}$ for the set of occupations in year $t \in \{1997, \dots, 2021\}$. The index is computed from the occupational model using OEWS-based labor cost shares and the estimated parameterization; formally, $\delta_{o,t} \equiv \tilde{\Lambda}_{o,t} / \Lambda_{o,t}$. *Mean* and *Std* are unweighted across occupations; *Min*/*Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively. *Corr_{income}* is the cross-sectional Pearson correlation between $\delta_{o,t}$ and mean OEWS earning $\bar{J}_{o,t}$ in year t (negative values indicate higher-income occupations tend to have lower $\delta_{o,t}$).

Table 8: Distortion Centrality — Capital

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max
1997	1.096	$1.084e^{-2}$	1.076	1.084	1.090	1.100	1.103	1.105	1.113
1998	1.092	$1.031e^{-2}$	1.073	1.082	1.086	1.095	1.098	1.100	1.109
1999	1.085	$8.094e^{-3}$	1.069	1.078	1.081	1.089	1.090	1.093	1.095
2000	1.079	$8.459e^{-3}$	1.066	1.070	1.072	1.080	1.086	1.089	1.091
2001	1.083	$8.888e^{-3}$	1.065	1.072	1.079	1.087	1.090	1.091	1.091
2002	1.080	$8.860e^{-3}$	1.062	1.071	1.076	1.083	1.086	1.089	1.091
2003	1.079	$8.207e^{-3}$	1.064	1.069	1.075	1.080	1.085	1.086	1.090
2004	1.083	$6.952e^{-3}$	1.074	1.076	1.077	1.086	1.088	1.091	1.093
2005	1.082	$7.088e^{-3}$	1.070	1.075	1.076	1.085	1.087	1.089	1.090
2006	1.082	$7.351e^{-3}$	1.068	1.075	1.078	1.085	1.088	1.090	1.090
2007	1.077	$7.140e^{-3}$	1.063	1.069	1.075	1.079	1.082	1.084	1.086
2008	1.066	$9.115e^{-3}$	1.053	1.055	1.061	1.064	1.073	1.076	1.080
2009	1.080	$1.033e^{-2}$	1.056	1.070	1.081	1.082	1.084	1.087	1.094
2010	1.079	$1.053e^{-2}$	1.058	1.066	1.078	1.081	1.083	1.087	1.095
2011	1.073	$1.051e^{-2}$	1.057	1.059	1.070	1.073	1.079	1.084	1.091
2012	1.082	$1.160e^{-2}$	1.064	1.066	1.079	1.082	1.087	1.091	1.102
2013	1.076	$9.040e^{-3}$	1.060	1.061	1.074	1.077	1.082	1.083	1.083
2014	1.072	$1.149e^{-2}$	1.053	1.058	1.066	1.070	1.081	1.084	1.088
2015	1.067	$1.249e^{-2}$	1.047	1.052	1.062	1.068	1.074	1.078	1.087
2016	1.068	$1.480e^{-2}$	1.044	1.049	1.063	1.070	1.076	1.080	1.094
2017	1.067	$1.647e^{-2}$	1.040	1.049	1.057	1.067	1.076	1.081	1.096
2018	1.068	$1.330e^{-2}$	1.044	1.046	1.066	1.073	1.075	1.077	1.080
2019	1.061	$2.023e^{-2}$	1.029	1.040	1.043	1.060	1.072	1.082	1.094
2020	1.036	$1.619e^{-2}$	1.016	1.019	1.023	1.036	1.043	1.054	1.067
2021	1.036	$1.439e^{-2}$	1.011	1.020	1.023	1.037	1.048	1.050	1.052

Notes: Each row reports cross-sectional summary statistics of the *capital asset distortion centrality* index $\delta_{k,t}$ for the nine capital asset classes in year $t \in \{1997, \dots, 2021\}$. The index is computed from the occupation model with capital using BEA ILPA (Extended Capital Detail) to firm asset-specific capital shares; formally, $\delta_{k,t} \equiv \tilde{\Lambda}_{k,t} / \Lambda_{k,t}$. *Mean* and *Std* are unweighted across occupations; *Min*/*Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively.

Table 9: Expenditure Centrality — Occupations

Year	Mean	Std	Min	P10	Q1	Median	Q3	P90	Max	Corr _{income}
1997	0.998	$2.160e^{-3}$	0.992	0.995	0.996	0.997	0.999	1.000	1.004	0.446
1998	0.997	$2.376e^{-3}$	0.991	0.995	0.996	0.998	0.999	1.000	1.004	0.462
1999	0.999	$2.456e^{-3}$	0.993	0.996	0.998	0.999	1.001	1.002	1.005	0.637
2000	0.999	$2.752e^{-3}$	0.992	0.995	0.997	0.999	1.001	1.002	1.006	0.677
2001	0.997	$1.696e^{-3}$	0.992	0.995	0.996	0.997	0.998	0.999	1.001	0.451
2002	0.997	$1.877e^{-3}$	0.991	0.995	0.996	0.997	0.998	0.999	1.004	0.317
2003	0.999	$1.822e^{-3}$	0.994	0.997	0.998	0.999	1.000	1.002	1.007	0.455
2004	0.995	$3.708e^{-3}$	0.987	0.990	0.993	0.996	0.998	1.000	1.004	0.657
2005	0.994	$2.596e^{-3}$	0.987	0.991	0.993	0.995	0.996	0.998	1.002	0.701
2006	0.994	$1.927e^{-3}$	0.990	0.992	0.993	0.994	0.995	0.997	1.000	0.580
2007	0.997	$2.158e^{-3}$	0.991	0.994	0.995	0.997	0.998	1.000	1.001	0.638
2008	1.002	$4.393e^{-3}$	0.990	0.995	0.999	1.002	1.005	1.008	1.012	0.582
2009	0.994	$2.871e^{-3}$	0.987	0.990	0.991	0.993	0.996	0.997	1.002	0.565
2010	0.993	$1.613e^{-3}$	0.988	0.991	0.992	0.993	0.994	0.995	0.998	0.697
2011	0.996	$1.660e^{-3}$	0.991	0.994	0.995	0.996	0.997	0.998	0.999	0.703
2012	0.989	$1.613e^{-3}$	0.985	0.986	0.988	0.989	0.990	0.991	0.993	0.619
2013	0.987	$1.447e^{-3}$	0.984	0.985	0.987	0.988	0.988	0.989	0.991	0.520
2014	0.988	$1.945e^{-3}$	0.983	0.985	0.986	0.988	0.989	0.990	0.993	0.761
2015	0.989	$1.841e^{-3}$	0.984	0.987	0.988	0.989	0.990	0.991	0.994	0.700
2016	0.988	$1.333e^{-3}$	0.983	0.986	0.987	0.988	0.988	0.989	0.991	0.432
2017	0.985	$1.234e^{-3}$	0.982	0.984	0.984	0.985	0.986	0.987	0.988	0.344
2018	0.986	$1.090e^{-3}$	0.984	0.985	0.985	0.986	0.987	0.987	0.989	0.564
2019	0.985	$1.187e^{-3}$	0.981	0.983	0.984	0.985	0.986	0.987	0.988	0.486
2020	0.983	$1.430e^{-3}$	0.980	0.981	0.982	0.983	0.984	0.985	0.987	0.390
2021	0.984	$1.780e^{-3}$	0.978	0.982	0.983	0.984	0.985	0.986	0.988	-0.028

Notes: Each row reports cross-sectional summary statistics of the *occupation-level expenditure centrality* index $M_{o,t}$ for the set of occupations in year $t \in \{1997, \dots, 2021\}$. $M_{o,t}$ is computed in the occupational model by mapping CE-based occupation bundles to the 66 industries (via MAKE), propagating through the USE-MAKE production network with sectoral inverse markups, and using OEWS-based factor cost shares where required. *Mean* and *Std* are unweighted across occupations; *Min/Max* are cross-sectional extremes; *P10*, *Q1*, *Median*, *Q3*, and *P90* denote the 10th percentile, 25th percentile, median, 75th percentile, and 90th percentile, respectively. *Corr_{income}* is the cross-sectional Pearson correlation between $M_{o,t}$ and mean OEWS earning $\bar{J}_{o,t}$ in year t (positive values indicate higher-income occupations tend to have higher $M_{o,t}$).

F Robustness

This appendix reports robustness checks for the empirical sufficient statistics and the transfer counterfactuals. The checks address four margins: treatment of loss-making industry-years, elasticity calibration, geographic heterogeneity, and placebo transfer rules.

F.1 Treatment of $\mu > 1$

Table 10 evaluates how sensitive the empirical results are to the treatment of measured inverse markups above one. The theory implies that wedge variation matters because it determines the centrality weights that map income-demand recomposition into TFP. Operating-loss observations can therefore have leverage even if they are infrequent, since they enter the sufficient statistics F , M , and δ , and these weights govern how reallocations across firms, households, and factors affect aggregate productivity.

Table 10: Robustness to Alternative Treatments of Industry-Years with $\mu > 1$

Treatment	$\mu > 1$	$\lambda_{\mu>1}$	Corr. F	Corr. M	Corr. δ	$\mathcal{R}_{\text{share}}$
Baseline	0.078	0.107	1.000	1.000	1.000	17.48
$\mu \leq 1$	0.078	0.107	0.974	0.950	0.992	3.96
Drop losses	0.078	0.107	0.956	0.936	0.985	-1.89
Winsorize 1/99	0.078	0.107	0.992	0.989	0.994	12.26
Winsorize 5/95	0.078	0.107	0.935	0.922	0.928	4.33
Treatment	$\overline{\Delta TFP}_{\text{med}}$	$\overline{\Delta TFP}_{\text{prog}}$	PTT-slope	S.E.	Years $_{>0,\text{med}}$	Years $_{>0,\text{prog}}$
Baseline	6.77×10^{-6}	2.75×10^{-6}	-0.0150	0.0015	20	22
$\mu \leq 1$	6.94×10^{-6}	3.09×10^{-6}	-0.0134	0.0018	20	22
Drop losses	7.03×10^{-6}	2.97×10^{-6}	-0.0131	0.0017	20	22
Winsorize 1/99	6.80×10^{-6}	3.05×10^{-6}	-0.0149	0.0016	20	22
Winsorize 5/95	1.38×10^{-6}	5.46×10^{-7}	-0.0161	0.0016	13	13

Notes: The baseline uses the measured inverse markups $\mu_{i,t} = \text{Cost}_{i,t} / \text{Sales}_{i,t}$ directly, including industry-years with $\mu_{i,t} > 1$. The row $\mu \leq 1$ caps measured μ at one. “Drop losses” treats observations with $\mu_{i,t} > 1$ as missing and linearly interpolates within industry over time. The winsorization rows winsorize the measured μ distribution within each year. The columns $\mu > 1$ and $\lambda_{\mu>1}$ report the unweighted and sales-weighted shares of raw industry-year observations with $\mu_{i,t} > 1$. Corr. F , Corr. M , and Corr. δ are correlations with the baseline revenue centrality, occupation expenditure centrality, and distortion centrality. $\mathcal{R}_{\text{share}}$ is the non-parametric distributional reallocation covariance share, $100 \times \text{Cov}(R_t, d \log TFP_t) / \text{Var}(d \log TFP_t)$, computed from the terms-of-trade decomposition. The columns $\overline{\Delta TFP}_{\text{med}}$ and $\overline{\Delta TFP}_{\text{prog}}$ come from the parametric first-order counterfactual model and report average TFP effects under the median and progressive redistribution rules. PTT-slope is the 2019 employment-weighted slope of the progressive change in occupation-level PTT on log mean earnings; S.E. is its standard error.

The first two columns show that industry-years with $\mu_{i,t} > 1$ account for 7.8% of observations and 10.7% of sales. The sales-weighted share exceeds the unweighted share, so operating-loss observations are not confined to negligible sectors. The next columns show that the centrality rankings are stable across treatments. Even when μ is capped at one, the correlations with the baseline remain high: Corr. $(F) = 0.974$, Corr. $(M) = 0.950$, and Corr. $(\delta) = 0.992$. Mild winsorization at the 1/99 percentiles leaves these correlations even closer to one. Thus,

alternative treatments of $\mu > 1$ do not substantially reorder firms, households, or factors in the centrality system.

The magnitude of the nonparametric distributional-reallocation component is more sensitive. The baseline covariance share is 17.48%. Capping $\mu \leq 1$ reduces it to 3.96%, and dropping/interpolating operating-loss observations reduces it to -1.89% . Mild winsorization leaves a sizable positive contribution, 12.26%, while stronger 5/95 winsorization reduces the share to 4.33%. These patterns indicate that the historical distributional-reallocation share is tail-sensitive. This sensitivity is not incidental: in the model, large wedge observations identify parts of the production network where reallocations have larger marginal TFP consequences.

The parametric transfer-counterfactual results are more robust. Under the baseline, the average TFP effects of the median and progressive transfer rules are positive but small. They remain similar under capping, interpolation, and 1/99 winsorization, and the number of positive years remains unchanged at 20 for the median rule and 22 for the progressive rule. The 5/95 winsorization is the exception: it weakens the average TFP effects and reduces the number of positive years to 13 under both rules. Thus, the transfer gains are not driven solely by observations with $\mu > 1$, but they do depend on broader tail variation in measured wedges.

Finally, the incidence result is stable. The PTT-income slope under the progressive rule is negative in every specification, ranging from -0.0131 to -0.0161 . Hence, even when aggressive tail compression weakens the aggregate TFP effect, the progressive rule continues to deliver larger PTT gains to lower-earning occupations. Overall, the robustness exercise shows that the level of the historical distributional-reallocation share is sensitive to tail variation in μ , while the qualitative counterfactual and incidence conclusions are robust to treatments of operating-loss observations except under very aggressive winsorization.

F.2 Alternative Elasticity Calibrations

The counterfactual exercises use the calibrated normalized-CES model to map the nonparametric sufficient statistics into finite changes in TFP and household positional terms of trade. The baseline elasticity vector is selected by the GMM procedure described in the main text. Because some estimated elasticities are close to limiting cases, especially the baseline value of θ , I assess whether the transfer-counterfactual results depend on the particular best-fit GMM vector.

In this robustness exercise, I hold fixed all empirical objects and vary only the elasticity vector $\theta = (\theta, \theta_\ell, \theta_x, \rho)$. Across rows, I do not change the raw data, sectoral productivity shocks $d \log A_t$, markup shocks $d \log \mu_t$, input-output matrices, OEWS occupation mappings, CEX expenditure mappings, transfer rules, transfer size, treatment of μ , or the M -based targeting rule used to assign transfers. The exercise therefore isolates the role of substitution elasticities

in the parametric counterfactual model.

I consider seven elasticity calibrations. The first two use the GMM estimates: the best-fit vector and the mean vector across multi-start runs. I then add three transparent benchmarks: Cobb–Douglas, low substitution, and high substitution. Finally, I include two literature-inspired specifications, labeled Boehm-like (Boehm et al., 2014) and Atalay-like (Atalay, 2017), which span common values used in production-network applications.

Table 11: Elasticity Calibrations Used in Robustness Analysis

Calibration	θ	θ_ℓ	θ_x	ρ	Description
Best GMM	0.001	0.946	2.320	0.065	Baseline best-fit GMM vector.
Mean GMM	0.079	1.466	1.960	0.482	Mean parameter vector across multi-start GMM runs.
Cobb–Douglas	1.000	1.000	1.000	1.000	Unit-elasticity benchmark in all nests.
Low substitution	0.200	0.500	0.200	0.500	Benchmark with limited substitution in production and consumption.
High substitution	1.500	1.500	1.500	1.500	Benchmark with high substitution in all nests.
Boehm-like	0.030	1.000	0.400	0.500	Literature-inspired calibration with strong production complementarities.
Atalay-like	0.850	1.000	0.200	0.500	Literature-inspired calibration with higher factor-intermediate substitution and low intermediate-input substitution.

Notes: The table reports the seven elasticity vectors used in Appendix D.2. The first two rows use the estimated GMM vectors; the next three rows are benchmark specifications; the final two rows are literature-inspired calibrations. Each vector is imposed directly in the normalized-CES model to evaluate the sensitivity of the counterfactual results to substitution patterns.

For each elasticity vector, I first compute the model-implied annual TFP growth series over the 1998–2021 growth sample and compare it with BLS aggregate TFP growth. Fit Corr. is the correlation between model-implied and observed TFP growth. Fit IA is the index of agreement. The full output also reports RMSE and mean bias.

I then rerun both transfer experiments. The transfer size remains \$1,000 per worker, with aggregate transfers expressed as a share of GDP and imposed to be budget neutral within each year. The first rule is the median-split M rule, which taxes occupations above median expenditure centrality and rebates to occupations below the median. The second rule is the progressive hierarchical rule, which restricts transfers to move from higher-income, higher- M occupations toward lower-income, lower- M occupations.

The counterfactuals are computed in two environments. Panel A considers pure redistribution: productivity and markup shocks are set to zero, $d \log A_t = d \log \mu_t = 0$, so the reported effects isolate the redistribution channel. Panel B considers the historical-shock environment: the model includes the observed sectoral productivity and markup shock sequences, and I compare the transfer counterfactual with the corresponding no-transfer baseline. In both panels, the reported TFP effect is $\Delta d \log TFP_t = d \log TFP_t^{\text{transfer}} - d \log TFP_t^{\text{baseline}}$. Finally, I compute

household incidence using PTTs. For each calibration, I form the 2019 occupation-level change $\Delta d \log PTT_o = d \log PTT_{o,2019}^{\text{prog}} - d \log PTT_{o,2019}^{\text{baseline}}$. I then estimate an employment-weighted cross-occupation regression of $\Delta d \log PTT_o$ on log mean occupation earnings. The compact table reports the slope from this regression. A negative slope means that the progressive transfer raises PTTs relatively more for lower-earning occupations. Bottom-Q gain reports the employment-weighted average $\Delta d \log PTT_o$ for occupations in the bottom earnings quintile.

Table 12: Robustness to Alternative Elasticity Calibrations

Panel A: Pure Redistribution					
Elasticities		$\overline{\Delta TFP}$	Years _{>0}	PTT slope	Bottom-Q gain
Best GMM		4.84×10^{-6}	24	-0.0151	0.02277
Mean GMM		5.39×10^{-6}	24	-0.0151	0.02288
Cobb–Douglas		5.76×10^{-6}	24	-0.0151	0.02290
Low substitution		5.56×10^{-6}	24	-0.0151	0.02270
High substitution		5.83×10^{-6}	24	-0.0151	0.02295
Boehm-like		5.72×10^{-6}	24	-0.0151	0.02281
Atalay-like		6.20×10^{-6}	24	-0.0151	0.02287

Panel B: Historical Shocks						
Elasticities	Fit Corr.	Fit IA	$\overline{\Delta TFP}$	Years _{>0}	PTT slope	Bottom-Q gain
Best GMM	0.849	0.907	2.75×10^{-6}	22	-0.0150	0.02260
Mean GMM	0.850	0.907	3.33×10^{-6}	22	-0.0150	0.02279
Cobb–Douglas	0.849	0.902	3.71×10^{-6}	22	-0.0151	0.02295
Low substitution	0.846	0.903	3.41×10^{-6}	22	-0.0151	0.02280
High substitution	0.852	0.902	3.86×10^{-6}	22	-0.0151	0.02300
Boehm-like	0.846	0.903	3.56×10^{-6}	22	-0.0151	0.02280
Atalay-like	0.844	0.899	4.02×10^{-6}	22	-0.0151	0.02289

Notes: The table reports the progressive M -guided redistribution counterfactual under alternative calibrations of the four CES elasticities. Panel A sets productivity and markup shocks to zero and isolates the redistribution channel. Panel B uses the historical productivity and markup shocks and compares the transfer counterfactual with the corresponding no-transfer baseline. Fit Corr. is the correlation between model-implied and observed aggregate TFP growth over 1998–2021; Fit IA is the index of agreement. ΔTFP is the average annual change in model-implied TFP growth generated by the progressive transfer rule relative to the no-transfer baseline. Years_{>0} counts years with positive $\Delta d \log TFP_t$. PTT slope is the 2019 employment-weighted slope from a regression of occupation-level PTT changes on log mean earnings. Bottom-Q gain is the 2019 employment-weighted average PTT gain for occupations in the bottom earnings quintile.

Table 12 shows that the transfer-counterfactual results are not driven by the best-fit GMM elasticity vector. Across all seven calibrations, model fit remains similar: in the historical-shock environment, the correlation between model-implied and observed TFP growth ranges from 0.844 to 0.852, and the index of agreement ranges from 0.899 to 0.907.

The TFP effects are also stable. In the pure redistribution environment, the progressive transfer rule raises model-implied TFP growth in every year under every calibration. Average annual ΔTFP ranges from 4.84×10^{-6} under the Best GMM calibration to 6.20×10^{-6} under the Atalay-like calibration. With historical productivity and markup shocks, average ΔTFP remains positive for all seven calibrations, ranging from 2.75×10^{-6} to 4.02×10^{-6} , and the effect is positive in 22 of 24 years in every case.

The incidence results are essentially unchanged. The 2019 PTT-income slope is approximately -0.015 in every row, and the bottom-quintile PTT gain remains close to 0.023 log points. Thus, the progressive incidence of the policy is not sensitive to whether the model uses the best GMM vector, the mean GMM vector, Cobb–Douglas elasticities, low- or high-substitution benchmarks, or the Boehm-like and Atalay-like alternatives. Overall, the main counterfactual conclusion is robust: M -guided redistribution raises TFP in most or all years and delivers larger purchasing-power gains to lower-earning occupations.

F.3 Geographic Specification

For each year $t \in [1997, 2021]$, I use QCEW to measure state s 's labor-cost share in industry i as $\alpha_{is,t} = \frac{\text{Labor Income}_{is,t}}{\sum_{s'} \text{Labor Income}_{is',t}}$. This creates 52 labor types: the 50 states, the District of Columbia, and a residual U.S. labor type capturing labor compensation not assigned to a state in the QCEW mapping. I stack these labor types with the nine capital-asset cost shares to form the industry-by-factor cost-share matrix α_t . I construct state-specific consumption bundles, $\beta_{s,t}^{PCE}$, from state PCE commodity expenditures mapped to the 66-industry classification. The residual non-labor bundle is $\beta_t^{NL} = (1 - \sum_s \Lambda_{s,t})^{-1} (\beta_t^{IO} - \sum_s \Lambda_{s,t} \beta_{s,t}^{PCE})$, which reconciles aggregate final demand with the Production Accounts. The state-level model spans 66 sectors and 62 household types: 52 labor-income households, nine capital-asset households, and one residual profit-income household. Capital and profit households receive the residual non-labor bundle, so aggregate final demand is preserved and household budgets exhaust GDP. Relative to the representative-household model, the index of agreement falls from 0.923 to 0.916, and the correlation with BLS aggregate TFP growth falls from 0.875 to 0.867. These small changes, $\Delta IA = -7e^{-3}$ and $\Delta \text{corr.} = -8e^{-3}$, indicate that introducing QCEW- and PCE-based state heterogeneity has negligible effects on aggregate TFP fit.

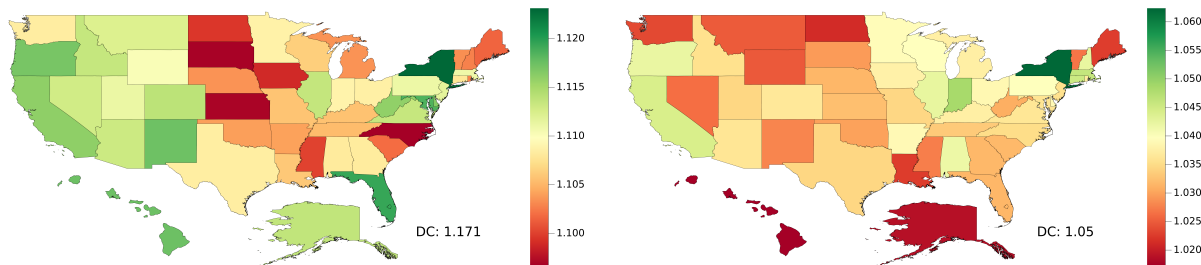
This geographic specification is not intended to replace the occupational model used in the counterfactuals. Its role is diagnostic. It asks whether the aggregate distributional-reallocation result survives a different household partition, one based on where labor income is earned and where consumption bundles are located rather than on occupations and income-conditioned CEX bundles. The near-identical aggregate fit indicates that introducing state-level heterogeneity does not distort the aggregate TFP accounting. At the same time, state-level centralities are less dispersed than occupational centralities. Thus, the geographic model provides a robustness check on the aggregate decomposition, while the occupational model remains the policy-relevant specification for targeted redistribution.

Figures 2 and 3 display state-level distortion centrality, $\delta_{s,t}$, and expenditure centrality, $M_{s,t}$, for 1997 and 2021. The maps illustrate that the centrality objects are not merely aggregate

residuals: they assign local reallocation weights to concrete household partitions. From **Theorem 2**, labor-income reallocation raises TFP when income moves from high- δ to low- δ labor types, while expenditure reallocation raises TFP when spending moves from high- M to low- M households. The state-level maps therefore show which geographic reallocations would be locally productivity-enhancing, holding centralities fixed.

In 1997, the three states with the highest δ are DC, New York, and Florida, while the lowest- δ states are North Carolina, South Dakota, and Kansas. The highest- M states are Arkansas, Wyoming, and DC, while the lowest- M states are Hawai'i, Alaska, and California. In 2021, the highest- δ states are New York, DC, and Indiana, while the lowest- δ states are Hawai'i, Alaska, and North Dakota. The highest- M states are DC, Hawai'i, and Maryland, while the lowest- M states are Wyoming, Arkansas, and Nebraska. Classifying states by within-year quartiles of δ and M , four groups emerge. **1. High- δ , high- M states:** DC and Maryland in 1997; California, DC, and New York in 2021. **2. High- δ , low- M states:** Alaska, California, Hawai'i, Oregon, and West Virginia in 1997; Illinois, New Hampshire, and Wisconsin in 2021. **3. Low- δ , high- M states:** Iowa, Kansas, North Carolina, and North Dakota in 1997; Hawai'i, New Mexico, and Vermont in 2021. **4. Low- δ , low- M states:** Maine, New Hampshire, and Rhode Island in 1997; Montana and Wyoming in 2021.

Figure 2: Distortion Centrality by State: 1997 and 2021

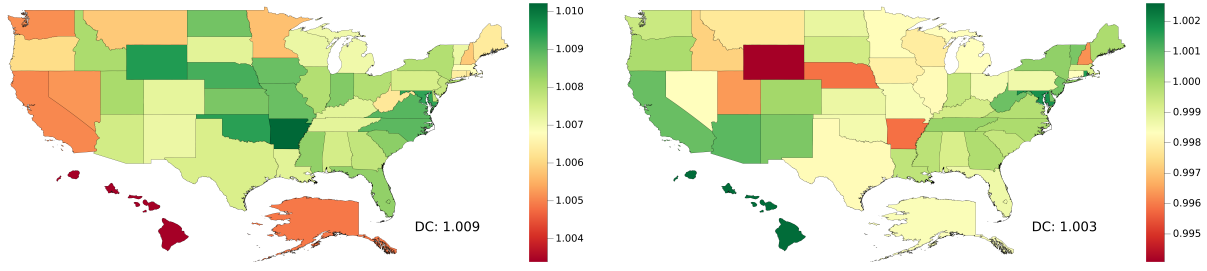


Notes: Panels show the distortion centrality index $\delta_{s,t}$ for each state s . Left: 1997; right: 2021. Distortion centrality is computed as the ratio of cost-based value-added exposure to factor-income exposure, $\tilde{\Lambda}_{s,t} / \Lambda_{s,t}$, under the state-level parameterization described above. Darker green (red) denotes high (low) distortion centrality. The label next to DC reports its value.

Figures 2–3 report state-level distortion and expenditure centralities for all years from 1997 to 2021. Year-to-year levels and rankings are relatively stable, but comparing 1997 with 2021 reveals several shifts: DC and New York remain consistently high- δ , high- M states; Alabama, Kansas, Maine, North and South Dakota, and the Carolinas remain consistently low- δ ; Alaska and Hawai'i move from high- δ , low- M to low- δ , high- M ; and Arkansas, Nebraska, and Wyoming move from high- M to low- M .

The state maps also show that M -rankings are empirical and time-varying. In 1997, the distributive terms of trade imply that TFP would rise locally by shifting expenditure from high- M Interior states, such as Arkansas, Nebraska, and Wyoming, toward lower- M Pacific

Figure 3: Expenditure Centrality by State: 1997 and 2021



Notes: Panels show the expenditure centrality index $M_{s,t}$ for each state s . Left: 1997; right: 2021. Expenditure centrality is $M_{s,t} = \sum_{h \in \mathcal{H}} \mathcal{C}_{sh,t} \delta_{h,t}$, the distortion-centrality exposure embedded in state s 's expenditure bundle. Darker green (red) denotes high (low) expenditure centrality. The label next to DC reports its value.

states, such as California, Hawai'i, and Oregon. By 2021, the ranking reverses: the Pacific states become relatively high- M , while several Interior states become relatively low- M . This reversal reflects changes in expenditure composition. Pacific-state expenditure shifts toward high- F sectors, especially housing, while Interior-state expenditure shifts relatively more toward other sectors such as ambulatory health care and securities, commodity contracts, and investments. Housing's share in the Pacific states rises by 1.4 percentage points, from 16.6% to 18.0%, while the Interior's housing share falls by 0.6 percentage points, from 14.1% to 13.5%. Ambulatory health care in the Interior rises by 1.4 percentage points, from 5.3% to 6.4%, while the Pacific states rise by 0.7 percentage points, from 7.6% to 8.3%. Securities, commodity contracts, and investments rise by 1.3 percentage points in the Interior, from 1.7% to 3.0%, compared with 0.5 percentage points in the Pacific states, from 1.0% to 1.5%. Hawai'i is the clearest case: its M moves from among the lowest in 1997 to among the highest in 2021, driven in part by its housing share rising from 16.9% to 20.1%.¹¹

Overall, the geographic specification supports the paper's main empirical interpretation. The aggregate distributional-reallocation component is stable under a household partition based on geography, so the result is not driven mechanically by the occupation-to-consumption mapping. However, geographic centralities are less dispersed and less directly tied to redistributive policy margins than occupational centralities. This is why the main counterfactuals use occupations: the geographic exercise validates the aggregate decomposition, while the occupational specification provides the sharper M -ranking for expenditure-targeted transfers.

¹¹Housing: Arkansas (13.9% → 12.5%); Nebraska (13.3% → 13.8%); Wyoming (15.6% → 14.3%); California (17.1% → 17.1%); Hawai'i (16.9% → 20.1%); Oregon (15.9% → 16.9%). Ambulatory health care services: Arkansas (6.5% → 7.6%); Nebraska (5.1% → 7.4%); Wyoming (4.4% → 5.2%); California (8.3% → 9.1%); Hawai'i (8.1% → 8.0%); Oregon (6.5% → 8.0%). Securities, commodity contracts, and investments: Arkansas (1.2% → 2.7%); Nebraska (1.9% → 2.7%); Wyoming (1.9% → 3.7%); California (1.3% → 2.0%); Hawai'i (0.9% → 1.4%); Oregon (0.9% → 1.3%).

F.4 Placebo Transfer Rules

This appendix evaluates whether the productivity effects of the redistribution counterfactual are specific to targeting households by expenditure centrality M , rather than reflecting progressivity or fiscal scale alone. The main counterfactuals transfer resources across occupational labor households using $M_{o,t}$, which summarizes how changes in occupation o 's purchasing power feed back into aggregate productivity through the production network. The theory predicts that transfers toward lower- M households should raise TFP, while transfers in the opposite direction should be weaker or negative.

All placebo exercises use the same calibrated model, occupation mapping, employment and earnings data, sectoral productivity shocks, and markup shocks as the baseline analysis. Transfers are defined at the worker level: a transfer of \$1,000 to occupation o in year t corresponds to an aggregate transfer of $1000 N_{o,t}$, where $N_{o,t}$ is occupation-year employment. Each rule is budget balanced within year. I report fiscal volume as the positive side of the transfer schedule, $V_t = \sum_o \max\{T_{o,t}, 0\}$, expressed as a share of GDP.

I compare four transfer rules. The first is the M -median rule, the main benchmark: occupations below the employment-weighted median of $M_{o,t}$ receive transfers, while occupations above the median finance them. The second is the progressive M rule: transfers move only from higher-income, higher- M occupations to lower-income, lower- M occupations, with matches formed sequentially along the income and M rankings. The third is reverse- M , which uses the same median split but reverses the direction of redistribution, taxing low- M occupations and transferring to high- M occupations. This is the direct placebo opposite of the theory's prediction. The fourth is income-only progressivity, which ignores $M_{o,t}$ and transfers from high-earning to low-earning occupations. This isolates whether the results are driven by progressivity rather than explicit M -targeting.

For comparability, the compact table reports same-volume results. In each year, every non-random rule is rescaled to match the fiscal volume of the M -median rule, $V_t^* = V_t^{M\text{-median}}$. Thus, differences across rows come from the allocation of transfers across occupations, not from differences in fiscal scale.

Each rule is evaluated in two environments. In the pure redistribution environment, productivity and markup shocks are set to zero, so the comparison isolates the general equilibrium effect of redistribution alone. In the historical-shock environment, the model uses the observed sequences of productivity and markup shocks, and each transfer counterfactual is compared with a no-transfer baseline exposed to the same shocks. For each rule and year, I compute $\Delta d \log TFP_t = d \log TFP_t(\text{transfer}) - d \log TFP_t(\text{no transfer})$. I also report the effect per billion

dollars transferred, $\Delta d \log TFP_t / (V_t / 10^9)$. Household incidence is measured using occupation-level PTT changes, $\Delta d \log PTT_{o,t} = d \log PTT_{o,t}(\text{transfer}) - d \log PTT_{o,t}(\text{no transfer})$. The table reports employment-weighted average PTT gains for occupations in the bottom and top earnings quintiles.

Table 13: Placebo Transfer Rules and the Role of M -Targeting

Panel A: Pure Redistribution						
Rule	Volume	$\overline{\Delta TFP}$	Years $_{>0}$	$\Delta TFP / \$1bn$	Bott-Q PTT	Top-Q PTT
M -median	0.822	2.68×10^{-5}	24	2.80×10^{-7}	1.86×10^{-2}	-1.26×10^{-2}
Prog. M	0.822	3.30×10^{-5}	24	3.36×10^{-7}	6.34×10^{-2}	-2.63×10^{-2}
Reverse- M	0.822	-2.68×10^{-5}	0	-2.80×10^{-7}	-2.58×10^{-2}	1.19×10^{-2}
Income-only	0.822	1.66×10^{-5}	23	1.81×10^{-7}	4.59×10^{-2}	-1.32×10^{-2}
Panel B: Historical Shocks						
Rule	Volume	$\overline{\Delta TFP}$	Years $_{>0}$	$\Delta TFP / \$1bn$	Bott-Q PTT	Top-Q PTT
M -median	0.822	1.33×10^{-5}	20	1.50×10^{-7}	1.89×10^{-2}	-1.25×10^{-2}
Prog. M	0.822	1.56×10^{-5}	20	1.75×10^{-7}	6.29×10^{-2}	-2.60×10^{-2}
Reverse- M	0.822	-1.33×10^{-5}	4	-1.50×10^{-7}	-2.60×10^{-2}	1.18×10^{-2}
Income-only	0.822	1.20×10^{-5}	18	1.42×10^{-7}	4.59×10^{-2}	-1.31×10^{-2}

Notes: The table compares the main M -guided transfer rules with placebo redistribution rules. Panel A reports pure redistribution, with productivity and markup shocks set to zero. Panel B reports the historical-shock environment, where observed productivity and markup shocks are included and each transfer counterfactual is compared with the corresponding no-transfer baseline. Transfers are budget balanced within year and are defined over occupational labor households. Volume is the average annual positive transfer volume, expressed as a percent of GDP. The table reports same-volume results: each non-random rule is rescaled within year to match the fiscal volume of the M -median rule. $\overline{\Delta TFP}$ is the average annual change in model-implied TFP growth relative to the no-transfer baseline. Years $_{>0}$ counts the number of years with positive $\Delta d \log TFP_t$. $\Delta TFP / \$1bn$ reports the average TFP effect per billion dollars transferred. Bottom-Q PTT and Top-Q PTT report the 2019 employment-weighted average change in occupation-level PTT for occupations in the bottom and top earnings quintiles. The reverse- M rule transfers resources from low- M to high- M occupations. The income-only rule ranks occupations by earnings only and does not use M .

Table 13 shows that the productivity and incidence effects are tied to the direction of M -targeting rather than to fiscal volume or progressivity alone. By construction, all rules in the compact table have the same average fiscal volume, equal to 0.822% of GDP. Hence, differences across rows reflect where transfers are allocated, not the aggregate size of the policy.

Panel A isolates the redistribution channel. The two M -guided rules raise TFP in every year. The M -median rule raises TFP by 2.68×10^{-5} on average, while the progressive M rule raises TFP by 3.30×10^{-5} . Reversing the direction of the M -median rule flips the sign: the reverse- M rule lowers TFP by 2.68×10^{-5} on average and is negative in every year. This is the sharpest placebo test. It holds fixed the fiscal volume and the median-split structure, but reverses only the direction of redistribution across M .

The income-only rule is positive but weaker than the progressive M rule. In the pure redistribution environment, income-only progressivity raises TFP by 1.66×10^{-5} , with positive effects in 23 out of 24 years. This is consistent with the fact that earnings and M are correlated in the data. However, the effect is roughly half as large as the progressive M rule, so progressivity alone does not reproduce the full productivity effect of explicit M -targeting.

Panel B repeats the comparison with historical productivity and markup shocks. The average effects are smaller, but the ranking is unchanged. The M -median and progressive M rules remain positive on average and in 20 out of 24 years. Reverse- M again has the opposite sign, lowering TFP by 1.33×10^{-5} on average and producing positive effects in only 4 years. Income-only progressivity remains positive but weaker than progressive M , with positive effects in 18 years.

The incidence results reinforce the mechanism. The M -guided rules raise PTT for bottom-earnings occupations and reduce PTT for top-earnings occupations. This pattern is strongest under the progressive M rule: in the pure redistribution environment, bottom-quintile occupations gain 6.34×10^{-2} , while top-quintile occupations lose 2.63×10^{-2} . The historical-shock panel gives nearly the same incidence profile. Reverse- M flips the incidence pattern, lowering PTT for the bottom quintile and raising it for the top quintile. Income-only progressivity remains progressive in incidence, but delivers smaller TFP gains than progressive M .

Overall, the placebo results support the mechanism in the paper. Holding fiscal volume fixed, moving resources toward lower- M occupations raises TFP, while moving resources in the opposite direction lowers TFP. Income progressivity by itself generates positive but weaker effects. The main counterfactual therefore does not arise mechanically from transfer size or from redistribution toward low-income households alone; it depends on targeting the occupations whose purchasing-power changes have the strongest productivity implications through the production network.